

## 1. Overview

The MCU integrates multiple series of software-compatible Arm<sup>®</sup>-based 32-bit cores that share a common set of Renesas peripherals to facilitate design scalability and efficient platform-based product development.

The MCU in this series incorporates a high-performance Arm<sup>®</sup> Cortex<sup>®</sup>-M85 core running up to 1 GHz and Arm<sup>®</sup> Cortex<sup>®</sup>-M33 core running up to 250 MHz with the following features:

- Up to 1 MB MRAM
- 2 MB SRAM (256 KB of CM85 TCM RAM, 128 KB CM33 TCM RAM, 1664 KB of user SRAM)
- Arm<sup>®</sup> Ethos<sup>™</sup>-U55 NPU
- Octal Serial Peripheral Interface (OSPI)
- Layer 3 Ethernet Switch Module (ESWM), USBFS, USBHS, SD/MMC Host Interface
- Graphics LCD Controller (GLCDC)
- 2D Drawing Engine (DRW)
- MIPI DSI/CSI interface
- Analog peripherals
- Security and safety features

### 1.1 Function Outline

**Table 1.1 Arm core**

| Feature                                        | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arm <sup>®</sup> Cortex <sup>®</sup> -M85 core | <ul style="list-style-type: none"> <li>• Maximum operating frequency: up to 1 GHz</li> <li>• Arm<sup>®</sup> Cortex<sup>®</sup>-M85 core <ul style="list-style-type: none"> <li>– Revision: (r1p1-00rel0)</li> <li>– ARMv8.1-M architecture profile</li> <li>– Armv8-M Security Extension</li> <li>– Floating Point Unit (FPU) compliant with the ANSI/IEEE Std 754-2008 Scalar half, single, and double-precision floating-point operation</li> <li>– M-profile Vector Extension (MVE) Integer, half-precision, and single-precision floating-point MVE (MVE-F)</li> </ul> </li> <li>• Arm<sup>®</sup> Memory Protection Unit (Arm MPU) <ul style="list-style-type: none"> <li>– Protected Memory System Architecture (PMSAv8)</li> <li>– Secure MPU (MPU_S): 8 regions</li> <li>– Non-secure MPU (MPU_NS): 8 regions</li> </ul> </li> <li>• SysTick timer <ul style="list-style-type: none"> <li>– Embeds two SysTick timers: Secure instance (SysTick_S) and Non-secure instance (SysTick_NS)</li> <li>– Driven by CPUCLK0 or MOCO divided by 8</li> </ul> </li> <li>• CoreSight<sup>™</sup> ETM-M85</li> </ul> |
| Arm <sup>®</sup> Cortex <sup>®</sup> -M33 core | <ul style="list-style-type: none"> <li>• Maximum operating frequency: up to 250 MHz</li> <li>• Arm<sup>®</sup> Cortex<sup>®</sup>-M33 core <ul style="list-style-type: none"> <li>– Revision: (r0p4-00rel2)</li> <li>– ARMv8-M architecture profile</li> <li>– Armv8-M Security Extension</li> <li>– Armv8-DSP Extension</li> <li>– Floating Point Unit (FPU) compliant with the ANSI/IEEE Std 754-2008 single-precision floating-point operation</li> </ul> </li> <li>• Arm<sup>®</sup> Memory Protection Unit (Arm MPU) <ul style="list-style-type: none"> <li>– Protected Memory System Architecture (PMSAv8)</li> <li>– Secure MPU (MPU_S): 8 regions</li> <li>– Non-secure MPU (MPU_NS): 8 regions</li> </ul> </li> <li>• SysTick timer <ul style="list-style-type: none"> <li>– Embeds two SysTick timers: Secure instance (SysTick_S) and Non-secure instance (SysTick_NS)</li> <li>– Driven by CPUCLK1 or MOCO divided by 8</li> </ul> </li> <li>• CoreSight<sup>™</sup> ETM-M33</li> </ul>                                                                                                                |

**Table 1.2 Memory**

| Feature               | Functional description                                                                                                             |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Code MRAM             | Maximum 1 MB of code MRAM.<br>See <a href="#">section 60, MRAM</a> .                                                               |
| Flash memory          | System in package (SiP) maximum 8 MB serial flash memory.<br>See <a href="#">section 45.4, SiP Product Configuration</a> .         |
| Option-setting memory | The option-setting memory determines the state of the MCU after a reset.<br>See <a href="#">section 7, Option-Setting Memory</a> . |
| SRAM                  | On-chip high-speed SRAM with Error Correction Code (ECC).<br>See <a href="#">section 59, SRAM</a> .                                |
| OTP                   | On-chip OTP contains First Stage Bootloader (FSBL)<br>General purpose 96-byte OTP                                                  |

**Table 1.3 System (1 of 2)**

| Feature                                            | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating modes                                    | Three operating modes: <ul style="list-style-type: none"> <li>• Single-chip mode</li> <li>• JTAG boot mode</li> <li>• SCI/USB boot mode</li> </ul> See <a href="#">section 4, Operating Modes</a> .                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Resets                                             | This MCU provides the following 21 types of reset.<br>See <a href="#">section 6, Resets</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Programmable Voltage Detection (PVD)               | The Programmable Voltage Detection (PVD) module monitors the voltage level input to the VCC pin. The detection level can be selected by register settings. The PVD module consists of five separate voltage level detectors (PVD0, PVD1, PVD2, PVD4, PVD5). These PVDs measure the voltage level input to the VCC pin. PVD registers allow your application to configure detection of VCC changes at various voltage thresholds.<br>See <a href="#">section 8, Programmable Voltage Detection (PVD)</a> .                                                                                                                    |
| Clocks                                             | <ul style="list-style-type: none"> <li>• Main clock oscillator (MOSC)</li> <li>• Sub-clock oscillator (SOSC)</li> <li>• High-speed on-chip oscillator (HOCO)</li> <li>• Middle-speed on-chip oscillator (MOCO)</li> <li>• Low-speed on-chip oscillator (LOCO)</li> <li>• PLL1/PLL2</li> <li>• Clock out support</li> </ul> See <a href="#">section 9, Clock Generation Circuit</a> .                                                                                                                                                                                                                                         |
| Clock Frequency Accuracy Measurement Circuit (CAC) | The Clock Frequency Accuracy Measurement Circuit (CAC) counts pulses of the clock to be measured (measurement target clock) within the time generated by the clock selected as the measurement reference (measurement reference clock), and determines the accuracy depending on whether the number of pulses is within the allowable range. When measurement is complete or the number of pulses within the time generated by the measurement reference clock is not within the allowable range, an interrupt request is generated.<br>See <a href="#">section 10, Clock Frequency Accuracy Measurement Circuit (CAC)</a> . |
| Interrupt Controller Unit (ICU)                    | The Interrupt Controller Unit (ICU) controls which event signals are linked to the Nested Vector Interrupt Controller (NVIC), DMA Controller (DMAC) module and the Data Transfer Controller (DTC) modules. The ICU also controls non-maskable interrupts.<br>See <a href="#">section 14, Interrupt Controller Unit (ICU)</a> .                                                                                                                                                                                                                                                                                               |
| Low power modes                                    | Power consumption can be reduced in multiple ways, including setting clock dividers, controlling EBCLK output, controlling SDCLK output, stopping modules, power gating control, selecting operating power control modes in normal operation, and transitioning to low power modes and processor low power modes.<br>See <a href="#">section 11, Low Power Mode</a> .                                                                                                                                                                                                                                                        |
| Battery backup function                            | A battery backup function is provided for partial powering by a battery. The battery-powered area includes the RTC, SOSC, backup register, tamper detection and VBATT_R voltage drop detection and switch between VCC and VBATT.<br>See <a href="#">section 12, Battery Backup Function</a> .                                                                                                                                                                                                                                                                                                                                |
| Register write protection                          | The register write protection function protects important registers from being overwritten due to software errors. The registers to be protected are set with the Protect Register (PRCR_S and PRCR_NS).<br>See <a href="#">section 13, Register Write Protection Function</a> .                                                                                                                                                                                                                                                                                                                                             |

**Table 1.3 System (2 of 2)**

| Feature                      | Functional description                                                                                                 |
|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Memory Protection Unit (MPU) | All bus masters have Memory Protection Units (MPUs).<br>See <a href="#">section 16, Memory Protection Unit (MPU)</a> . |

**Table 1.4 Event link**

| Feature                     | Functional description                                                                                                                                                                                                                                                                      |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event Link Controller (ELC) | The Event Link Controller (ELC) uses the event requests generated by various peripheral modules as source signals to connect them to different modules, allowing direct link between the modules without CPU intervention.<br>See <a href="#">section 20, Event Link Controller (ELC)</a> . |

**Table 1.5 Direct memory access**

| Feature                        | Functional description                                                                                                                                                                                                                                                                                                 |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Transfer Controller (DTC) | A Data Transfer Controller (DTC) module is provided for transferring data when activated by an interrupt request.<br>See <a href="#">section 18, Data Transfer Controller (DTC)</a> .                                                                                                                                  |
| DMA Controller (DMAC)          | The 8-channel direct memory access controller (DMAC) that can transfer data without intervention from the CPU. When a DMA transfer request is generated, the DMAC transfers data stored at the transfer source address to the transfer destination address.<br>See <a href="#">section 17, DMA Controller (DMAC)</a> . |

**Table 1.6 External bus interface**

| Feature        | Functional description                                                                                                                                                                                                                                                                                                                                                             |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| External buses | <ul style="list-style-type: none"> <li>CS area (ECBI): Connected to the external devices (external memory interface)</li> <li>SDRAM area (ECBI): Connected to the SDRAM (external memory interface)</li> <li>OSPI0 area (OSPI0BI): Connected to the OSPI0 (external device interface)</li> <li>OSPI1 area (OSPI1BI): Connected to the OSPI1 (external device interface)</li> </ul> |

**Table 1.7 Timers (1 of 2)**

| Feature                                            | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General PWM Timer (GPT)                            | The General PWM Timer (GPT) is a 32-bit timer with GPT32 × 14 channels. PWM waveforms can be generated by controlling the up-counter, down-counter, or the up- and down-counter. In addition, PWM waveforms can be generated for controlling brushless DC motors. The GPT can also be used as a general-purpose timer.<br>See <a href="#">section 23, General PWM Timer (GPT)</a> .                                                                                                                                  |
| PWM Delay Generation Circuit (PDG)                 | The PWM Delay Generation circuit (PDG) has 4 channels delay circuits that can connect to the GPT. The PDG can control the rise and fall edge timing with which the PWM output for the GPT320 through the GPT323.<br>See <a href="#">section 24, PWM Delay Generation Circuit (PDG)</a> .                                                                                                                                                                                                                             |
| Port Output Enable for GPT (POEG)                  | The Port Output Enable (POEG) function can place the General PWM Timer (GPT) output pins in the output disable state<br>See <a href="#">section 22, Port Output Enable for GPT (POEG)</a> .                                                                                                                                                                                                                                                                                                                          |
| Low Power Asynchronous General Purpose Timer (AGT) | The Low Power Asynchronous General Purpose Timer (AGT) is a 16-bit timer that can be used for pulse output, external pulse width or period measurement, and counting external events. This timer consists of a reload register and a down counter. The reload register and the down counter are allocated to the same address, and can be accessed with the AGT register.<br>See <a href="#">section 25, Low Power Asynchronous General Purpose Timer (AGT)</a> .                                                    |
| Ultra-Low-Power Timer (ULPT)                       | The Ultra-Low-Power Timer (ULPT) is a 32-bit timer which can be used for outputting pulses or counting external events. This 32-bit timer consists of reload registers and a down-counter. The reload registers and the down-counter are allocated to the same address and can be accessed through the ULPTCNT register.<br>See <a href="#">section 26, Ultra-Low-Power Timer (ULPT)</a> .                                                                                                                           |
| Realtime Clock (RTC)                               | The realtime clock (RTC) has two counting modes, calendar count mode and binary count mode, that are used by switching register settings. For calendar count mode, the RTC has a 100-year calendar from 2000 to 2099 and automatically adjusts dates for leap years. For binary count mode, the RTC counts seconds and retains the information as a serial value. Binary count mode can be used for calendars other than the Gregorian (Western) calendar.<br>See <a href="#">section 27, Realtime Clock (RTC)</a> . |

**Table 1.7 Timers (2 of 2)**

| Feature                           | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Watchdog Timer (WDT)              | The Watchdog Timer (WDT) is a 14-bit down counter that can be used to reset the MCU when the counter underflows because the system has run out of control and is unable to refresh the WDT. In addition, the WDT can be used to generate a non-maskable interrupt or an underflow interrupt.<br>See <a href="#">section 28, Watchdog Timer (WDT)</a> .                                                                                                                                                                                                                                                                                |
| Independent Watchdog Timer (IWDT) | The Independent Watchdog Timer (IWDT) has a 14-bit down-counter, which resets the MCU by a reset output when the down-counter underflows. Alternatively, generation of an interrupt request when the counter underflows can be selected. This enables detection of a program runaway taking the refresh interval into account. The IWDT has two start modes: auto start mode, in which counting automatically starts after release from the reset state, and register start mode, in which counting is started by refreshing (writing to a specific register).<br>See <a href="#">section 29, Independent Watchdog Timer (IWDT)</a> . |

**Table 1.8 Communication interfaces (1 of 2)**

| Feature                                                     | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serial Communications Interface (SCI)                       | The Serial Communications Interface (SCI) × 10 channels have asynchronous and synchronous serial interfaces: <ul style="list-style-type: none"> <li>• Asynchronous interfaces (UART and Asynchronous Communications Interface Adapter (ACIA))</li> <li>• 8-bit clock synchronous interface</li> <li>• Simple IIC (master-only)</li> <li>• Simple SPI</li> <li>• Smart card interface</li> <li>• Manchester interface</li> <li>• Simple LIN interface</li> </ul> The smart card interface complies with the ISO/IEC 7816-3 standard for electronic signals and transmission protocol. All channels have FIFO buffers to enable continuous and full-duplex communication, and the data transfer speed can be configured independently using an on-chip baud rate generator.<br>The maximum rate supported on this MCU. Refer to the electrical characteristics for the actual rate.<br>See <a href="#">section 39, Serial Communications Interface (SCI)</a> . |
| I <sup>2</sup> C Bus interface (IIC)                        | The I <sup>2</sup> C Bus interface (IIC) has 3 channels. The IIC module conforms with and provides a subset of the NXP I <sup>2</sup> C (Inter-Integrated Circuit) bus interface functions.<br>See <a href="#">section 40, I<sup>2</sup>C Bus Interface (IIC)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| I3C Bus Interface (I3C)                                     | The I3C Bus Interface (I3C) has 1 channel. The I3C module conform with and provide a subset of the NXP I <sup>2</sup> C (Inter-Integrated Circuit) bus interface functions and a subset of the MIPI I3C.<br>See <a href="#">section 41, I3C Bus Interface (I3C)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Serial Peripheral Interface (SPI)                           | The Serial Peripheral Interface (SPI) provides high-speed full-duplex synchronous serial communications with multiple processors and peripheral devices.<br>The maximum rate supported on this MCU. Refer to the electrical characteristics for the actual rate.<br>See <a href="#">section 44, Serial Peripheral Interface (SPI)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Control Area Network with Flexible Data-Rate Module (CANFD) | The CAN with Flexible Data-Rate (CANFD) module can handle classical CAN frames and CANFD frames complied with ISO 11898-1 standard.<br>The module supports 4 transmit buffers per channel and 16 receive buffers per channel.<br>See <a href="#">section 42, CAN with Flexible Data-rate (CANFD)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| USB 2.0 Full-Speed module (USBFS)                           | The USB 2.0 Full-Speed module (USBFS) can operate as a host controller or device controller. The module supports full-speed and low-speed (host controller only) transfer as defined in Universal Serial Bus Specification 2.0. The module has an internal USB transceiver and supports all of the transfer types defined in Universal Serial Bus Specification 2.0. The USB has buffer memory for data transfer, providing a maximum of 10 pipes. Pipes 1 to 9 can be assigned any endpoint number based on the peripheral devices used for communication or based on your system.<br>See <a href="#">section 37, USB 2.0 Full-Speed Module (USBFS)</a> .                                                                                                                                                                                                                                                                                                   |

**Table 1.8 Communication interfaces (2 of 2)**

| Feature                                    | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USB 2.0 High-speed Module (USBHS)          | The USB 2.0 High-Speed Module (USBHS) that operates as a host or a device controller compliant with the Universal Serial Bus (USB) Specification revision 2.0. The host controller supports USB 2.0 high-speed, full speed, and low-speed transfers, and the device controller supports USB 2.0 high-speed and full-speed transfers.<br>The USBHS has an internal USB transceiver and supports all of the transfer types defined in the USB 2.0 specification.<br>The USBHS has FIFO buffer for data transfers, providing a maximum of 10 pipes.<br>See <a href="#">section 38, USB 2.0 High-Speed Module (USBHS)</a> .                                                                                                                                             |
| Octal Serial Peripheral Interface (OSPI)   | The Octal Serial Peripheral Interface (OSPI) is a memory controller that supports Expanded Serial Peripheral Interface (xSPI) (JEDEC Standard JESD251, JESD251-1 and JESD252). The OSPI supports 1-bit, 2-bit, 4-bit and 8-bit protocols.<br>JESD251 specifies two interface profiles where profile 1.0 is Octal SPI and profile 2.0 is HyperBus™ (HyperRAM™ and HyperFlash™). OSPI supports QSPI protocol.<br>See <a href="#">section 45, Octal Serial Peripheral Interface (OSPI)</a> .                                                                                                                                                                                                                                                                           |
| Serial Sound Interface Enhanced (SSIE)     | The Serial Sound Interface Enhanced (SSIE) peripheral provides functionality to interface with digital audio devices for transmitting I <sup>2</sup> S/Monaural/TDM audio data over a serial bus. The SSIE supports an audio clock frequency of up to 50 MHz, and can be operated as a slave or master receiver, transmitter, or transceiver to suit various applications. The SSIE includes 32-stage FIFO buffers in the receiver and transmitter, and supports interrupts and DMA-driven data reception and transmission.<br>See <a href="#">section 47, Serial Sound Interface Enhanced (SSIE)</a> .                                                                                                                                                             |
| SD/MMC Host Interface (SDHI)               | The Secure Digital (SD) Card and Multi Media Card (MMC) Host Interface provides the functionality required to connect a variety of external memory cards to the MCU. The SDHI supports both 1- and 4-bit buses for connecting memory cards that support SD, SDHC, and SDXC formats. When developing host devices that are compliant with the SD Specifications, you must comply with the SD Host/Ancillary Product License Agreement (SD HALA). The MMC interface supports 1-bit, 4-bit, and 8-bit MMC buses that provide eMMC 4.51 (JEDEC Standard JESD 84-B451) device access. This interface also provides backward compatibility and supports high-speed SDR transfer modes.<br>See <a href="#">section 48, SD/MMC Host Interface (SDHI)</a> .                  |
| Layer 3 Ethernet Switch Module (ESWM)      | The Layer 3 Ethernet Switch Module (ESWM) consists of two channels of Gigabit Ethernet controller, an Ethernet switch with high level routing capability, and multi-protocol interface support. The Gigabit Ethernet controller conforms to the definition of the Ethernet MAC (Media Access Control) layer in the IEEE 802.3 standard. This can transmit and receive Ethernet (IEEE 802.3) frames by connecting with an external physical-layer LSI chip (PHY-LSI) which complies with the standard. The Ethernet switch allows autonomous frame routing within a same network interface protocol, or between different network interfaces protocols or optimized gateway applications.<br>See <a href="#">section 30, Layer 3 Ethernet Switch Module (ESWM)</a> . |
| Pulse Density Modulation Interface (PDMIF) | PDM-IF has maximum three channels that are connectable with external microphone which outputs the pulse density modulated (PDM) signal. PDM-IF is connectable with up to three external microphones. PDM-IF can filter and convert 1-bit digital data streams that were pulse density modulated at a high sampling rate into 20-bit or 16-bit digital data at a lower sampling rate.<br>See <a href="#">section 50, Pulse Density Modulation Interface (PDM-IF)</a> .                                                                                                                                                                                                                                                                                               |

**Table 1.9 Analog (1 of 2)**

| Feature                       | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16-bit A/D Converter (ADC16H) | A 16-bit A/D Converter is provided. Up to 23 analog input channels are selectable. Temperature sensor output, and internal reference voltage and VBATT 1/6 voltage monitor are selectable for conversion.<br>See <a href="#">section 54, 16-bit A/D Converter (ADC16H)</a> .                                                                                                                                                                                                                                                                                                                                                           |
| 12-bit D/A Converter (DAC12)  | A 12-bit D/A Converter (DAC12) is provided.<br>See <a href="#">section 55, 12-Bit D/A Converter (DAC12)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Temperature Sensor (TSN)      | The on-chip Temperature Sensor (TSN) determines and monitors the die temperature for reliable operation of the device. The sensor outputs a voltage directly proportional to the die temperature, and the relationship between the die temperature and the output voltage is fairly linear. The output voltage is provided to the ADC16H for conversion and can be further used by the end application. The sensor outputs an abnormal temperature detection signal to the reset control circuit and can be used to prevent the malfunction due to abnormal temperature.<br>See <a href="#">section 56, Temperature Sensor (TSN)</a> . |

**Table 1.9 Analog (2 of 2)**

| Feature                               | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High-Speed Analog Comparator (ACMPHS) | The High-Speed Analog Comparator (ACMPHS) can be used to compare an analog input voltage with a reference voltage and to provide a digital output based on the result of conversion. Both the analog input voltage and the reference voltage can be provided to the ACMPHS from internal sources (D/A converter output or internal reference voltage) and an external source. Such flexibility is useful in applications that require go/no-go comparisons to be performed between analog signals without necessarily requiring A/D conversion.<br>See <a href="#">section 57, High-Speed Analog Comparator (ACMPHS)</a> . |

**Table 1.10 Human machine interfaces**

| Feature                         | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Graphics LCD Controller (GLCDC) | The Graphics LCD Controller (GLCDC) provides multiple functions and supports various data formats and panels. Key GLCDC features include: <ul style="list-style-type: none"> <li>GLCDC0BI/GLCDC1BI master function for accessing graphics data</li> <li>Superimposition of three planes (single-color background plane, graphic 1-plane, and graphic 2-plane)</li> <li>Support for many types of 32-bit or 16-bit per pixel graphics data and 8-bit, 4-bit, or 1-bit LUT data format</li> <li>Digital interface signal output supporting a video image size of WXGA.</li> </ul> See <a href="#">section 64, Graphics LCD Controller (GLCDC)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2D Drawing Engine (DRW)         | The 2D Drawing Engine (DRW) provides flexible functions that can support almost any object geometry rather than being bound to only a few specific geometries such as lines, triangles, or circles. The edges of every object can be independently blurred or antialiased. Rasterization is executed at one pixel per clock on the bounding box of the object from left to right and top to bottom. The DRW can also raster from bottom to top to optimize the performance in certain cases. In addition, optimization methods are available to avoid rasterization of many empty pixels of the bounding box.<br>The distances to the edges of the object are calculated by a set of edge equations for every pixel of the bounding box. These edge equations can be combined to describe the entire object. If a pixel is inside the object, it is selected for rendering. If it is outside, it is discarded. If it is on the edge, an alpha value can be chosen proportional to the distance of the pixel to the nearest edge for antialiasing.<br>Every pixel that is selected for rendering can be textured. The resulting ARGB quadruple can be modified by a general raster operation approach independently for each of the four channels. The ARGB quadruples can then be blended with one of the multiple blend modes of the DRW. The DRW provides two inputs (texture read and framebuffer read), and one output (framebuffer write). The internal color format is always ARGB (8888). The color formats from the inputs are converted to the internal format on read and a conversion back is made on write.<br>See <a href="#">section 63, 2D Drawing Engine (DRW)</a> . |
| Capture Engine Unit (CEU)       | The Capture Engine Unit (CEU) is a capture module that fetches image data externally input and transfers it to the memory.<br>See <a href="#">section 61, Capture Engine Unit (CEU)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| MIPI DSI interface              | The MIPI DSI interface module has a Transmitter function for MIPI Alliance Specification for Display Serial Interface 2 (DSI-2). This module supports MIPI Alliance Specification for Display Serial Interface 2 (DSI-2) Specification. And it works with MIPI Alliance Specification for D-PHY Specification. This module provides a solution for transmitting MIPI DSI-2 compliant digital video and packets.<br>See <a href="#">section 65, MIPI PHY</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| MIPI CSI interface              | MIPI CSI-2 can receive signals conforming to the MIPI CSI-2 standard, extracts video data from various packets, and sends them to Video Input Module in the subsequent stage.<br>See <a href="#">section 67, MIPI CSI Interface</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Video Input Module (VIN)        | Video Input Module (VIN) can receive video data received from MIPI CSI-2, and perform appropriate image processing for each. The image-processed data is temporarily stored in the FIFO and transferred to an external memory.<br>See <a href="#">section 68, Video Input Module (VIN)</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Table 1.11 Neural processing**

| Feature             | Functional description                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arm® Ethos™-U55 NPU | <ul style="list-style-type: none"> <li>Maximum operating frequency: 500 MHz</li> <li>Arm® Ethos™-U55 NPU               <ul style="list-style-type: none"> <li>Revision: r2p0_01eac0</li> <li>Number of 8x8 MACs: 256 units</li> <li>Network: 8-bit and 16-bit integer quantized Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN)</li> <li>Compression: 8-bits weights</li> </ul> </li> </ul> |

**Table 1.12 Data processing**

| Feature                                  | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyclic Redundancy Check (CRC) calculator | The Cyclic Redundancy Check (CRC) calculator generates CRC codes to detect errors in the data. The bit order of CRC calculation results can be switched for LSB-first or MSB-first communication. Additionally, various CRC-generation polynomials are available. The snoop function allows monitoring reads from and writes to specific addresses. This function is useful in applications that require CRC code to be generated automatically in certain events, such as monitoring writes to the serial transmit buffer and reads from the serial receive buffer. See <a href="#">section 49, Cyclic Redundancy Check (CRC)</a> . |
| Data Operation Circuit (DOC)             | The Data Operation Circuit (DOC) compares, adds, and subtracts 32-bits data. When a selected condition applies, 32-bit data is compared and an interrupt can be generated. See <a href="#">section 58, Data Operation Circuit (DOC)</a> .                                                                                                                                                                                                                                                                                                                                                                                            |

**Table 1.13 Security**

| Feature                       | Functional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security function             | <ul style="list-style-type: none"> <li>ARMv8-M TrustZone security</li> <li>Privileged control</li> <li>Device lifecycle management</li> <li>Authentication Level (AL)</li> <li>Key injection</li> <li>Secure pin multiplexing</li> <li>HUK zeroization</li> <li>VBATT backup registers zeroization</li> <li>Secure boot</li> <li>Secure factory programming</li> </ul> See <a href="#">section 52, Security Features</a> .                                                                                          |
| Renesas Secure IP (RSIP-E50D) | <ul style="list-style-type: none"> <li>Symmetric cryptography: AES and ChaCha20-Poly1305</li> <li>Asymmetric cryptography: RSA and ECC</li> <li>Message digest computation: HASH, HMAC</li> <li>128-bit true random number generation circuit</li> <li>256-bit Hardware Unique Key (HUK)</li> <li>128-bit unique ID</li> <li>OEM boot loader version</li> <li>Key data for the decryption on-the-fly (DOTF)</li> <li>SPA/DPA Protections</li> </ul> See <a href="#">section 53, Renesas Secure IP (RSIP-E50D)</a> . |
| Decryption on-the-fly (DOTF)  | Decryption on-the-fly (DOTF) decrypts the encrypted content stored in the external memory in real-time. See <a href="#">section 46, Decryption On The Fly (DOTF)</a> .                                                                                                                                                                                                                                                                                                                                              |

## 1.2 Block Diagram

Figure 1.1 shows a block diagram of the MCU superset. Some individual devices within the group have a subset of the features.

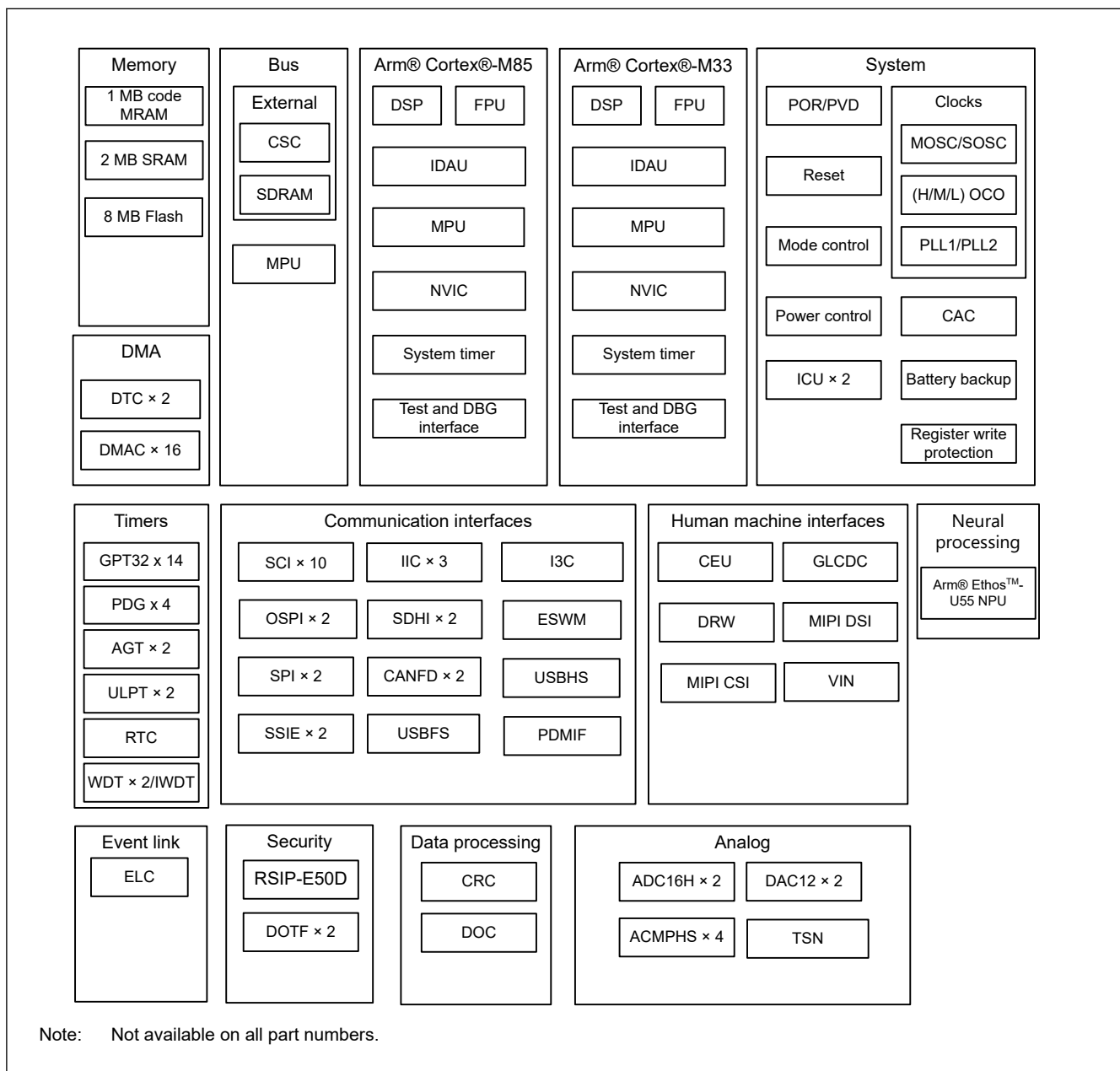


Figure 1.1 Block diagram

## 1.3 Part Numbering

Figure 1.2 shows the product part number information, including memory capacity and package type. Table 1.14 shows a list of products.

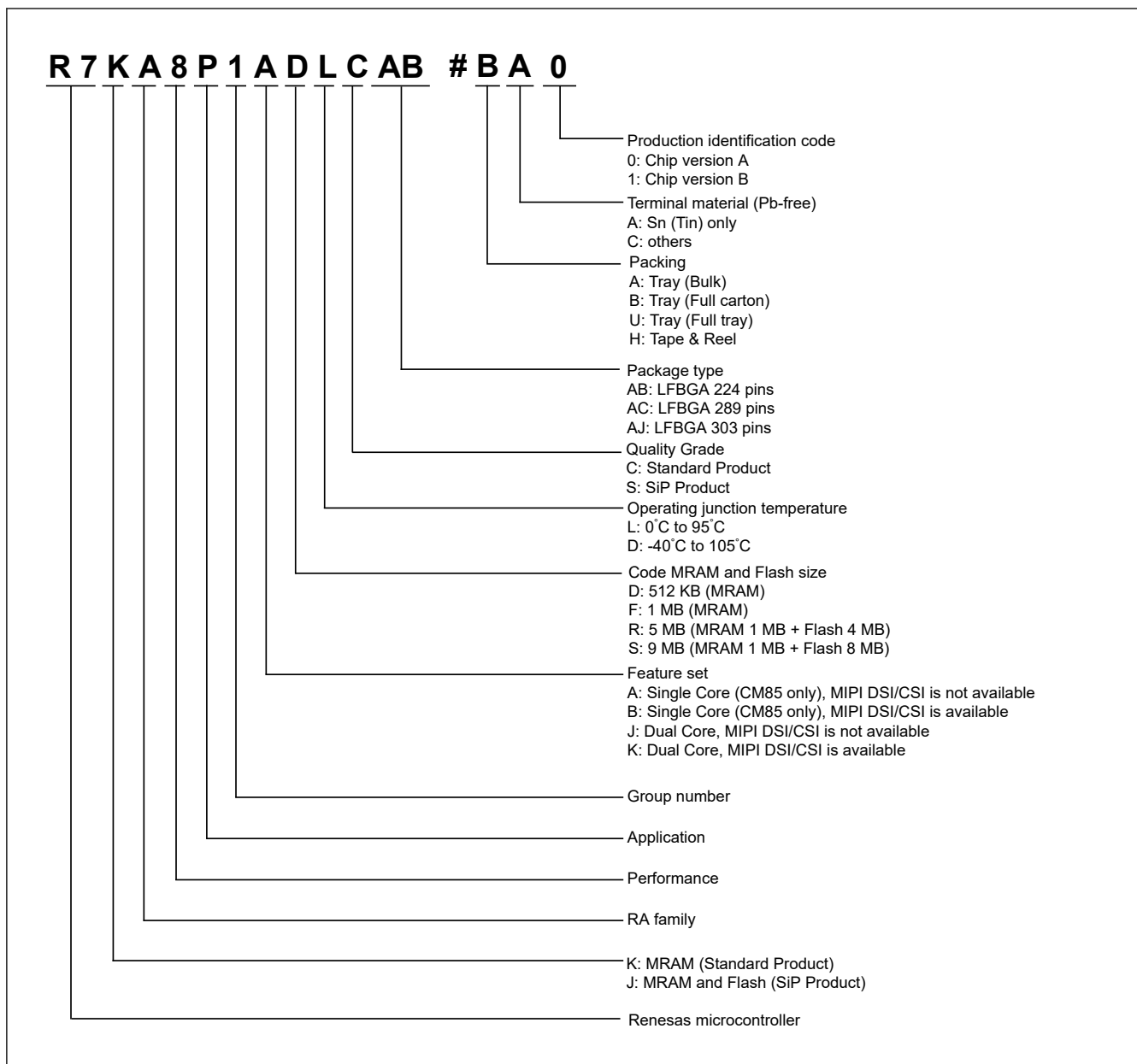


Figure 1.2 Part numbering scheme

Table 1.14 Product list

| Product part number | Product group | CPU    | MIPI DSI/CSI | Package code  | Code MRAM  | SRAM | Flash | Operating junction temperature |               |  |  |               |  |  |
|---------------------|---------------|--------|--------------|---------------|------------|------|-------|--------------------------------|---------------|--|--|---------------|--|--|
| R7KA8P1ADLCAB       | A             | single | —            | PLBG0224JA-A  | 512 KB     | 2 MB | —     | 0 to 95 °C                     |               |  |  |               |  |  |
| R7KA8P1ADLCAC       |               |        |              | PLBG0289JA-A  |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1ADDCAB       | B             |        |              | PLBG0224JA-A  | 1 MB       |      |       | -40 to 105 °C                  |               |  |  |               |  |  |
| R7KA8P1ADDCAC       |               |        |              | PLBG0289JA-A  |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1AFLCAB       | A             |        |              | PLBG0224JA-A  |            |      |       | 0 to 95 °C                     |               |  |  |               |  |  |
| R7KA8P1AFLCAC       |               |        |              | PLBG0289JA-A  |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1AFDCAB       | B             |        |              | PLBG0224JA-A  |            |      |       | -40 to 105 °C                  |               |  |  |               |  |  |
| R7KA8P1AFDCAC       |               |        |              | PLBG0289JA-A  |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BDLCAB       | A             | ✓      | PLBG0224JA-A | 512 KB        | 0 to 95 °C |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BDLCAC       | PLBG0289JA-A  |        |              |               |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BDDCAB       | B             |        | PLBG0224JA-A | -40 to 105 °C |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BDDCAC       |               |        | PLBG0289JA-A |               |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BFLCAB       | A             |        | PLBG0224JA-A | 1 MB          | 0 to 95 °C |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BFLCAC       | PLBG0289JA-A  |        |              |               |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BFDCAB       | B             |        | PLBG0224JA-A | -40 to 105 °C |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1BFDCAC       |               |        | PLBG0289JA-A |               |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1JFLCAB       | A             | dual   | —            | PLBG0224JA-A  | 1 MB       |      | 4 MB  |                                | 0 to 95 °C    |  |  |               |  |  |
| R7KA8P1JFLCAC       | PLBG0289JA-A  |        |              |               |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1JFDCAB       | B             |        |              | PLBG0224JA-A  |            |      |       |                                | -40 to 105 °C |  |  |               |  |  |
| R7KA8P1JFDCAC       |               |        |              | PLBG0289JA-A  |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1KFLCAB       | A             |        | ✓            | PLBG0224JA-A  |            |      |       |                                | 0 to 95 °C    |  |  |               |  |  |
| R7KA8P1KFLCAC       | PLBG0289JA-A  |        |              |               |            |      |       |                                |               |  |  |               |  |  |
| R7KA8P1KFDCAB       | B             |        |              | PLBG0224JA-A  |            |      |       |                                |               |  |  | -40 to 105 °C |  |  |
| R7KA8P1KFDCAC       |               |        |              | PLBG0289JA-A  |            |      |       |                                |               |  |  |               |  |  |
| R7JA8P1JRLSAJ       | A             | dual   | —            | PLBG0303GA-A  | 1 MB       |      | 4 MB  | 0 to 95 °C                     |               |  |  |               |  |  |
| R7JA8P1JSLSAJ       |               |        |              |               |            |      | 8 MB  |                                |               |  |  |               |  |  |
| R7JA8P1JRDSAJ       | B             |        |              |               |            |      | 4 MB  | -40 to 105 °C                  |               |  |  |               |  |  |
| R7JA8P1JSDSAJ       |               |        |              |               |            |      | 8 MB  |                                |               |  |  |               |  |  |
| R7JA8P1KRLSAJ       | A             |        | ✓            |               |            |      | 4 MB  | 0 to 95 °C                     |               |  |  |               |  |  |
| R7JA8P1KSLSAJ       |               |        |              |               |            |      | 8 MB  |                                |               |  |  |               |  |  |
| R7JA8P1KRDSAJ       | B             |        |              |               |            |      | 4 MB  | -40 to 105 °C                  |               |  |  |               |  |  |
| R7JA8P1KSDSAJ       |               |        |              |               |            |      |       | 8 MB                           |               |  |  |               |  |  |

## 1.4 Function Comparison

**Table 1.15 Function Comparison (1 of 2)**

| Parts number      |                    | R7KA8P<br>1AxxCA<br>C         | R7KA8P<br>1BxxCA<br>C | R7KA8P<br>1JxxCA<br>C | R7KA8P<br>1KxxCA<br>C | R7KA8P<br>1AxxCA<br>B | R7KA8P<br>1BxxCA<br>B | R7KA8P<br>1JxxCA<br>B | R7KA8P<br>1KxxCA<br>B | R7JA8P1<br>JxxSAJ      | R7JA8P1<br>KxxSAJ |
|-------------------|--------------------|-------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|-------------------|
| Pin count         |                    | 289                           |                       |                       |                       | 224                   |                       |                       |                       | 303                    |                   |
| Package           |                    | BGA                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| I/O Port          |                    | 208                           | 199                   | 208                   | 199                   | 149                   | 142                   | 149                   | 142                   | 195                    | 186               |
| Code MRAM         |                    | 1 MB, 512 KB                  |                       |                       |                       |                       |                       |                       |                       | 1 MB                   |                   |
| CPU0 TCM          |                    | 256 KB                        |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| CPU1 TCM          |                    | No                            |                       | 128 KB                |                       | No                    |                       | 128 KB                |                       |                        |                   |
| CPU0 I/D Caches   |                    | 32 KB                         |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| CPU1 C/S Caches   |                    | No                            |                       | 32 KB                 |                       | No                    |                       | 32 KB                 |                       |                        |                   |
| SRAM              |                    | 1792 KB                       |                       | 1664 KB               |                       | 1792 KB               |                       | 1664 KB               |                       |                        |                   |
| Flash             |                    | No                            |                       |                       |                       |                       |                       |                       |                       | 8 MB, 4 MB             |                   |
| DMA               | DTC                | 1                             |                       | 2                     |                       | 1                     |                       | 2                     |                       |                        |                   |
|                   | DMAC               | 8                             |                       | 16                    |                       | 8                     |                       | 16                    |                       |                        |                   |
| BUS               | External bus       | 32-bit bus                    |                       |                       |                       | 16-bit bus            |                       |                       |                       |                        |                   |
|                   | SDRAM              | 32-bit bus                    |                       |                       |                       | 16-bit bus            |                       |                       |                       |                        |                   |
| System            | CPU0 clock         | 1 GHz (max.)                  |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | CPU1 clock         | No                            |                       | 250 MHz (max.)        |                       | No                    |                       | 250 MHz (max.)        |                       |                        |                   |
|                   | CPUs clock sources | MOSC, SOSC, HOCO, MOCO, PLL1P |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | CAC                | Yes                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | WDT                | 1                             |                       | 2                     |                       | 1                     |                       | 2                     |                       |                        |                   |
|                   | IWDT               | Yes                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | Backup register    | 128 B                         |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| Communic<br>ation | SCI                | 10                            |                       |                       |                       | 9                     |                       |                       |                       | 10                     |                   |
|                   | IIC                | 3                             |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | I3C                | Yes                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | SPI                | 2                             |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | CANFD              | 2                             |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | USBFS              | Yes                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | USBHS              | Yes                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | OSPI               | 2                             |                       |                       |                       | 1                     |                       |                       |                       | 2*2                    |                   |
|                   | SSIE               | 2                             |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | SDHI/MM C          | 2                             |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                   | ESWM               | MII, RMII, GMII, RGMII        |                       |                       |                       | MII, RMII, RGMII      |                       |                       |                       | MII, RMII, GMII, RGMII |                   |
|                   | PDMIF              | Yes                           |                       |                       |                       |                       |                       |                       |                       |                        |                   |

**Table 1.15 Function Comparison (2 of 2)**

| Parts number             |                 | R7KA8P<br>1AxxCA<br>C                                                                    | R7KA8P<br>1BxxCA<br>C | R7KA8P<br>1JxxCA<br>C | R7KA8P<br>1KxxCA<br>C | R7KA8P<br>1AxxCA<br>B | R7KA8P<br>1BxxCA<br>B | R7KA8P<br>1JxxCA<br>B | R7KA8P<br>1KxxCA<br>B | R7JA8P1<br>JxxSAJ      | R7JA8P1<br>KxxSAJ |
|--------------------------|-----------------|------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|-------------------|
| Timers                   | GPT32*1         | 14                                                                                       |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | PDG             | 4                                                                                        |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | AGT*1           | 2                                                                                        |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | ULPT*1          | 2                                                                                        |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | RTC             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| Analog                   | ADC16H          | Unit 0: 15, Unit 1: 15                                                                   |                       |                       |                       | Unit 0: 7, Unit 1: 5  |                       |                       |                       | Unit 0: 15, Unit 1: 15 |                   |
|                          | DAC12           | 2                                                                                        |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | ACMPHS          | 4                                                                                        |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | TSN             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| HMI                      | GLCDC           | RGB888                                                                                   |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | DRW             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | MIPI<br>DSI/CSI | No                                                                                       | Yes                   | No                    | Yes                   | No                    | Yes                   | No                    | Yes                   | No                     | Yes               |
|                          | VIN             | No                                                                                       | Yes                   | No                    | Yes                   | No                    | Yes                   | No                    | Yes                   | No                     | Yes               |
|                          | CEU             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| Neural<br>processin<br>g | NPU             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| Data<br>processin<br>g   | CRC             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
|                          | DOC             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| Event<br>control         | ELC             | Yes                                                                                      |                       |                       |                       |                       |                       |                       |                       |                        |                   |
| Security                 |                 | RSIP-E50D, Decryption on-the-fly, Secure Debug, OTP, TrustZone, and Lifecycle management |                       |                       |                       |                       |                       |                       |                       |                        |                   |

Note: The product name differs depending on the supported memory size. See [section 1.3. Part Numbering](#).

Note: The maximum frequency varies by product group. See [section 70.3.1. Frequency](#) for details.

Note 1. Available pins depend on the Pin count, about details see [section 1.7. Pin Lists](#).

Note 2. OSPI1 is connected to the serial Flash in the SiP product.

## 1.5 Pin Functions

**Table 1.16 Pin functions (1 of 7)**

| Function               | Signal                                          | I/O    | Description                                                                                                                                                                                                                                                                |
|------------------------|-------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power supply           | VCC_01 to VCC_10,<br>VCC2_11 to VCC2_15         | Input  | Power supply pin. Connect it to the system power supply. Connect this pin to the same numbered VSS_01 to VSS_15 by a 0.1-μF capacitor. The capacitor should be placed close to the pin.<br>In the SiP product, connect VCC2_11 to VCC2_15 to the 1.8V system power supply. |
|                        | VCC2_16 to<br>VCC2_19                           | Input  | Dedicated power supply pin for the SiP product. Connect it to the 1.8V system power supply. Connect this pin to the same numbered VSS_16 to VSS_19 by a 0.1-μF capacitor. The capacitor should be placed close to the pin.                                                 |
|                        | VCC_DCDC                                        | Input  | Switching regulator power supply pin.                                                                                                                                                                                                                                      |
|                        | VLO                                             | I/O    | Switching regulator pin.                                                                                                                                                                                                                                                   |
|                        | VCL0 to VCL11                                   | Input  | Connect this pin to the same numbered VSS0 to VSS11 pin by the smoothing capacitor used to stabilize the internal power supply.<br>Place the capacitor close to the pin.                                                                                                   |
|                        | VBATT                                           | Input  | Battery Backup power pin                                                                                                                                                                                                                                                   |
|                        | VSS_01 to VSS_15,<br>VSS0 to VSS11,<br>VSS_DCDC | Input  | Ground pin. Connect it to the system power supply (0 V).                                                                                                                                                                                                                   |
|                        | VSS_16 to VSS_19,<br>VSS                        | Input  | Dedicated ground pin for the SiP product. Connect it to the system power supply (0 V).                                                                                                                                                                                     |
|                        | Vpp                                             | Input  | Power supply pin for serial Flash programming operation. See the ISSI serial Flash IS25WX064 datasheet for the details. If not used, the Vpp pin can be floated. Note that the deep power down current varies depending on the power supply status of the Vpp pin.         |
| Clock                  | XTAL                                            | Output | Pins for a crystal resonator. An external clock signal can be input through the EXTAL pin.                                                                                                                                                                                 |
|                        | EXTAL                                           | Input  |                                                                                                                                                                                                                                                                            |
|                        | XCIN                                            | Input  | Input/output pins for the sub-clock oscillator. Connect a crystal resonator between XCOUT and XCIN.                                                                                                                                                                        |
|                        | XCOUT                                           | Output |                                                                                                                                                                                                                                                                            |
|                        | EXCIN                                           | Input  | External sub-clock input                                                                                                                                                                                                                                                   |
|                        | CLKOUT                                          | Output | Clock output pin                                                                                                                                                                                                                                                           |
| Operating mode control | MD                                              | Input  | Pin for setting the operating mode. The signal level on this pin must not be changed during operation mode transition on release from the reset state.                                                                                                                     |
| System control         | RES                                             | Input  | Reset signal input pin. The MCU enters the reset state when this signal goes low.                                                                                                                                                                                          |
|                        | PUP                                             | Input  | Connect to VCC2 through a resistor.                                                                                                                                                                                                                                        |
| CAC                    | CACREF                                          | Input  | Measurement reference clock input pin                                                                                                                                                                                                                                      |
| On-chip emulator       | TMS                                             | Input  | On-chip emulator or boundary scan pins                                                                                                                                                                                                                                     |
|                        | TDI                                             | Input  |                                                                                                                                                                                                                                                                            |
|                        | TCK                                             | Input  |                                                                                                                                                                                                                                                                            |
|                        | TDO                                             | Output |                                                                                                                                                                                                                                                                            |
|                        | TCLK                                            | Output | Output clock for synchronization with the trace data                                                                                                                                                                                                                       |
|                        | TDATA0 to TDATA3                                | Output | Trace data output                                                                                                                                                                                                                                                          |
|                        | SWO                                             | Output | Serial wire trace output pin                                                                                                                                                                                                                                               |
|                        | SWDIO                                           | I/O    | Serial wire debug data input/output pin                                                                                                                                                                                                                                    |
|                        | SWCLK                                           | Input  | Serial wire clock pin                                                                                                                                                                                                                                                      |

**Table 1.16 Pin functions (2 of 7)**

| Function               | Signal             | I/O    | Description                                                                                                                                                                                                |
|------------------------|--------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interrupt              | NMI                | Input  | Non-maskable interrupt request pin                                                                                                                                                                         |
|                        | IRQn               | Input  | Maskable interrupt request pins                                                                                                                                                                            |
|                        | IRQn-DS            | Input  | Maskable interrupt request pins that can also be used in Deep Software Standby mode                                                                                                                        |
| External bus interface | EBCLK              | Output | Outputs the external bus clock for external devices                                                                                                                                                        |
|                        | RD                 | Output | Strobe signal indicating that reading from the external bus interface space is in progress, active-low                                                                                                     |
|                        | WR                 | Output | Strobe signal indicating that writing to the external bus interface space is in progress, in 1-write strobe mode, active-low                                                                               |
|                        | WRn                | Output | Strobe signals indicating that either group of data bus pins (D07 to D00, D15 to D08, D23 to D16 or D31 to D24) is valid in writing to the external bus interface space, in byte strobe mode, active-low   |
|                        | BCn                | Output | Strobe signals indicating that either group of data bus pins (D07 to D00, D15 to D08, D23 to D16 or D31 to D24) is valid in access to the external bus interface space, in 1-write strobe mode, active-low |
|                        | ALE                | Output | Address latch signal when address/data multiplexed bus is selected                                                                                                                                         |
|                        | WAIT               | Input  | Input pin for wait request signals in access to the external space, active-low                                                                                                                             |
|                        | CSn                | Output | Select signals for CS areas, active-low                                                                                                                                                                    |
|                        | A00 to A23         | Output | Address bus                                                                                                                                                                                                |
|                        | D00 to D31         | I/O    | Data bus                                                                                                                                                                                                   |
|                        | A00/D00 to A15/D15 | I/O    | Address/data multiplexed bus                                                                                                                                                                               |
| SDRAM interface        | SDCLK              | Output | Outputs the SDRAM-dedicated clock                                                                                                                                                                          |
|                        | CKE                | Output | SDRAM clock enable signal                                                                                                                                                                                  |
|                        | SDCS               | Output | SDRAM chip select signal, active low                                                                                                                                                                       |
|                        | RAS                | Output | SDRAM low address strobe signal, active low                                                                                                                                                                |
|                        | CAS                | Output | SDRAM column address strobe signal, active low                                                                                                                                                             |
|                        | WE                 | Output | SDRAM write enable signal, active low                                                                                                                                                                      |
|                        | DQMn               | Output | SDRAM I/O data mask enable signal for DQ07 to DQ00, DQ15 to DQ08, DQ23 to DQ16 or DQ31 to DQ24                                                                                                             |
|                        | A00 to A16         | Output | Address bus                                                                                                                                                                                                |
|                        | DQ00 to DQ31       | I/O    | Data bus                                                                                                                                                                                                   |

**Table 1.16 Pin functions (3 of 7)**

| Function | Signal                                                                                         | I/O    | Description                                                                         |
|----------|------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------|
| GPT      | GTETRG <sub>A</sub> ,<br>GTETRG <sub>B</sub> ,<br>GTETRG <sub>C</sub> ,<br>GTETRG <sub>D</sub> | Input  | External trigger input pins                                                         |
|          | GTIOCN <sub>A</sub> , GTIOCN <sub>B</sub>                                                      | I/O    | Input capture, output compare, or PWM output pins                                   |
|          | GTADSM <sub>0</sub> ,<br>GTADSM <sub>1</sub>                                                   | Output | A/D conversion start request monitoring output pins                                 |
|          | GTCPPOn                                                                                        | Output | Toggle output synchronized with PWM period                                          |
|          | GTIU                                                                                           | Input  | Hall sensor input pin U                                                             |
|          | GTIV                                                                                           | Input  | Hall sensor input pin V                                                             |
|          | GTIW                                                                                           | Input  | Hall sensor input pin W                                                             |
|          | GTOUUP                                                                                         | Output | 3-phase PWM output for BLDC motor control (positive U phase)                        |
|          | GTOULO                                                                                         | Output | 3-phase PWM output for BLDC motor control (negative U phase)                        |
|          | GTOVUP                                                                                         | Output | 3-phase PWM output for BLDC motor control (positive V phase)                        |
|          | GTOVLO                                                                                         | Output | 3-phase PWM output for BLDC motor control (negative V phase)                        |
|          | GTOWUP                                                                                         | Output | 3-phase PWM output for BLDC motor control (positive W phase)                        |
|          | GTOWLO                                                                                         | Output | 3-phase PWM output for BLDC motor control (negative W phase)                        |
| AGT      | AGTEEn                                                                                         | Input  | External event input enable signals                                                 |
|          | AGTIO <sub>n</sub>                                                                             | I/O    | External event input and pulse output pins                                          |
|          | AGTO <sub>n</sub>                                                                              | Output | Pulse output pins                                                                   |
|          | AGTOA <sub>n</sub>                                                                             | Output | Output compare match A output pins                                                  |
|          | AGTOB <sub>n</sub>                                                                             | Output | Output compare match B output pins                                                  |
| ULPT     | ULPTEEn                                                                                        | Input  | External count control input                                                        |
|          | ULPTEVIn                                                                                       | Input  | External event input                                                                |
|          | ULPTEEn-DS                                                                                     | Input  | External count control input that can also be used in Deep Software Standby mode 1  |
|          | ULPTEVIn-DS                                                                                    | Input  | External event input that can also be used in Deep Software Standby mode 1          |
|          | ULPTOn                                                                                         | Output | Pulse output                                                                        |
|          | ULPTOA <sub>n</sub>                                                                            | Output | Output compare match A output                                                       |
|          | ULPTOB <sub>n</sub>                                                                            | Output | Output compare match B output                                                       |
|          | ULPTOn-DS                                                                                      | Output | Pulse output that can also be used in Deep Software Standby mode 1                  |
|          | ULPTOA <sub>n</sub> -DS                                                                        | Output | Output compare match A output that can also be used in Deep Software Standby mode 1 |
|          | ULPTOB <sub>n</sub> -DS                                                                        | Output | Output compare match B output that can also be used in Deep Software Standby mode 1 |
| RTC      | RTCOUT                                                                                         | Output | Output pin for 1-Hz or 64-Hz clock                                                  |
|          | RTCIC <sub>n</sub>                                                                             | Input  | Time capture event input pins                                                       |

**Table 1.16 Pin functions (4 of 7)**

| Function | Signal                             | I/O    | Description                                                                                                                                                                                                                                        |
|----------|------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI      | SCKn                               | I/O    | Input/output pins for the clock (clock synchronous mode)                                                                                                                                                                                           |
|          | RXDn                               | Input  | Input pins for received data (asynchronous mode/clock synchronous mode)                                                                                                                                                                            |
|          | TXDn                               | Output | Output pins for transmitted data (asynchronous mode/clock synchronous mode)                                                                                                                                                                        |
|          | CTS <sub>n</sub> _RTS <sub>n</sub> | I/O    | Input/output pins for controlling the start of transmission and reception (asynchronous mode/clock synchronous mode), active-low.                                                                                                                  |
|          | CTS <sub>n</sub>                   | Input  | Input for the start of transmission.                                                                                                                                                                                                               |
|          | DEn                                | Output | Driver enable signal for RS-485                                                                                                                                                                                                                    |
|          | SCLn                               | I/O    | Input/output pins for the IIC clock (simple IIC mode)                                                                                                                                                                                              |
|          | SDAn                               | I/O    | Input/output pins for the IIC data (simple IIC mode)                                                                                                                                                                                               |
|          | SCKn                               | I/O    | Input/output pins for the clock (simple SPI mode)                                                                                                                                                                                                  |
|          | MISO <sub>n</sub>                  | I/O    | Input/output pins for slave transmission of data (simple SPI mode)                                                                                                                                                                                 |
|          | MOSI <sub>n</sub>                  | I/O    | Input/output pins for master transmission of data (simple SPI mode)                                                                                                                                                                                |
|          | SS <sub>n</sub>                    | Input  | Chip-select input pins (simple SPI mode), active-low                                                                                                                                                                                               |
| IIC      | SCLn                               | I/O    | Input/output pins for the clock                                                                                                                                                                                                                    |
|          | SDAn                               | I/O    | Input/output pins for data                                                                                                                                                                                                                         |
| I3C      | I3C_SCL0                           | I/O    | Input/output pins for the clock                                                                                                                                                                                                                    |
|          | I3C_SDA0                           | I/O    | Input/output pins for data                                                                                                                                                                                                                         |
| SPI      | RSPCKA, RSPCKB                     | I/O    | Clock input/output pin                                                                                                                                                                                                                             |
|          | MOSIA, MOSIB                       | I/O    | Input or output pins for data output from the master                                                                                                                                                                                               |
|          | MISOA, MISOB                       | I/O    | Input or output pins for data output from the slave                                                                                                                                                                                                |
|          | SSLA0, SSLB0                       | I/O    | Input or output pin for slave selection                                                                                                                                                                                                            |
|          | SSLA1 to SSLA3, SSLB1 to SSLB3     | Output | Output pins for slave selection                                                                                                                                                                                                                    |
| CANFD    | CRX <sub>n</sub>                   | Input  | Receive data                                                                                                                                                                                                                                       |
|          | CTX <sub>n</sub>                   | Output | Transmit data                                                                                                                                                                                                                                      |
| USBFS    | VCC_USB                            | Input  | Power supply pin                                                                                                                                                                                                                                   |
|          | VSS_USB                            | Input  | Ground pin                                                                                                                                                                                                                                         |
|          | USB_DP                             | I/O    | D+ pin of the USB on-chip transceiver. Connect this pin to the D+ pin of the USB bus.                                                                                                                                                              |
|          | USB_DM                             | I/O    | D- pin of the USB on-chip transceiver. Connect this pin to the D- pin of the USB bus.                                                                                                                                                              |
|          | USB_VBUS                           | Input  | USB cable connection monitor pin. Connect this pin to VBUS of the USB bus. The VBUS pin status (connected or disconnected) can be detected when the USB module is operating as a function controller.                                              |
|          | USB_EXICEN                         | Output | Low-power control signal for external power supply (OTG) chip                                                                                                                                                                                      |
|          | USB_VBUSEN                         | Output | VBUS (5 V) supply enable signal for external power supply chip                                                                                                                                                                                     |
|          | USB_OVRCURA, USB_OVRCURB           | Input  | Connect the external overcurrent detection signals to these pins. Connect the VBUS comparator signals to these pins when the OTG power supply chip is connected.                                                                                   |
|          | USB_OVRCURA-DS, USB_OVRCURB-DS     | Input  | Overcurrent pins for USBFS that can also be used in Deep Software Standby mode 1. Connect the external overcurrent detection signals to these pins. Connect the VBUS comparator signals to these pins when the OTG power supply chip is connected. |
|          | USB_ID                             | Input  | Connect the MicroAB connector ID input signal to this pin during operation in OTG mode                                                                                                                                                             |

**Table 1.16 Pin functions (5 of 7)**

| Function | Signal                                        | I/O    | Description                                                                                                          |
|----------|-----------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------|
| USBHS    | VCC_USBHS                                     | Input  | Power supply pin                                                                                                     |
|          | VSS1_USBHS,<br>VSS2_USBHS                     | Input  | Ground pin                                                                                                           |
|          | AVCC_USBHS                                    | Input  | Analog power supply                                                                                                  |
|          | USBHS_RREF                                    | I/O    | Reference current source pin for the USBHS. Must be connected to the VSS2_USBHS pin through a 2.2-kΩ (±1%) resistor. |
|          | USBHS_DP                                      | I/O    | Input/output pin for the D+ data line of the USB bus                                                                 |
|          | USBHS_DM                                      | I/O    | Input/output pin for the D- data line of the USB bus                                                                 |
|          | USBHS_EXICEN                                  | Output | Must be connected to the OTG power supply IC                                                                         |
|          | USBHS_ID                                      | input  | Must be connected to the OTG power supply IC                                                                         |
|          | USBHS_VBUSEN                                  | Output | VBUS power supply enable pin for the USBHS                                                                           |
|          | USBHS_OVRCURA,<br>USBHS_OVRCURB               | Input  | Overcurrent pin for the USBHS                                                                                        |
|          | USBHS_OVRCURA-<br>DS,<br>USBHS_OVRCURB-<br>DS | Input  | Overcurrent pin for the USBHS that can also be used in Deep Software Standby mode 1.                                 |
|          | USBHS_VBUS                                    | Input  | USB cable connection monitor input pin                                                                               |
| OSPI     | OM_n_SCLK                                     | Output | Clock output (OCTACLK divided by 2)                                                                                  |
|          | OM_n_SCLKN                                    | Output | Inverted clock output (OCTACLK divided by 2)                                                                         |
|          | OM_n_CSn                                      | Output | Chip select signal for an OctaFlash device, active-low                                                               |
|          | OM_n_DQS                                      | I/O    | Read data strobe/write data mask signal                                                                              |
|          | OM_n_SIO <sub>n</sub>                         | I/O    | Data input/output                                                                                                    |
|          | OM_n_RESET                                    | Output | Reset signal for both slave devices, active-low                                                                      |
|          | OM_n_ECSINT1                                  | Input  | Error Correction Status and Interrupt for slave1                                                                     |
|          | OM_n_RSTO1                                    | Input  | Slave reset status for slave1                                                                                        |
|          | OM_n_WP1                                      | Output | Write Protect for slave1, active-low                                                                                 |
| SSIE     | SSIBCK0, SSIBCK1                              | I/O    | SSIE serial bit clock pins                                                                                           |
|          | SSILRCK0/SSIFS0,<br>SSILRCK1/SSIFS1           | I/O    | LR clock/frame synchronization pins                                                                                  |
|          | SSITXD0                                       | Output | Serial data output pin                                                                                               |
|          | SSIRXD0                                       | Input  | Serial data input pin                                                                                                |
|          | SSIDATA1                                      | I/O    | Serial data input/output pins                                                                                        |
|          | AUDIO_CLK                                     | Input  | External clock pin for audio (input oversampling clock)                                                              |
| SDHI/MMC | SDnCLK                                        | Output | SD clock output pins                                                                                                 |
|          | SDnCMD                                        | I/O    | Command output pin and response input signal pins                                                                    |
|          | SDnDAT0 to<br>SDnDAT7                         | I/O    | SD and MMC data bus pins                                                                                             |
|          | SDnCD                                         | Input  | SD card detection pins                                                                                               |
|          | SDnWP                                         | Input  | SD write-protect signals                                                                                             |

**Table 1.16 Pin functions (6 of 7)**

| Function | Signal                                               | I/O    | Description                                                          |
|----------|------------------------------------------------------|--------|----------------------------------------------------------------------|
| ESWM     | ETn_GTX_CLK                                          | Output | 1000 Mb/s transmit clock                                             |
|          | ETn_TX_CLK                                           | Input  | 100 Mb/s, 10 Mb/s transmit clock                                     |
|          | ETn_RX_CLK                                           | Input  | Receive clock                                                        |
|          | ETn_TX_EN                                            | Output | Transmit enable                                                      |
|          | ETn_TXD0 to ETn_TXD7                                 | Output | Transmit data                                                        |
|          | ETn_TX_ER                                            | Output | Transmit coding error                                                |
|          | ETn_RX_DV                                            | Input  | Receive data valid                                                   |
|          | ETn_RXD0 to ETn_RXD7                                 | Input  | Receive data                                                         |
|          | ETn_RX_ER                                            | Input  | Receive error                                                        |
|          | ETn_MDC                                              | Output | Management data clock                                                |
|          | ETn_MDIO                                             | I/O    | Management data input/output                                         |
|          | RGMII <sub>n</sub> _TXC                              | Output | Transmit clock                                                       |
|          | RGMII <sub>n</sub> _RXC                              | Input  | Receive clock                                                        |
|          | RGMII <sub>n</sub> _TX_CTL                           | Output | Transmit control                                                     |
|          | RGMII <sub>n</sub> _TXD0 to RGMII <sub>n</sub> _TXD3 | Output | Transmit data                                                        |
|          | RGMII <sub>n</sub> _RX_CTL                           | Input  | Receive control                                                      |
|          | RGMII <sub>n</sub> _RXD0 to RGMII <sub>n</sub> _RXD3 | Input  | Receive data                                                         |
|          | RMII <sub>n</sub> _REF50CK                           | Input  | Synchronous clock reference                                          |
|          | RMII <sub>n</sub> _TX_EN                             | Output | Transmit enable                                                      |
|          | RMII <sub>n</sub> _TXD0 to RMII <sub>n</sub> _TXD1   | Output | Transmit data                                                        |
|          | RMII <sub>n</sub> _CRS_DV                            | Input  | Carrier sense/Receive data valid                                     |
|          | RMII <sub>n</sub> _RXD0 to RMII <sub>n</sub> _RXD1   | Input  | Receive data                                                         |
|          | RMII <sub>n</sub> _RX_ER                             | Input  | Receive error                                                        |
|          | ETn_LINKSTA                                          | Input  | PHY Link Status                                                      |
|          | ETn_INT                                              | Input  | PHY interrupt                                                        |
|          | ETn_WOL                                              | Output | Wake-on-LAN. This signal indicates that a Magic Packet was received. |
|          | GPTP_CAPTURE <sub>n</sub>                            | Input  | Media clock capture input                                            |
|          | GPTP_MATCH <sub>n</sub>                              | Output | Media clock recovery output                                          |
|          | GPTP_PPS <sub>n</sub>                                | Output | PPS signal                                                           |
|          | GPTP_PTPOUT0 to GPTP_PTPOUT3                         | Output | PTP Pulse generator signal                                           |
|          | ET_TAS_STA0 to ET_TAS_STA3                           | Output | TAS status monitor                                                   |
|          | ETHPHYCLK                                            | Output | Clock output for PHY                                                 |
| PDMIF    | PDMCLK0 to PDMCLK2                                   | Output | Clock output pin                                                     |
|          | PDMDAT0 to PDMDAT2                                   | Input  | Data input pin                                                       |

**Table 1.16 Pin functions (7 of 7)**

| Function            | Signal                   | I/O    | Description                                                                                                                                                    |
|---------------------|--------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analog power supply | AVCC0                    | Input  | Analog voltage supply pin. This is used as the analog power supply for the respective modules.                                                                 |
|                     | AVSS0                    | Input  | Analog ground pin. This is used as the analog ground for the respective modules. Supply this pin with the same voltage as the VSS pin.                         |
|                     | VREFH                    | Input  | Analog reference voltage supply pin for the ADC16H (unit 1) and D/A Converter. Connect this pin to AVCC0 when not using the ADC16H (unit 1) and D/A Converter. |
|                     | VREFL                    | Input  | Analog reference ground pin for the ADC16H and D/A Converter. Connect this pin to AVSS0 when not using the ADC16H (unit 1) and D/A Converter.                  |
|                     | VREFH0                   | Input  | Analog reference voltage supply pin for the ADC16H (unit 0). Connect this pin to AVCC0 when not using the ADC16H (unit 0).                                     |
|                     | VREFL0                   | Input  | Analog reference ground pin for the ADC16H. Connect this pin to AVSS0 when not using the ADC16H (unit 0).                                                      |
| ADC16H              | ANxxx                    | Input  | Input pins for the analog signals to be processed by the A/D converter.                                                                                        |
|                     | ADTRGm                   | Input  | Input pins for the external trigger signals that start the A/D conversion, active-low.                                                                         |
|                     | ADSTm                    | Output | AD conversion start                                                                                                                                            |
|                     | ADmFLAG1                 | Output | AD conversion end                                                                                                                                              |
|                     | ADSYNC                   | Output | Synchronization signal between units                                                                                                                           |
| DAC12               | DAn                      | Output | Output pins for the analog signals processed by the D/A converter.                                                                                             |
| ACMPHS              | VCOUT                    | Output | Comparator output pin                                                                                                                                          |
|                     | IVREFn                   | Input  | Reference voltage input pins for comparator                                                                                                                    |
|                     | IVCMPn                   | Input  | Analog voltage input pins for comparator                                                                                                                       |
| I/O ports           | Pmn                      | I/O    | General-purpose input/output pins (m: port number, n: pin number)                                                                                              |
|                     | P200                     | Input  | General-purpose input pin                                                                                                                                      |
| GLCDC               | LCD_DATA23 to LCD_DATA00 | Output | Data output pins for panel                                                                                                                                     |
|                     | LCD_TCON3 to LCD_TCON0   | Output | Output pins for panel timing adjustment                                                                                                                        |
|                     | LCD_CLK                  | Output | Panel clock output pin                                                                                                                                         |
|                     | LCD_EXTCLK               | Input  | Panel clock source input pin                                                                                                                                   |
| MIPI                | VCC18_MIPI               | Input  | Power supply pin                                                                                                                                               |
|                     | AVCC_MIPI                | Input  | Analog power supply                                                                                                                                            |
|                     | VSS_MIPI                 | Input  | Ground pin                                                                                                                                                     |
|                     | MIPI_CL_P                | Output | DSI/CSI Clock Lane positive pin                                                                                                                                |
|                     | MIPI_CL_N                | Output | DSI/CSI Clock Lane negative pin                                                                                                                                |
|                     | MIPI_DL0_P               | I/O    | DSI/CSI Data Lane 0 positive pin                                                                                                                               |
|                     | MIPI_DL0_N               | I/O    | DSI/CSI Data Lane 0 negative pin                                                                                                                               |
|                     | MIPI_DL1_P               | Output | DSI/CSI Data Lane 1 positive pin                                                                                                                               |
|                     | MIPI_DL1_N               | Output | DSI/CSI Data Lane 1 negative pin                                                                                                                               |
|                     | DSI_TE                   | Input  | DSI Tearing Effect pin                                                                                                                                         |
| CEU                 | VIO_D15 to VIO_D0        | Input  | CEU data bus pins                                                                                                                                              |
|                     | VIO_CLK                  | Input  | CEU clock pins                                                                                                                                                 |
|                     | VIO_VD                   | Input  | CEU vertical sync pins                                                                                                                                         |
|                     | VIO_HD                   | Input  | CEU horizontal sync pins                                                                                                                                       |
|                     | VIO_FLD                  | Input  | Field signal pins                                                                                                                                              |

## 1.6 Pin Assignments

The following figures show the pin assignments from the top view.

|   | 1           | 2           | 3           | 4          | 5    | 6                | 7         | 8        | 9         | 10        | 11        | 12      | 13   | 14      | 15           | 16           | 17                |   |
|---|-------------|-------------|-------------|------------|------|------------------|-----------|----------|-----------|-----------|-----------|---------|------|---------|--------------|--------------|-------------------|---|
| A | P609        | P113        | P115        | P112       | P302 | P915             | VLO       | VLO      | VSS_D CDC | VCC_D CDC | VCC_D CDC | P309    | P906 | P905    | P907         | P904         | P207              | A |
| B | P813        | PA12        | P114        | PA11       | P300 | P303             | VLO       | VLO      | VSS_D CDC | VCC_D CDC | VCC_D CDC | P311    | P908 | P909    | P206         | PD01         | PD02              | B |
| C | PA06        | P613        | PA13        | P301       | P200 | P210/T MS/S WDIO | P208/T DI | P110     | P308      | P305      | P307      | P911    | P312 | PD04    | PD03         | PD05         | PD06              | C |
| D | PA04        | P611        | P610        | PA14       | RES  | P211/T CK/S WCLK | P109      | P108     | P903      | P304      | P306      | P912    | PB04 | PB07    | PB05         | PB03         | PB01              | D |
| E | PA15        | P615        | P614        | P612       | P914 | P201/ MD         | P209/T DO | P111     | P902      | P310      | P910      | P913    | PB02 | PB06    | PD07         | PB00         | P706              | E |
| F | PA02        | PA10        | PA08        | PA09       | PC14 | VCC_0 8          | VSS_0 8   | VSS3     | VCL3      | VSS_0 7   | VCC_0 7   | P700    | P702 | P406    | P701         | P707         | P705              | F |
| G | PA00        | PA03        | PA05        | PA07       | PC12 | VCC_0 9          | VSS_0 9   | VSS4     | VCL4      | VSS_0 6   | VCC_0 6   | P405    | P704 | P703    | VSS_0 3      | VCC_0 5      | VSS_0 5           | G |
| H | P504        | P503        | P505        | PA01       | PC11 | VCC_1 0          | VSS_1 0   | VSS7     | VCL5      | VSS5      | VCC_0 4   | VSS_0 4 | P403 | VCC_0 3 | VCC_U SBHS   | USBH S_DP    | USBH S_DM         | H |
| J | P506        | P507        | P508        | P509       | PC13 | VCC2_ 11         | VSS_1 1   | VCL7     | VCL6      | VSS6      | VCL2      | VSS2    | P404 | VSS_0 2 | USBH S_RRE F | VSS2_ USBH S | VSS1_ USBH S      | J |
| K | PC15        | P608        | P510        | PD00       | PC07 | VSS_1 2          | VSS9      | VCL9     | VCL8      | VSS8      | VCL1      | VSS1    | P410 | VCC_0 2 | AVCC_ USBH S | P213/X TAL   | P212/E XTAL       | K |
| L | PC03        | PC02        | PC04        | PC09       | PC05 | VCC2_ 12         | VSS_1 4   | VSS_1 5  | VSS10     | VCL10     | VCL0      | VSS0    | P414 | P402    | VCC_0 1      | P214/X COUT  | P215/X CIN/EX CIN | L |
| M | PC00        | P607        | PC01        | PC08       | PC10 | P104             | VCC2_ 14  | VCC2_ 15 | P810      | VSS11     | VCL11     | P412    | P710 | P411    | P408         | VBATT        | VSS_0 1           | M |
| N | P605        | P604        | P606        | PC06       | P107 | P106             | P105      | P811     | P013      | P011      | P807      | P708    | P712 | P714    | P711         | P713         | P401              | N |
| P | P603        | P602        | P600        | P601       | P102 | P801             | P803      | P812     | P012      | P010      | P009      | P805    | P512 | P413    | P515         | P709         | P400              | P |
| R | VCC2_ 13    | VCC18_ MIPI | VSS_ MIPI   | P103       | P101 | P802             | P804      | P501     | AVCC0     | AVSS0     | P005      | P003    | P513 | P514    | P415         | P409         | P407              | R |
| T | MIPI_D L0_P | MIPI_C L_P  | MIPI_D L1_P | AVCC_ MIPI | P809 | P800             | P502      | P014     | VREFL 0   | VREFL 0   | P004      | P007    | P001 | P806    | P715         | P815/U SB_DP | VSS_U SB          | T |
| U | MIPI_D L0_N | MIPI_C L_N  | MIPI_D L1_N | VSS_1 3    | P808 | P100             | P500      | P015     | VREF H    | VREF H0   | P008      | P006    | P000 | P002    | P511         | P814/U SB_DP | VCC_U SB          | U |
|   | 1           | 2           | 3           | 4          | 5    | 6                | 7         | 8        | 9         | 10        | 11        | 12      | 13   | 14      | 15           | 16           | 17                |   |

Figure 1.3 Pin assignment for BGA 289-pin

|   | 1        | 2    | 3    | 4       | 5    | 6                | 7         | 8        | 9         | 10        | 11        | 12      | 13   | 14      | 15           | 16            | 17                |   |
|---|----------|------|------|---------|------|------------------|-----------|----------|-----------|-----------|-----------|---------|------|---------|--------------|---------------|-------------------|---|
| A | P609     | P113 | P115 | P112    | P302 | P915             | VLO       | VLO      | VSS_D CDC | VCC_D CDC | VCC_D CDC | P309    | P906 | P905    | P907         | P904          | P207              | A |
| B | P813     | PA12 | P114 | PA11    | P300 | P303             | VLO       | VLO      | VSS_D CDC | VCC_D CDC | VCC_D CDC | P311    | P908 | P909    | P206         | PD01          | PD02              | B |
| C | PA06     | P613 | PA13 | P301    | P200 | P210/T MS/S WDIO | P208/T DI | P110     | P308      | P305      | P307      | P911    | P312 | PD04    | PD03         | PD05          | PD06              | C |
| D | PA04     | P611 | P610 | PA14    | RES  | P211/T CK/S WCLK | P109      | P108     | P903      | P304      | P306      | P912    | PB04 | PB07    | PB05         | PB03          | PB01              | D |
| E | PA15     | P615 | P614 | P612    | P914 | P201/ MD         | P209/T DO | P111     | P902      | P310      | P910      | P913    | PB02 | PB06    | PD07         | PB00          | P706              | E |
| F | PA02     | PA10 | PA08 | PA09    | PC14 | VCC_0 8          | VSS_0 8   | VSS3     | VCL3      | VSS_0 7   | VCC_0 7   | P700    | P702 | P406    | P701         | P707          | P705              | F |
| G | PA00     | PA03 | PA05 | PA07    | PC12 | VCC_0 9          | VSS_0 9   | VSS4     | VCL4      | VSS_0 6   | VCC_0 6   | P405    | P704 | P703    | VSS_0 3      | VCC_0 5       | VSS_0 5           | G |
| H | P504     | P503 | P505 | PA01    | PC11 | VCC_1 0          | VSS_1 0   | VSS7     | VCL5      | VSS5      | VCC_0 4   | VSS_0 4 | P403 | VCC_0 3 | VCC_U SBHS   | USBH S_DP     | USBH S_DM         | H |
| J | P506     | P507 | P508 | P509    | PC13 | VCC2_ 11         | VSS_1 1   | VCL7     | VCL6      | VSS6      | VCL2      | VSS2    | P404 | VSS_0 2 | USBH S_RRE F | VSS2_ USBH S  | VSS1_ USBH S      | J |
| K | PC15     | P608 | P510 | PD00    | PC07 | VSS_1 2          | VSS9      | VCL9     | VCL8      | VSS8      | VCL1      | VSS1    | P410 | VCC_0 2 | AVCC_ USBH S | P213/X TAL    | P212/E XTAL       | K |
| L | PC03     | PC02 | PC04 | PC09    | PC05 | VCC2_ 12         | VSS_1 4   | VSS_1 5  | VSS10     | VCL10     | VCL0      | VSS0    | P414 | P402    | VCC_0 1      | P214/X COUT   | P215/X CIN/EX CIN | L |
| M | PC00     | P607 | PC01 | PC08    | PC10 | P104             | VCC2_ 14  | VCC2_ 15 | P810      | VSS11     | VCL11     | P412    | P710 | P411    | P408         | VBATT         | VSS_0 1           | M |
| N | P605     | P604 | P606 | PC06    | P107 | P106             | P105      | P811     | P013      | P011      | P807      | P708    | P712 | P714    | P711         | P713          | P401              | N |
| P | P603     | P602 | P600 | P601    | P102 | P801             | P803      | P812     | P012      | P010      | P009      | P805    | P512 | P413    | P515         | P709          | P400              | P |
| R | VCC2_ 13 | P315 | P900 | P103    | P101 | P802             | P804      | P501     | AVCC0     | AVSS0     | P005      | P003    | P513 | P514    | P415         | P409          | P407              | R |
| T | P205     | P203 | P313 | P901    | P809 | P800             | P502      | P014     | VREFL     | VREFL 0   | P004      | P007    | P001 | P806    | P715         | P815/U SB_D M | VSS_U SB          | T |
| U | P204     | P202 | P314 | VSS_1 3 | P808 | P100             | P500      | P015     | VREF H    | VREF H 0  | P008      | P006    | P000 | P002    | P511         | P814/U SB_DP  | VCC_U SB          | U |
|   | 1        | 2    | 3    | 4       | 5    | 6                | 7         | 8        | 9         | 10        | 11        | 12      | 13   | 14      | 15           | 16            | 17                |   |

Figure 1.4 Pin assignment for without\_MIPi\_BGA 289-pin

|   | 1           | 2           | 3           | 4          | 5        | 6         | 7                | 8                | 9         | 10        | 11   | 12      | 13           | 14            | 15                |   |
|---|-------------|-------------|-------------|------------|----------|-----------|------------------|------------------|-----------|-----------|------|---------|--------------|---------------|-------------------|---|
| A | NC          | PA11        | P114        | P112       | P300     | VLO       | VLO              | VSS_D CDC        | VCC_D CDC | VCC_D CDC | P309 | P312    | P908         | P905          | P206              | A |
| B | P610        | PA12        | P115        | P113       | P302     | VLO       | VLO              | VSS_D CDC        | VCC_D CDC | VCC_D CDC | P311 | P310    | P906         | P907          | P909              | B |
| C | P612        | P611        | PA13        | P609       | P301     | RES       | P210/T MS/S WDIO | P211/T CK/S WCLK | P304      | P306      | P305 | P307    | PB03         | PB00          | PB01              | C |
| D | P615        | P613        | P614        | PA14       | P200     | P208/T DI | P201/ MD         | P209/T DO        | P902      | P308      | PB02 | PB04    | P705         | P707          | P706              | D |
| E | PA15        | PA08        | P813        | PA09       | VCC_0 8  | VSS_0 8   | VSS5             | VCL5             | VSS_0 7   | VCC_0 7   | P405 | P702    | P704         | P406          | P701              | E |
| F | PA06        | PA10        | PA05        | PA07       | VCC_0 9  | VSS_0 9   | VSS6             | VCL6             | VCL4      | VSS4      | P700 | P703    | VSS_0 3      | VCC_0 5       | VSS_0 5           | F |
| G | PA04        | PA02        | PA01        | PA03       | VCC_1 0  | VSS_1 0   | VSS7             | VCL7             | VCL3      | VSS3      | P404 | VCC_0 3 | VCC_U SBHS   | USBH S_DP     | USBH S_DM         | G |
| H | PA00        | P504        | P503        | P505       | PC14     | VSS_1 5   | VSS8             | VCL8             | VCL2      | VSS2      | P403 | VSS_0 2 | USBH S_RRE F | VSS2_ USBH S  | VSS1_ USBH S      | H |
| J | P506        | P510        | P507        | P508       | PC12     | VCC2_ 15  | VSS9             | VCL9             | VCL1      | VSS1      | P402 | VCC_0 2 | AVCC_ USBH S | P213/X TAL    | P212/E XTAL       | J |
| K | PC15        | P608        | PD00        | P509       | VCC2_ 14 | VSS_1 4   | VSS10            | VCL10            | VCL0      | VSS0      | P410 | P407    | VCC_0 1      | P214/X COUT   | P215/X CIN/EX CIN | K |
| L | PC13        | P604        | P603        | P107       | P106     | P104      | P105             | VSS11            | VCL11     | P409      | P414 | P408    | P415         | VBATT         | VSS_0 1           | L |
| M | PC11        | P602        | P600        | P601       | P102     | P801      | P803             | P009             | P007      | P708      | P411 | P710    | P709         | P711          | P401              | M |
| N | VCC2_ 12    | VCC18_ MIPI | VSS_ MIPI   | P103       | P101     | P802      | P804             | AVCC0            | AVSS0     | P005      | P001 | P712    | P714         | P713          | P400              | N |
| P | MIPI_D LO_P | MIPI_C L_P  | MIPI_D L1_P | AVCC_ MIPI | P809     | P800      | P015             | VREFL            | VREFL 0   | P006      | P002 | P003    | P512         | P815/U SB_D M | VSS_U SB          | P |
| R | MIPI_D LO_N | MIPI_C L_N  | MIPI_D L1_N | VSS_1 2    | P808     | P100      | P014             | VREF H           | VREF H0   | P008      | P004 | P000    | P511         | P814/U SB_DP  | VCC_U SB          | R |
|   | 1           | 2           | 3           | 4          | 5        | 6         | 7                | 8                | 9         | 10        | 11   | 12      | 13           | 14            | 15                |   |

Figure 1.5 Pin assignment for BGA 224-pin

|   | 1        | 2    | 3       | 4        | 5        | 6         | 7                | 8                | 9         | 10        | 11   | 12      | 13           | 14           | 15                |   |
|---|----------|------|---------|----------|----------|-----------|------------------|------------------|-----------|-----------|------|---------|--------------|--------------|-------------------|---|
| A | NC       | PA11 | P114    | P112     | P300     | VLO       | VLO              | VSS_D CDC        | VCC_D CDC | VCC_D CDC | P309 | P312    | P908         | P905         | P206              | A |
| B | P610     | PA12 | P115    | P113     | P302     | VLO       | VLO              | VSS_D CDC        | VCC_D CDC | VCC_D CDC | P311 | P310    | P906         | P907         | P909              | B |
| C | P612     | P611 | PA13    | P609     | P301     | RES       | P210/T MS/S WDIO | P211/T CK/S WCLK | P304      | P306      | P305 | P307    | PB03         | PB00         | PB01              | C |
| D | P615     | P613 | P614    | PA14     | P200     | P208/T DI | P201/ MD         | P209/T DO        | P902      | P308      | PB02 | PB04    | P705         | P707         | P706              | D |
| E | PA15     | PA08 | P813    | PA09     | VCC_0 8  | VSS_0 8   | VSS5             | VCL5             | VSS_0 7   | VCC_0 7   | P405 | P702    | P704         | P406         | P701              | E |
| F | PA06     | PA10 | PA05    | PA07     | VCC_0 9  | VSS_0 9   | VSS6             | VCL6             | VCL4      | VSS4      | P700 | P703    | VSS_0 3      | VCC_0 5      | VSS_0 5           | F |
| G | PA04     | PA02 | PA01    | PA03     | VCC_1 0  | VSS_1 0   | VSS7             | VCL7             | VCL3      | VSS3      | P404 | VCC_0 3 | VCC_U SBHS   | USBH S_DP    | USBH S_DM         | G |
| H | PA00     | P504 | P503    | P505     | PC14     | VSS_1 5   | VSS8             | VCL8             | VCL2      | VSS2      | P403 | VSS_0 2 | USBH S_RRE F | VSS2_ USBH S | VSS1_ USBH S      | H |
| J | P506     | P510 | P507    | P508     | PC12     | VCC2_ 15  | VSS9             | VCL9             | VCL1      | VSS1      | P402 | VCC_0 2 | AVCC_ USBH S | P213/X TAL   | P212/E XTAL       | J |
| K | PC15     | P608 | PD00    | P509     | VCC2_ 14 | VSS_1 4   | VSS10            | VCL10            | VCL0      | VSS0      | P410 | P407    | VCC_0 1      | P214/X COUT  | P215/X CIN/EX CIN | K |
| L | PC13     | P604 | P603    | P107     | P106     | P104      | P105             | VSS11            | VCL11     | P409      | P414 | P408    | P415         | VBATT        | VSS_0 1           | L |
| M | PC11     | P602 | P600    | P601     | P102     | P801      | P803             | P009             | P007      | P708      | P411 | P710    | P709         | P711         | P401              | M |
| N | VCC2_ 12 | P315 | VSS_1 3 | P103     | P101     | P802      | P804             | AVCC0            | AVSS0     | P005      | P001 | P712    | P714         | P713         | P400              | N |
| P | P205     | P203 | P313    | VCC2_ 13 | P809     | P800      | P015             | VREFL            | VREFL 0   | P006      | P002 | P003    | P512         | P815/U SB_DP | VSS_U SB          | P |
| R | P204     | P202 | P314    | VSS_1 2  | P808     | P100      | P014             | VREF H           | VREF H0   | P008      | P004 | P000    | P511         | P814/U SB_DP | VCC_U SB          | R |
|   | 1        | 2    | 3       | 4        | 5        | 6         | 7                | 8                | 9         | 10        | 11   | 12      | 13           | 14           | 15                |   |

Figure 1.6 Pin assignment for without\_MIPi\_BGA 224-pin

|   | 1       | 2              | 3              | 4              | 5         | 6            | 7                      | 8       | 9       | 10           | 11           | 12           | 13     | 14   | 15     | 16              | 17                  | 18                      |   |
|---|---------|----------------|----------------|----------------|-----------|--------------|------------------------|---------|---------|--------------|--------------|--------------|--------|------|--------|-----------------|---------------------|-------------------------|---|
| A | VSS     | P114           | P609           | P113           | P301      | P208/<br>TDI | P210/<br>TMS/<br>SWDIO | VLO     | VLO     | VSS_D<br>CDC | VCC_D<br>CDC | VCC_D<br>CDC | P309   | P906 | P905   | P907            | P207                | VSS                     | A |
| B | P813    | PA12           | P115           | PA11           | P112      | P209/<br>TDO | P211/<br>TCK/<br>SWCLK | VLO     | VLO     | VSS_D<br>CDC | VCC_D<br>CDC | VCC_D<br>CDC | P311   | P908 | P909   | P904            | PD01                | PD02                    | B |
| C | PA06    | P613           | PA13           | P300           | P302      | P200         | RES                    | P110    | P903    | P308         | P305         | P307         | P911   | P206 | PD04   | PD03            | PD05                | PD06                    | C |
| D | PA04    | P611           | P610           | PA14           | P303      | P915         | P108                   | P111    | P109    | P310         | P304         | P306         | P912   | PB04 | PB07   | PB05            | PB03                | PB01                    | D |
| E | PA15    | P615           | P614           | P612           | P914      | P201/<br>MD  |                        |         |         | P902         | P312         | P910         | P913   | PB02 | PB06   | PD07            | PB00                | P706                    | E |
| F | PA02    | PA10           | PA08           | PA09           | PC14      |              | VCC_08                 | VSS_08  | VSS3    | VCL3         | VSS_07       | VCC_07       | P700   | P702 | P406   | P701            | P707                | P705                    | F |
| G | PA00    | PA03           | PA05           | PA07           | PC12      |              | VCC_09                 | VSS_09  | VSS4    | VCL4         | VSS_06       | VCC_06       | P405   | P704 | P703   | VSS_03          | VCC_05              | VSS_05                  | G |
| H | P504    | P503           | P505           | PA01           | PC11      |              | VCC_10                 | VSS_10  | VSS7    | VCL5         | VSS5         | VCC_04       | VSS_04 |      | VCC_03 | VCC_U<br>SBHS   | USBHS<br>_DP        | USBHS<br>_DM            | H |
| J | P506    | P507           | P508           | P509           | PC13      |              | VCC2_11                | VSS_11  | VCL7    | VCL6         | VSS6         | VCL2         | VSS2   |      | VSS_02 | USBHS<br>_RREF  | VSS2_1<br>USBHS     | VSS1_1<br>USBHS         | J |
| K | PC15    | P608           | P510           | PD00           | VSS       | VSS          | VSS_12                 | VSS9    | VCL9    | VCL8         | VSS8         | VCL1         | VSS1   |      | VCC_02 | AVCC_1<br>USBHS | P213/<br>XTAL       | P212/<br>EXTAL          | K |
| L | PC10    | VSS            | PUP            | VCC2_16        | VSS_16    |              | VCC2_12                | VSS_14  | VSS_15  | VSS10        | VCL10        | VCL0         | VSS0   | P403 | P404   | VCC_01          | P214/<br>XCIN       | P215/<br>XCIN/<br>EXCIN | L |
| M | PC09    | VSS            | VSS            | VCC2_17        | VSS_17    |              |                        | VCC2_14 | VCC2_15 |              | VSS11        | VCL11        |        | P414 | P402   | P410            | VBATT               | VSS_01                  | M |
| N | PC08    | VSS            | VSS            | VCC2_18        | VSS_18    |              | P105                   |         |         | P810         |              |              |        | P710 | P411   | P408            | P412                | P401                    | N |
| P | VSS     | VSS            | VSS            | VCC2_19        | VSS_19    | P104         | P107                   | P106    | P811    | P013         | P011         | P807         | P708   | P712 | P714   | P711            | P713                | P400                    | P |
| R | P602    | VSS            | VSS            | P600           | P601      | P102         | P801                   | P803    | P812    | P012         | P010         | P009         | P805   | P512 | P413   | P515            | P709                | P407                    | R |
| T | Vpp     | VSS            | VCC18<br>_MIPI | VSS_MI<br>PI   | P103      | P101         | P802                   | P804    | P501    | AVCC0        | AVSS0        | P005         | P003   | P513 | P514   | P415            | P409                | VCC_U<br>SB             | T |
| U | VCC2_13 | MIPI_D<br>L0_P | MIPI_C<br>L_P  | MIPI_D<br>L1_P | AVCC_MIPI | P809         | P800                   | P502    | P014    | VREFL        | VREFL<br>0   | P004         | P007   | P001 | P806   | P715            | P815/<br>USB_D<br>M | VSS_U<br>SB             | U |
| V | VSS     | MIPI_D<br>L0_N | MIPI_C<br>L_N  | MIPI_D<br>L1_N | VSS_13    | P808         | P100                   | P500    | P015    | VREFH        | VREFH<br>0   | P008         | P006   | P000 | P002   | P511            | P814/<br>USB_D<br>P | VSS                     | V |
|   | 1       | 2              | 3              | 4              | 5         | 6            | 7                      | 8       | 9       | 10           | 11           | 12           | 13     | 14   | 15     | 16              | 17                  | 18                      |   |

Figure 1.7 Pin assignment for BGA 303-pin

|   | 1       | 2    | 3    | 4       | 5      | 6            | 7                      | 8       | 9       | 10           | 11           | 12           | 13     | 14   | 15     | 16              | 17                  | 18                      |   |
|---|---------|------|------|---------|--------|--------------|------------------------|---------|---------|--------------|--------------|--------------|--------|------|--------|-----------------|---------------------|-------------------------|---|
| A | VSS     | P114 | P609 | P113    | P301   | P208/<br>TDI | P210/<br>TMS/<br>SWDIO | VLO     | VLO     | VSS_D<br>CDC | VCC_D<br>CDC | VCC_D<br>CDC | P309   | P906 | P905   | P907            | P207                | VSS                     | A |
| B | P813    | PA12 | P115 | PA11    | P112   | P209/<br>TDO | P211/<br>TCK/<br>SWCLK | VLO     | VLO     | VSS_D<br>CDC | VCC_D<br>CDC | VCC_D<br>CDC | P311   | P908 | P909   | P904            | PD01                | PD02                    | B |
| C | PA06    | P613 | PA13 | P300    | P302   | P200         | RES                    | P110    | P903    | P308         | P305         | P307         | P911   | P206 | PD04   | PD03            | PD05                | PD06                    | C |
| D | PA04    | P611 | P610 | PA14    | P303   | P915         | P108                   | P111    | P109    | P310         | P304         | P306         | P912   | PB04 | PB07   | PB05            | PB03                | PB01                    | D |
| E | PA15    | P615 | P614 | P612    | P914   | P201/<br>MD  |                        |         |         | P902         | P312         | P910         | P913   | PB02 | PB06   | PD07            | PB00                | P706                    | E |
| F | PA02    | PA10 | PA08 | PA09    | PC14   |              | VCC_08                 | VSS_08  | VSS3    | VCL3         | VSS_07       | VCC_07       | P700   | P702 | P406   | P701            | P707                | P705                    | F |
| G | PA00    | PA03 | PA05 | PA07    | PC12   |              | VCC_09                 | VSS_09  | VSS4    | VCL4         | VSS_06       | VCC_06       | P405   | P704 | P703   | VSS_03          | VCC_05              | VSS_05                  | G |
| H | P504    | P503 | P505 | PA01    | PC11   |              | VCC_10                 | VSS_10  | VSS7    | VCL5         | VSS5         | VCC_04       | VSS_04 |      | VCC_03 | VCC_U<br>SBHS   | USBHS<br>_DP        | USBHS<br>_DM            | H |
| J | P506    | P507 | P508 | P509    | PC13   |              | VCC2_11                | VSS_11  | VCL7    | VCL6         | VSS6         | VCL2         | VSS2   |      | VSS_02 | USBHS<br>_RREF  | VSS2_1<br>USBHS     | VSS1_1<br>USBHS         | J |
| K | PC15    | P608 | P510 | PD00    | VSS    | VSS          | VSS_12                 | VSS9    | VCL9    | VCL8         | VSS8         | VCL1         | VSS1   |      | VCC_02 | AVCC_1<br>USBHS | P213/<br>XTAL       | P212/<br>EXTAL          | K |
| L | PC10    | VSS  | PUP  | VCC2_16 | VSS_16 |              | VCC2_12                | VSS_14  | VSS_15  | VSS10        | VCL10        | VCL0         | VSS0   | P403 | P404   | VCC_01          | P214/<br>XCIN       | P215/<br>XCIN/<br>EXCIN | L |
| M | PC09    | VSS  | VSS  | VCC2_17 | VSS_17 |              |                        | VCC2_14 | VCC2_15 |              | VSS11        | VCL11        |        | P414 | P402   | P410            | VBATT               | VSS_01                  | M |
| N | PC08    | VSS  | VSS  | VCC2_18 | VSS_18 |              | P105                   |         |         | P810         |              |              |        | P710 | P411   | P408            | P412                | P401                    | N |
| P | VSS     | VSS  | VSS  | VCC2_19 | VSS_19 | P104         | P107                   | P106    | P811    | P013         | P011         | P807         | P708   | P712 | P714   | P711            | P713                | P400                    | P |
| R | P602    | VSS  | VSS  | P600    | P601   | P102         | P801                   | P803    | P812    | P012         | P010         | P009         | P805   | P512 | P413   | P515            | P709                | P407                    | R |
| T | Vpp     | VSS  | P315 | P900    | P103   | P101         | P802                   | P804    | P501    | AVCC0        | AVSS0        | P005         | P003   | P513 | P514   | P415            | P409                | VCC_U<br>SB             | T |
| U | VCC2_13 | P205 | P203 | P313    | P901   | P809         | P800                   | P502    | P014    | VREFL        | VREFL_0      | P004         | P007   | P001 | P806   | P715            | P815/<br>USB_D<br>M | VSS_U<br>SB             | U |
| V | VSS     | P204 | P202 | P314    | VSS_13 | P808         | P100                   | P500    | P015    | VREFH        | VREFH_0      | P008         | P006   | P000 | P002   | P511            | P814/<br>USB_D<br>P | VSS                     | V |
|   | 1       | 2    | 3    | 4       | 5      | 6            | 7                      | 8       | 9       | 10           | 11           | 12           | 13     | 14   | 15     | 16              | 17                  | 18                      |   |

Figure 1.8 Pin assignment for without\_MIP1\_BGA 303-pin

## 1.7 Pin Lists

Table 1.17 Pin list for the Standard product (1 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC           | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU        |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|----------------------------|---------------------|-----------------------|
| A1     | A1                  | C4     | C4                  | —                                | P609      | D7/DQ7      | IRQ29        | TXD0_C/SDA0_C/MOSI0_C/MISOA_B/CTX1                                                      | GTIU/GTIOC5B/ULPTOA1-DS    | AD1FLAG1            | LCD_DA TA6_A          |
| A2     | A2                  | B4     | B4                  | —                                | P113      | D4/DQ4      | IRQ28        | RXD0_A/SCL0_A/MISO0_A/SSLA1_B/SSILRCK0_B/SSIFS0_B/SD0DAT5_B                             | GTETRGB/GTIOC2A/ULPTOA0-DS | ADST1               | LCD_DA TA9_A          |
| A3     | A3                  | B3     | B3                  | —                                | P115      | D6/DQ6      | IRQ31-DS     | CTS0_A/MOSIA_B/SSITXD0_B/SD0DAT7_B                                                      | GTETRGD/GTIOC5A            | AD0FLAG1            | LCD_DA TA7_A          |
| A4     | A4                  | A4     | A4                  | —                                | P112      | D3/DQ3      | IRQ27        | TXD0_A/SDA0_A/MOSI0_A/SSLA2_B/SSIBCK0_B/SD0DAT4_B                                       | GTETRGA/GTIOC3B/ULPTOB0-DS | ADST0               | LCD_DA TA10_A         |
| A5     | A5                  | B5     | B5                  | —                                | P302      | D0/DQ0      | IRQ5         | RXD6_B/SCL6_B/MISO6_B/SD0DAT1_B                                                         | GTOUUP/GTIOC4A/ULPTO0-DS   | —                   | LCD_DA TA13_A         |
| A6     | A6                  | —      | —                   | —                                | P915      | —           | IRQ8         | CTS6_B                                                                                  | GTIOC5A                    | —                   | LCD_DA TA1_B          |
| A7     | A7                  | A6     | A6                  | VLO                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| A8     | A8                  | A7     | A7                  | VLO                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| A9     | A9                  | A8     | A8                  | VSS_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| A10    | A10                 | A9     | A9                  | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| A11    | A11                 | A10    | A10                 | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| A12    | A12                 | A11    | A11                 | —                                | P309      | —           | IRQ25-DS     | CTS9_B/ET1 GTX_CLK/RGMII1_TXC                                                           | GTCPPO8                    | VCOUT               | LCD_DA TA15_A/VIO_D10 |
| A13    | A13                 | B13    | B13                 | —                                | P906      | —           | IRQ9         | CTS6_A/USB_ID/SSILRCK1_A/SSIFS1_A/ET1_RXD0/RGMII1_RXD0/RMII1_RXD0/PDMDAT0               | GTIOC13B/ULPTO1            | AD0FLAG1            | LCD_DA TA20_A/VIO_D5  |
| A14    | A14                 | A14    | A14                 | —                                | P905      | —           | IRQ8         | RXD3_B/SCL3_B/MISO3_B/ET1_RX_CLK/RGMII1_RXC/RMII1_REF50CK/PDMDAT1                       | GTCPPO13                   | AD1FLAG1            | LCD_DA TA19_A/VIO_D6  |
| A15    | A15                 | B14    | B14                 | —                                | P907      | —           | IRQ10        | SCK6_A/DE6/USB_EXICEN/SSIBCK1_A/ET1_RXD1/RGMII1_RXD1/RMII1_RXD1/PDMCLK2                 | GTIOC13A/ULPTEE1           | ADSYNC              | LCD_DA TA21_A/VIO_D4  |
| A16    | A16                 | —      | —                   | —                                | P904      | —           | IRQ2         | ET1_RXD4                                                                                | GTIOC11B                   | —                   | LCD_DA TA8_B          |
| A17    | A17                 | —      | —                   | —                                | P207      | —           | IRQ25        | ET1_RXD5                                                                                | GTCPPO3                    | —                   | LCD_DA TA9_B          |
| B1     | B1                  | E3     | E3                  | —                                | P813      | SDCS        | IRQ15        | SCK7_A/DE7/PDMCLK2                                                                      | GTETRGA/GTIOC7B            | —                   | VIO_D13               |
| B2     | B2                  | B2     | B2                  | —                                | PA12      | D9/DQ9      | IRQ11        | RXD9_C/SCL9_C/MISO9_C                                                                   | GTIW/GTIOC6B               | —                   | —                     |
| B3     | B3                  | A3     | A3                  | —                                | P114      | D5/DQ5      | IRQ30-DS     | CTS_RTS0_A/SS0_A/DE0/SSLA0_B/SSIRXD0_B/SD0DAT6_B                                        | GTETRGD/GTIOC2B            | ADSYNC              | LCD_DA TA8_A          |
| B4     | B4                  | A2     | A2                  | —                                | PA11      | D8/DQ8      | IRQ10        | SCK9_C/DE9                                                                              | GTIV/GTIOC6A               | —                   | —                     |
| B5     | B5                  | A5     | A5                  | —                                | P300      | D2/DQ2      | IRQ4         | SCK0_A/DE0/SSLA3_B/SD0DAT3_B                                                            | GTIOC3A/ULPTEVI0-DS        | —                   | LCD_DA TA11_A         |
| B6     | B6                  | —      | —                   | —                                | P303      | —           | IRQ29-DS     | SCK6_B/DE6                                                                              | GTIOC7B                    | —                   | LCD_DA TA14_A         |
| B7     | B7                  | B6     | B6                  | VLO                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| B8     | B8                  | B7     | B7                  | VLO                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| B9     | B9                  | B8     | B8                  | VSS_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| B10    | B10                 | B9     | B9                  | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| B11    | B11                 | B10    | B10                 | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                     |
| B12    | B12                 | B11    | B11                 | —                                | P311      | —           | IRQ23-DS     | SCK3_B/DE3/CRX0/ET1_TX_CLK                                                              | GTADSM1/GTCPPO6/AGTOB1     | —                   | LCD_DA TA17_A/VIO_D8  |
| B13    | B13                 | A13    | A13                 | —                                | P908      | —           | IRQ11        | TXD6_A/SDA6_A/MOSI6_A/CRX1/USB_OVRCURB/USBHS_ID/ET1_RXD2/RGMII1_RXD2/PDMCLK1            | GTIOC12B/ULPTEVI1          | ADST1               | LCD_DA TA22_A/VIO_D3  |
| B14    | B14                 | B15    | B15                 | —                                | P909      | —           | IRQ21-DS     | RXD6_A/SCL6_A/MISO6_A/CTX1/USB_OVRCURA/USBHS_EXICEN/ET1_RXD3/RGMII1_RXD3/PDMCLK0        | GTIOC12A/ULPTOA1           | ADST0               | LCD_DA TA23_A/VIO_D2  |

Table 1.17 Pin list for the Standard product (2 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC                 | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU       |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|----------------------------------|---------------------|----------------------|
| B15    | B15                 | A15    | A15                 | CLKOUT                           | P206      | CS7         | IRQ0-DS      | USB_VBUSEN/SSIDATA1_A/SD0DAT7_C/ET1_RX_DV/RGMII1_RX_CTL/RMII1_CRS_DV                    | GTIU/GTCCPO0/ULPTOB1             | —                   | VIO_D0               |
| B16    | B16                 | —      | —                   | —                                | PD01      | —           | IRQ22        | SCK8_C/DE8/SD0DAT2_C/ET1_RXD6                                                           | GTCCPO2                          | —                   | —                    |
| B17    | B17                 | —      | —                   | —                                | PD02      | —           | IRQ21        | TXD8_C/SDA8_C/MOSI8_C/SD0DAT1_C/ET1_RXD7                                                | GTCCPO1                          | —                   | —                    |
| C1     | C1                  | F1     | F1                  | —                                | PA06      | CS1/CKE     | IRQ17        | CTS2_C/SD0DAT1_A/PDMDAT1                                                                | GTETRG/GTIOC7B                   | —                   | VIO_D11              |
| C2     | C2                  | D2     | D2                  | —                                | P613      | D15/DQ15    | IRQ19        | CTS0_C/USBHS_OVRCURB                                                                    | GTETRG/GTIOC9B/AGTO1             | —                   | LCD_DATA2_A          |
| C3     | C3                  | C3     | C3                  | —                                | PA13      | D10/DQ10    | IRQ12        | CTS_RTS9_C/SS9_C/DE9                                                                    | GTOVUP/GTIOC10A                  | —                   | —                    |
| C4     | C4                  | C5     | C5                  | —                                | P301      | D1/DQ1      | IRQ6         | TXD6_B/SDA6_B/MOSI6_B/SD0DAT2_B                                                         | GTOULO/GTIOC4B/AGTIO0/ULPTEE0-DS | —                   | LCD_DATA12_A         |
| C5     | C5                  | D5     | D5                  | —                                | P200      | —           | NMI          | —                                                                                       | —                                | —                   | —                    |
| C6     | C6                  | C7     | C7                  | TMS/SWDIO                        | P210      | —           | IRQ24        | CTS_RTS9_B/SS9_B/DE9                                                                    | GTOULO/GTIOC0B                   | —                   | —                    |
| C7     | C7                  | D6     | D6                  | TDI                              | P208      | —           | IRQ3         | RXD9_B/SCL9_B/MISO9_B/CRX1                                                              | GTOVLO/GTIOC1B                   | VCOOUT              | —                    |
| C8     | C8                  | —      | —                   | —                                | P110      | —           | IRQ20        | SD0DAT4_C                                                                               | GTIOC9B                          | —                   | —                    |
| C9     | C9                  | D10    | D10                 | TCLK                             | P308      | —           | IRQ26-DS     | CTS3_B/SD0CLK_B/ET1_TX_ER/ETHPHYCLK                                                     | GTIU/GTCCPO9/ULPTOB1             | —                   | VIO_D11              |
| C10    | C10                 | C11    | C11                 | TDATA2                           | P305      | —           | IRQ8         | SD0WP/ET1_TXD2/RGMII1_TXD2                                                              | GTOVUP/GTCCPO12/ULPTEE1          | —                   | VIO_D14              |
| C11    | C11                 | C12    | C12                 | TDATA0                           | P307      | —           | IRQ27-DS     | CTS_RTS6_A/SS6_A/DE6/SD0CMD_B/ET1_TXD0/RGMII1_TXD0/RMII1_TXD0                           | GTIV/GTCCPO10/ULPTOA1            | —                   | VIO_D12              |
| C12    | C12                 | —      | —                   | —                                | P911      | —           | IRQ6         | ET1_TXD5                                                                                | GTIOC3B                          | —                   | LCD_DATA5_B          |
| C13    | C13                 | A12    | A12                 | —                                | P312      | —           | IRQ22-DS     | CTS_RTS3_B/SS3_B/DE3/CTX0/ET1_RX_ER/RMII1_RX_ER/PDMDAT2                                 | GTADSM0/GTCCPO5/AGTOA1           | —                   | LCD_DATA18_A/VIO_D7  |
| C14    | C14                 | —      | —                   | —                                | PD04      | —           | IRQ20        | CTS_RTS8_C/SS8_C/DE8/USBHS_ID/SD0CMD_C/ET0_RXD5                                         | GTIOC3A                          | —                   | —                    |
| C15    | C15                 | —      | —                   | —                                | PD03      | —           | IRQ21        | RXD8_C/SCL8_C/MISO8_C/USBHS_EXICEN/SD0DAT0_C/ET0_RXD4                                   | GTIOC3B                          | —                   | —                    |
| C16    | C16                 | —      | —                   | —                                | PD05      | —           | IRQ19        | CTS8_C/USBHS_OVRCURB/SD0CLK_C/ET0_RXD6                                                  | GTIOC2B                          | —                   | —                    |
| C17    | C17                 | —      | —                   | —                                | PD06      | —           | IRQ18        | USBHS_OVRCURA/SD0WP/ET0_RXD7                                                            | GTIOC2A                          | —                   | —                    |
| D1     | D1                  | G1     | G1                  | —                                | PA04      | A1/DQM3     | IRQ19        | SCK2_C/DE2/SD0DAT3_A                                                                    | GTIU/GTIOC4B                     | ADST0               | VIO_D9               |
| D2     | D2                  | C2     | C2                  | CACREF/CLKOUT                    | P611      | D13/DQ13    | IRQ17        | SCK0_C/DE0/MOSIA_B/USBHS_VBUSEN                                                         | GTOULO/GTIOC4B                   | —                   | LCD_DATA4_A          |
| D3     | D3                  | B1     | B1                  | —                                | P610      | D12/DQ12    | IRQ16        | RXD0_C/SCL0_C/MISO0_C/RSPCKA_B/CRX1                                                     | GTOUUP/GTIOC4A/ULPTOB1-DS        | —                   | LCD_DATA5_A          |
| D4     | D4                  | D4     | D4                  | —                                | PA14      | D11/DQ11    | IRQ13        | TXD9_C/SDA9_C/MOSI9_C                                                                   | GTOVLO/GTIOC10B                  | —                   | —                    |
| D5     | D5                  | C6     | C6                  | RES                              | —         | —           | —            | —                                                                                       | —                                | —                   | —                    |
| D6     | D6                  | C8     | C8                  | TCK/SWCLK                        | P211      | —           | IRQ23        | SCK9_B/DE9                                                                              | GTOUUP/GTIOC0A                   | —                   | —                    |
| D7     | D7                  | —      | —                   | —                                | P109      | —           | IRQ23        | SD0DAT5_C                                                                               | GTIOC10A                         | —                   | —                    |
| D8     | D8                  | —      | —                   | —                                | P108      | —           | IRQ24        | SD0DAT6_C                                                                               | GTIOC10B                         | —                   | —                    |
| D9     | D9                  | —      | —                   | —                                | P903      | —           | IRQ1         | —                                                                                       | GTIOC11A                         | —                   | LCD_DATA2_B          |
| D10    | D10                 | C9     | C9                  | TDATA3                           | P304      | —           | IRQ9         | SD0DAT0_B/ET1_TXD3/RGMII1_TXD3                                                          | GTOVLO/GTIOC7A/ULPTO1            | —                   | VIO_D15              |
| D11    | D11                 | C10    | C10                 | TDATA1                           | P306      | —           | IRQ28-DS     | SD0CD/ET1_TXD1/RGMII1_TXD1/RMII1_TXD1                                                   | GTIW/GTCCPO11/ULPTEV1            | —                   | VIO_D13              |
| D12    | D12                 | —      | —                   | —                                | P912      | —           | IRQ5         | ET1_TXD6                                                                                | GTIOC3A                          | —                   | LCD_DATA6_B          |
| D13    | D13                 | D12    | D12                 | —                                | PB04      | —           | IRQ9         | SCK5_C/DE5/ET0_TXD3/RGMII0_TXD3                                                         | GTCCPO3                          | AD0FLAG1            | LCD_DATA14_B/VIO_CLK |

Table 1.17 Pin list for the Standard product (3 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM  | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC      | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU        |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|--------------|--------------|-----------------------------------------------------------------------------------------|-----------------------|---------------------|-----------------------|
| D14    | D14                 | —      | —                   | —                                | PB07      | —            | IRQ1         | ET0_TXD5                                                                                | GTIOC9B               | —                   | LCD_DA TA10_B         |
| D15    | D15                 | —      | —                   | —                                | PB05      | —            | IRQ15        | CTS5_C/ET0_TXD7                                                                         | GTCPPO4               | —                   | LCD_DA TA12_B         |
| D16    | D16                 | C13    | C13                 | —                                | PB03      | —            | IRQ13        | TXD5_C/SDA5_C/MOSI5_C/ET0_TXD2/RGMII0_TXD2                                              | GTCPPO1               | ADSYNC              | LCD_DA TA15_B/VIO_HD  |
| D17    | D17                 | C15    | C15                 | —                                | PB01      | ALE          | IRQ12        | CTS_RTS1_B/SS1_B/DE1/ET0_TX_CLK                                                         | GTCPPO2               | AD1FLAG1            | LCD_DA TA13_B/VIO_FLD |
| E1     | E1                  | E1     | E1                  | —                                | PA15      | EBCLK/SDCLK  | IRQ14        | CTS9_C/PDMCLK1                                                                          | GTIOC7A               | —                   | VIO_D14               |
| E2     | E2                  | D1     | D1                  | —                                | P615      | WR2/BC2/DQM2 | IRQ7         | TXD7_A/SDA7_A/MOSI7_A/USBHS_EXICEN                                                      | GTETRG/GTCTPP010      | —                   | LCD_DA TA0_A          |
| E3     | E3                  | D3     | D3                  | —                                | P614      | WR/WR0/DQM0  | IRQ20        | RXD7_A/SCL7_A/MISO7_A/USBHS_ID                                                          | GTETRG/GTCTPP09/AGT00 | —                   | LCD_DA TA1_A          |
| E4     | E4                  | C1     | C1                  | —                                | P612      | D14/DQ14     | IRQ18        | CTS_RTS0_C/SS0_C/DE0/SSLA0_B/USBHS_OVRCURA                                              | GTIOC9A               | —                   | LCD_DA TA3_A          |
| E5     | E5                  | —      | —                   | —                                | P914      | —            | IRQ9         | CTS_RTS6_B/SS6_B/DE6                                                                    | GTIOC5B               | —                   | LCD_DA TA0_B          |
| E6     | E6                  | D7     | D7                  | MD                               | P201      | —            | IRQ4         | —                                                                                       | —                     | —                   | —                     |
| E7     | E7                  | D8     | D8                  | TDO/SWO/CLKOUT                   | P209      | —            | IRQ25        | TXD9_B/SDA9_B/MOSI9_B/CTX1                                                              | GTOVUP/GTIOC1A        | —                   | —                     |
| E8     | E8                  | —      | —                   | —                                | P111      | —            | IRQ19        | SD0DAT3_C                                                                               | GTIOC9A               | —                   | —                     |
| E9     | E9                  | D9     | D9                  | —                                | P902      | ALE          | IRQ0         | AUDIO_CLK/ETHPHYCLK                                                                     | GTCPPO13              | —                   | LCD_DA TA3_B/VIO_D1   |
| E10    | E10                 | B12    | B12                 | —                                | P310      | —            | IRQ24-DS     | TXD3_B/SDA3_B/MOSI3_B/ET1_TX_EN/RGMII1_TX_CTL/RMII1_TX_EN                               | GTCPPO7/AGTEE1        | —                   | LCD_DA TA16_A/VIO_D9  |
| E11    | E11                 | —      | —                   | —                                | P910      | —            | IRQ7         | ET1_TXD4                                                                                | GTCPPO12              | —                   | LCD_DA TA4_B          |
| E12    | E12                 | —      | —                   | CLKOUT                           | P913      | —            | IRQ3         | ET1_TXD7                                                                                | GTCPPO11              | —                   | LCD_DA TA7_B          |
| E13    | E13                 | D11    | D11                 | —                                | PB02      | —            | IRQ11        | RXD5_C/SCL5_C/MISO5_C/ET0_TXD1/RGMII0_TXD1/RMII0_TXD1                                   | GTCPPO0               | ADST1               | LCD_DA TA16_B/VIO_VD  |
| E14    | E14                 | —      | —                   | —                                | PB06      | —            | IRQ0         | CTS_RTS5_C/SS5_C/DE5/ET0_TXD6                                                           | GTIOC9A               | —                   | LCD_DA TA11_B         |
| E15    | E15                 | —      | —                   | —                                | PD07      | —            | IRQ17        | USBHS_VBUSEN/SD0CD/ET0_TXD4                                                             | GTCPPO0               | —                   | —                     |
| E16    | E16                 | C14    | C14                 | —                                | PB00      | —            | IRQ10        | SCK1_B/DE1/USBHS_VBUSEN/ET0_TXD0/RGMII0_TXD0/RMII0_TXD0/PDMDAT2                         | GTCPPO4               | ADST0               | LCD_DA TA17_B         |
| E17    | E17                 | D15    | D15                 | —                                | P706      | —            | IRQ7         | RXD1_B/SCL1_B/MISO1_B/USBHS_OVRCURB-DS/ET0_GTX_CLK/RGMII0_TXC/ETHPHYCLK/PDMDAT0         | GTCPPO2/AGTIO0        | —                   | VIO_D10               |
| F1     | F1                  | G2     | G2                  | —                                | PA02      | A3           | IRQ31        | RXD2_C/SCL2_C/MISO2_C/SD0DAT5_A                                                         | GTIW/GTCTPP09         | ADSYNC              | VIO_D7                |
| F2     | F2                  | F2     | F2                  | —                                | PA10      | CS2/RAS      | IRQ4         | SCK5_B/DE5/PDMCLK0                                                                      | GTCPPO13              | —                   | LCD_TC ON1_A/VIO_D15  |
| F3     | F3                  | E2     | E2                  | —                                | PA08      | CS0/WE       | IRQ6         | RXD5_B/SCL5_B/MISO5_B                                                                   | GTETRG/GTCTPP011      | —                   | LCD_TC ON3_A          |
| F4     | F4                  | E4     | E4                  | —                                | PA09      | CS3/CAS      | IRQ5         | TXD5_B/SDA5_B/MOSI5_B                                                                   | GTCPPO12              | —                   | LCD_TC ON2_A          |
| F5     | F5                  | H5     | H5                  | —                                | PC14      | D16/DQ16     | IRQ0         | TXD6_C/SDA6_C/MOSI6_C/ET0_WOL                                                           | GTADSM1/GTCTPP09      | —                   | —                     |
| F6     | F6                  | E5     | E5                  | VCC_08                           | —         | —            | —            | —                                                                                       | —                     | —                   | —                     |
| F7     | F7                  | E6     | E6                  | VSS_08                           | —         | —            | —            | —                                                                                       | —                     | —                   | —                     |
| F8     | F8                  | G10    | G10                 | VSS3                             | —         | —            | —            | —                                                                                       | —                     | —                   | —                     |
| F9     | F9                  | G9     | G9                  | VCL3                             | —         | —            | —            | —                                                                                       | —                     | —                   | —                     |
| F10    | F10                 | E9     | E9                  | VSS_07                           | —         | —            | —            | —                                                                                       | —                     | —                   | —                     |
| F11    | F11                 | E10    | E10                 | VCC_07                           | —         | —            | —            | —                                                                                       | —                     | —                   | —                     |
| F12    | F12                 | F11    | F11                 | —                                | P700      | —            | IRQ16-DS     | RXD2_B/SCL2_B/MISO2_B/MISOA_C/SSIDATA1_B/SD1WP/ET0_RXD2/RGMII0_RXD2                     | GTIOC5A               | —                   | VIO_D4                |

Table 1.17 Pin list for the Standard product (4 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF    | GPT/AGT/ULPT/RTC       | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU       |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|--------------------------------------------------------------------------------------------|------------------------|---------------------|----------------------|
| F13    | F13                 | E12    | E12                 | —                                | P702      | —           | IRQ18-DS     | CTS2_B/RSPCKA_C/SSIBCK1_B/SD1DAT5_B/ET0_RXD0/RGMII0_RXD0/RMII0_RXD0                        | GTIOC6A/ULPT00         | —                   | VIO_D6               |
| F14    | F14                 | E14    | E14                 | —                                | P406      | —           | IRQ31        | TXD2_B/SDA2_B/MOSI2_B/SSLA3_C/SSIRXD0_A/SD1CD/ET0_RXD3/RGMII0_RXD3                         | GTIOC1B                | —                   | VIO_D3               |
| F15    | F15                 | E15    | E15                 | —                                | P701      | —           | IRQ17-DS     | CTS_RTS2_B/SS2_B/DE2/MOSIA_C/SSILRCK1_B/SSIFS1_B/SD1DAT4_B/ET0_RXD1/RGMII0_RXD1/RMII0_RXD1 | GTIOC5B/ULPT01         | —                   | VIO_D5               |
| F16    | F16                 | D14    | D14                 | —                                | P707      | —           | IRQ8         | TXD1_B/SDA1_B/MOSI1_B/USBHS_OVRCURA-DS/ET0_TX_ER/ETHPHYCLK/PDMDAT1                         | GTCPP03                | —                   | LCD_DATA18_B/VIO_D11 |
| F17    | F17                 | D13    | D13                 | —                                | P705      | —           | IRQ19        | CTS1_B/SSLA2_C/CRX0/ET0_TX_EN/RGMII0_TX_CTL/RMII0_TX_EN/PDMCLK2                            | GTADSM1/GTCPP01/AGTIO0 | —                   | VIO_D9               |
| G1     | G1                  | H1     | H1                  | —                                | PA00      | A5          | IRQ22        | CTS_RTS5_B/SS5_B/DE5/SD0DAT7_A                                                             | GTOVLO/GTCPP07         | AD1FLAG1            | LCD_CLK_A/VIO_D5     |
| G2     | G2                  | G4     | G4                  | —                                | PA03      | A2          | IRQ20        | TXD2_C/SDA2_C/MOSI2_C/SD0DAT4_A                                                            | GTIV/GTCPP010          | ADST1               | VIO_D8               |
| G3     | G3                  | F3     | F3                  | —                                | PA05      | A0/BC0/DQM1 | IRQ18        | CTS_RTS2_C/SS2_C/DE2/SD0DAT2_A/PDMDAT2                                                     | GTETRGD/GTIOC4A        | —                   | VIO_D10              |
| G4     | G4                  | F4     | F4                  | —                                | PA07      | RD          | IRQ16        | CTS7_A/SD0DAT0_A/PDMDAT0                                                                   | GTETRGB/GTIOC7A        | VCOUT               | VIO_D12              |
| G5     | G5                  | J5     | J5                  | —                                | PC12      | D18/DQ18    | IRQ2         | SCK6_C/DE6/ET0_MDIO                                                                        | GTCPP011               | —                   | —                    |
| G6     | G6                  | F5     | F5                  | VCC_09                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G7     | G7                  | F6     | F6                  | VSS_09                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G8     | G8                  | F10    | F10                 | VSS4                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G9     | G9                  | F9     | F9                  | VCL4                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G10    | G10                 | —      | —                   | VSS_06                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G11    | G11                 | —      | —                   | VCC_06                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G12    | G12                 | E11    | E11                 | —                                | P405      | —           | IRQ30        | SCK2_B/DE2/SSITXD0_A/SD1DAT3_B/ET0_RX_DV/RGMII0_RX_CTL/RMII0_CRS_DV                        | GTIOC1A/AGTIO1         | —                   | VIO_D2               |
| G13    | G13                 | E13    | E13                 | —                                | P704      | —           | IRQ26        | SSLA1_C/CTX0/SD1DAT7_B/ET0_RX_ER/RMII0_RX_ER/PDMCLK1                                       | GTADSM0/GTCPP00/AGTIO0 | —                   | VIO_D8               |
| G14    | G14                 | F12    | F12                 | —                                | P703      | —           | IRQ19-DS     | SSLA0_C/SD1DAT6_B/ET0_RX_CLK/RGMII0_RXC/RMII0_REF50CK/PDMCLK0                              | GTIOC6B/AGTIO1         | VCOUT               | VIO_D7               |
| G15    | G15                 | F13    | F13                 | VSS_03                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G16    | G16                 | F14    | F14                 | VCC_05                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| G17    | G17                 | F15    | F15                 | VSS_05                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H1     | H1                  | H2     | H2                  | —                                | P504      | A7          | IRQ7         | SD0WP                                                                                      | GTOULO/GTCPP01         | —                   | VIO_D3               |
| H2     | H2                  | H3     | H3                  | —                                | P503      | A6          | IRQ6         | SD0CD                                                                                      | GTOUUP/GTCPP06         | —                   | VIO_D4               |
| H3     | H3                  | H4     | H4                  | —                                | P505      | A8          | IRQ8         | SD0CLK_A                                                                                   | GTOWUP/GTCPP02         | —                   | VIO_D2               |
| H4     | H4                  | G3     | G3                  | —                                | PA01      | A4          | IRQ21        | CTS5_B/SD0DAT6_A                                                                           | GTOVUP/GTCPP08         | AD0FLAG1            | LCD_TCON0_A/VIO_D6   |
| H5     | H5                  | M1     | M1                  | —                                | PC11      | D19/DQ19    | IRQ3         | CTS_RTS6_C/SS6_C/DE6/ET0_MDC                                                               | GTCPP012               | —                   | —                    |
| H6     | H6                  | G5     | G5                  | VCC_10                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H7     | H7                  | G6     | G6                  | VSS_10                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H8     | H8                  | G7     | G7                  | VSS7                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H9     | H9                  | E8     | E8                  | VCL5                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H10    | H10                 | E7     | E7                  | VSS5                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H11    | H11                 | —      | —                   | VCC_04                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H12    | H12                 | —      | —                   | VSS_04                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                    |
| H13    | H13                 | H11    | H11                 | —                                | P403      | —           | IRQ14-DS     | CTS_RTS1_A/SS1_A/DE1/SSIBCK0_A/SD1DAT1_B/ET1_WOL                                           | GTIOC3A/RTCIC1         | AD0FLAG1            | —                    |

**Table 1.17 Pin list for the Standard product (5 of 9)**

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM  | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC            | ADC16H/DAC12/ACMPHS | MIPI/ GLCDC/ CEU            |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|--------------|--------------|-----------------------------------------------------------------------------------------|-----------------------------|---------------------|-----------------------------|
| H14    | H14                 | G12    | G12                 | VCC_03                           | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| H15    | H15                 | G13    | G13                 | VCC_USBHS                        | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| H16    | H16                 | G14    | G14                 | USBHS_DP                         | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| H17    | H17                 | G15    | G15                 | USBHS_DM                         | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J1     | J1                  | J1     | J1                  | —                                | P506      | A9           | IRQ9         | SD0CMD_A                                                                                | GTOWLO/GTCPPO3              | —                   | VIO_D1                      |
| J2     | J2                  | J3     | J3                  | —                                | P507      | A10          | IRQ10        | CTS_RTS7_A/SS7_A/DE7/ET_TAS_STA0                                                        | GTADSM0/GTIOC0A             | —                   | LCD_EX<br>TCLK_A/<br>VIO_D0 |
| J3     | J3                  | J4     | J4                  | —                                | P508      | A11          | IRQ1         | CTS5_A/ET_TAS_STA1                                                                      | GTADSM1/GTIOC0B             | —                   | VIO_VD                      |
| J4     | J4                  | K4     | K4                  | —                                | P509      | A12          | IRQ2         | CTS_RTS5_A/SS5_A/DE5/ET_TAS_STA2                                                        | GTIOC1A/ULPTEV11            | —                   | VIO_HD                      |
| J5     | J5                  | L1     | L1                  | —                                | PC13      | D17/<br>DQ17 | IRQ1         | RXD6_C/SCL6_C/MISO6_C/<br>ET0_INT                                                       | GTCPPO10                    | —                   | —                           |
| J6     | J6                  | —      | —                   | VCC2_11                          | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J7     | J7                  | —      | —                   | VSS_11                           | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J8     | J8                  | G8     | G8                  | VCL7                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J9     | J9                  | F8     | F8                  | VCL6                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J10    | J10                 | F7     | F7                  | VSS6                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J11    | J11                 | H9     | H9                  | VCL2                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J12    | J12                 | H10    | H10                 | VSS2                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J13    | J13                 | G11    | G11                 | —                                | P404      | —            | IRQ15-<br>DS | CTS1_A/SSILRCK0_A/SSIFS0_A/<br>SD1DAT2_B/ET0_WOL                                        | GTIOC3B/RTCIC2              | AD1FLAG<br>1        | —                           |
| J14    | J14                 | H12    | H12                 | VSS_02                           | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J15    | J15                 | H13    | H13                 | USBHS_RREF                       | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J16    | J16                 | H14    | H14                 | VSS2_USBHS                       | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| J17    | J17                 | H15    | H15                 | VSS1_USBHS                       | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K1     | K1                  | K1     | K1                  | —                                | PC15      | A16          | IRQ30        | CTS6_C/CRX1                                                                             | GTADSM0                     | —                   | —                           |
| K2     | K2                  | K2     | K2                  | CACREF                           | P608      | A14          | IRQ22        | TXD5_A/SDA5_A/MOSI5_A                                                                   | GTOWUP/GTCPPO4              | —                   | VIO_FLD                     |
| K3     | K3                  | J2     | J2                  | —                                | P510      | A13          | IRQ3         | RXD5_A/SCL5_A/MISO5_A/<br>ET_TAS_STA3                                                   | GTIOC1B/ULPTEV10            | —                   | VIO_CL<br>K                 |
| K4     | K4                  | K3     | K3                  | —                                | PD00      | A15          | IRQ23        | SCK5_A/DE5/CTX1                                                                         | GTOWLO/GTCPPO5              | —                   | —                           |
| K5     | K5                  | —      | —                   | —                                | PC07      | D23/<br>DQ23 | IRQ21        | OM_1_RESET                                                                              | GTCPPO0                     | —                   | —                           |
| K6     | K6                  | R4     | R4                  | VSS_12                           | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K7     | K7                  | J7     | J7                  | VSS9                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K8     | K8                  | J8     | J8                  | VCL9                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K9     | K9                  | H8     | H8                  | VCL8                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K10    | K10                 | H7     | H7                  | VSS8                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K11    | K11                 | J9     | J9                  | VCL1                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K12    | K12                 | J10    | J10                 | VSS1                             | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K13    | K13                 | K11    | K11                 | —                                | P410      | A19          | IRQ5         | SCK3_A/DE3/SCL0_A*1/<br>USB_OVRCURB-DS/<br>USBHS_OVRCURB/<br>GPTP_MATCH0                | GTVOLO/GTIOC9B/<br>AGTOB1   | ADST0               | —                           |
| K14    | K14                 | J12    | J12                 | VCC_02                           | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K15    | K15                 | J13    | J13                 | AVCC_USBHS                       | —         | —            | —            | —                                                                                       | —                           | —                   | —                           |
| K16    | K16                 | J14    | J14                 | XTAL                             | P213      | —            | IRQ2         | TXD1_C/SDA1_C/MOSI1_C                                                                   | GTETRG0/GTIOC0A/<br>ULPTEE0 | ADTRG1              | —                           |
| K17    | K17                 | J15    | J15                 | EXTAL                            | P212      | —            | IRQ3         | RXD1_C/SCL1_C/MISO1_C                                                                   | GTETRGD/GTIOC0B/<br>AGTEE1  | —                   | —                           |
| L1     | L1                  | —      | —                   | —                                | PC03      | D27/<br>DQ27 | IRQ25        | TXD7_C/SDA7_C/MOSI7_C/<br>OM_1_SIO4                                                     | GTCPPO4                     | —                   | —                           |
| L2     | L2                  | —      | —                   | —                                | PC02      | D28/<br>DQ28 | IRQ26        | SCK7_C/DE7/OM_1_SIO3                                                                    | GTCPPO5                     | —                   | —                           |
| L3     | L3                  | —      | —                   | —                                | PC04      | D26/<br>DQ26 | IRQ24        | RXD7_C/SCL7_C/MISO7_C/<br>OM_1_SIO2                                                     | GTCPPO3                     | —                   | —                           |
| L4     | L4                  | —      | —                   | —                                | PC09      | D21/<br>DQ21 | IRQ5         | OM_1_RST01                                                                              | —                           | —                   | —                           |

Table 1.17 Pin list for the Standard product (6 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC                  | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU        |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|-----------------------------------|---------------------|-----------------------|
| L5     | L5                  | —      | —                   | —                                | PC05      | D25/DQ25    | IRQ23        | OM_1_CS1                                                                                | GTCPPO2                           | —                   | —                     |
| L6     | L6                  | N1     | N1                  | VCC2_12                          | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L7     | L7                  | K6     | K6                  | VSS_14                           | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L8     | L8                  | H6     | H6                  | VSS_15                           | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L9     | L9                  | K7     | K7                  | VSS10                            | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L10    | L10                 | K8     | K8                  | VCL10                            | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L11    | L11                 | K9     | K9                  | VCL0                             | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L12    | L12                 | K10    | K10                 | VSS0                             | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L13    | L13                 | L11    | L11                 | —                                | P414      | A23         | IRQ9         | RXD4_B/SCL4_B/MISO4_B/SSLB0_B/CRX1/ET1_MDIO                                             | GTIOC0B                           | —                   | VIO_CLK               |
| L14    | L14                 | J11    | J11                 | CACREF                           | P402      | —           | IRQ4-DS      | SCK1_A/DE1/CRX0/AUDIO_CLK/SD1DAT0_B/ET0_LINKSTA                                         | RTICIC0                           | —                   | —                     |
| L15    | L15                 | K13    | K13                 | VCC_01                           | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| L16    | L16                 | K14    | K14                 | XCOUT                            | P214      | —           | IRQ21        | —                                                                                       | —                                 | —                   | —                     |
| L17    | L17                 | K15    | K15                 | XCIN/EXCIN                       | P215      | —           | IRQ20        | —                                                                                       | —                                 | —                   | —                     |
| M1     | M1                  | —      | —                   | —                                | PC00      | D30/DQ30    | IRQ28        | CTS_RTS7_C/SS7_C/DE7/OM_1_SIO5                                                          | GTCPPO7                           | —                   | —                     |
| M2     | M2                  | —      | —                   | —                                | P607      | D31/DQ31    | IRQ23        | OM_1_DQS                                                                                | —                                 | —                   | —                     |
| M3     | M3                  | —      | —                   | —                                | PC01      | D29/DQ29    | IRQ27        | CTS7_C/OM_1_SIO0                                                                        | GTCPPO6                           | —                   | —                     |
| M4     | M4                  | —      | —                   | —                                | PC08      | D22/DQ22    | IRQ29        | OM_1_CS0                                                                                | GTCPPO8                           | —                   | —                     |
| M5     | M5                  | —      | —                   | —                                | PC10      | D20/DQ20    | IRQ4         | OM_1_WP1                                                                                | GTCPPO13                          | —                   | —                     |
| M6     | M6                  | L6     | L6                  | —                                | P104      | —           | IRQ1         | CTS9_A/SSLB1_A/OM_0_CS1/GPTP_MATCH0                                                     | GTETRGB/GTIOC1B                   | AD0FLAG1            | —                     |
| M7     | M7                  | K5     | K5                  | VCC2_14                          | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| M8     | M8                  | J6     | J6                  | VCC2_15                          | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| M9     | M9                  | —      | —                   | —                                | P810      | —           | IRQ21        | SCK7_B/DE7/SD1DAT2_A/PDMCLK0                                                            | GTIOC10A/ULPTOA0                  | —                   | —                     |
| M10    | M10                 | L8     | L8                  | VSS11                            | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| M11    | M11                 | L9     | L9                  | VCL11                            | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| M12    | M12                 | —      | —                   | —                                | P412      | A21         | IRQ20-DS     | CTS3_A/USB_EXICEN/USBHS_EXICEN/GPTP_PTPOUT0                                             | GTOULO/GTCPPO8/AGTEE1             | —                   | —                     |
| M13    | M13                 | M12    | M12                 | —                                | P710      | CS5         | IRQ17        | CTS4_B/SSLB3_B/ET0_LINKSTA                                                              | GTIOC11B                          | —                   | LCD_EX_TCLK_B/VIO_D12 |
| M14    | M14                 | M11    | M11                 | CACREF                           | P411      | A20         | IRQ4         | CTS_RTS3_A/SS3_A/DE3/USB_ID/USBHS_ID/GPTP_PTPOUT1                                       | GTOVUP/GTIOC9A/AGTOA1             | —                   | DSL_TE                |
| M15    | M15                 | L12    | L12                 | —                                | P408      | A17         | IRQ7         | RXD3_A/SCL3_A/MISO3_A/SCL0_B*/USB_VBUSEN/USBHS_VBUS/GPTP_PTPOUT2                        | GTOWLO/GTIOC10A/ULPTOB0           | ADSYNC              | —                     |
| M16    | M16                 | L14    | L14                 | VBATT                            | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| M17    | M17                 | L15    | L15                 | VSS_01                           | —         | —           | —            | —                                                                                       | —                                 | —                   | —                     |
| N1     | N1                  | —      | —                   | —                                | P605      | —           | IRQ25        | CTS0_B/OM_1_SIO1                                                                        | GTIOC8A                           | —                   | —                     |
| N2     | N2                  | L2     | L2                  | —                                | P604      | —           | IRQ26        | CTS_RTS0_B/SS0_B/DE0/OM_1_SIO7                                                          | GTIOC8B                           | —                   | —                     |
| N3     | N3                  | —      | —                   | —                                | P606      | WR3/BC3     | IRQ24        | OM_1_SIO6                                                                               | —                                 | —                   | —                     |
| N4     | N4                  | —      | —                   | —                                | PC06      | D24/DQ24    | IRQ22        | OM_1_ECSINT1                                                                            | GTCPPO1                           | —                   | —                     |
| N5     | N5                  | L4     | L4                  | —                                | P107      | —           | IRQ31        | CTS4_A/OM_0_CS0/ET1_INT                                                                 | GTOWUP/GTIOC8A/AGTOA0             | ADST0               | —                     |
| N6     | N6                  | L5     | L5                  | —                                | P106      | —           | IRQ16        | CTS8_B/SSLB3_A/OM_0_RESET/ET1_LINKSTA                                                   | GTOWLO/GTIOC8B/AGTOB0/ULPTTEE1-DS | ADST1               | —                     |
| N7     | N7                  | L7     | L7                  | —                                | P105      | —           | IRQ0         | CTS_RTS8_B/SS8_B/DE8/SSLB2_A/OM_0_ECSINT1/GPTP_CAPTURE0                                 | GTIOC1A/ULPTO1-DS                 | ADSYNC              | —                     |

Table 1.17 Pin list for the Standard product (7 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC         | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU        |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|--------------------------|---------------------|-----------------------|
| N8     | N8                  | —      | —                   | —                                | P811      | —           | IRQ22        | CTS7_B/USB_ID/SD1DAT3_A/PDMCLK1                                                         | GTIOC10B/ULPTOB0         | —                   | —                     |
| N9     | N9                  | —      | —                   | —                                | P013      | —           | IRQ14        | —                                                                                       | —                        | AN013               | —                     |
| N10    | N10                 | —      | —                   | —                                | P011      | —           | IRQ16        | —                                                                                       | —                        | AN011               | —                     |
| N11    | N11                 | —      | —                   | —                                | P807      | —           | IRQ11        | —                                                                                       | GTIOC13A                 | —                   | LCD_TC ON2_B          |
| N12    | N12                 | M10    | M10                 | CACREF                           | P708      | WR1/BC1     | IRQ11        | SCK4_B/DE4/SDA2_A <sup>1</sup> /MOSIB_B/AUDIO_CLK/ET0_MDC                               | GTCPPO6                  | —                   | VIO_VD                |
| N13    | N13                 | N12    | N12                 | —                                | P712      | —           | IRQ2         | CTS1_C/SSLB1_B/GPTP_CAPTURE1                                                            | GTIOC2B/AGTOB0           | —                   | LCD_DA TA20_B         |
| N14    | N14                 | N13    | N13                 | —                                | P714      | —           | IRQ13        | TXD4_C/SDA4_C/MOSI4_C/GPTP_PPS1                                                         | GTIOC12B                 | —                   | DSL_TE/ LCD_DA TA22_B |
| N15    | N15                 | M14    | M14                 | —                                | P711      | —           | IRQ3         | CTS_RTS1_C/SS1_C/DE1/SSLB2_B/GPTP_PPS0                                                  | GTIOC11A/AGTEE0          | —                   | LCD_DA TA19_B         |
| N16    | N16                 | N14    | N14                 | —                                | P713      | —           | IRQ14        | CTS4_C/GPTP_MATCH1                                                                      | GTIOC2A/AGTOA0           | —                   | LCD_DA TA21_B         |
| N17    | N17                 | M15    | M15                 | —                                | P401      | —           | IRQ5-DS      | RXD1_A/SCL1_A/MISO1_A/I3C_SDA0/CTX0/SD1CMD_B                                            | GTETRG6/GTIOC6B          | —                   | VIO_D1                |
| P1     | P1                  | L3     | L3                  | —                                | P603      | —           | IRQ27        | TXD0_B/SDA0_B/MOSI0_B/OM_1_SCLK                                                         | GTIOC7A/ULPTO0           | —                   | —                     |
| P2     | P2                  | M2     | M2                  | —                                | P602      | —           | IRQ28        | RXD0_B/SCL0_B/MISO0_B/OM_1_SCLKN                                                        | GTIOC7B/ULPTEE0          | —                   | —                     |
| P3     | P3                  | M3     | M3                  | CACREF                           | P600      | —           | IRQ30        | OM_0_RST01/ET1_WOL                                                                      | GTIOC6B/ULPTEVI1-DS      | —                   | —                     |
| P4     | P4                  | M4     | M4                  | —                                | P601      | —           | IRQ29        | SCK0_B/DE0/OM_0_WP1                                                                     | GTIOC6A/ULPTEVI0/RTCOUNT | —                   | —                     |
| P5     | P5                  | M5     | M5                  | —                                | P102      | —           | IRQ17        | TXD9_A/SDA9_A/MOSI9_A/RSPCKB_A/CRX0/OM_0_SIO4                                           | GTOWLO/GTIOC2B/AGTO0     | ADTRG0              | —                     |
| P6     | P6                  | M6     | M6                  | —                                | P801      | —           | IRQ12        | TXD2_A/SDA2_A/MOSI2_A/OM_0_DQS/GPTP_PPS1                                                | GTIV/GTIOC11B/AGTOB0     | —                   | —                     |
| P7     | P7                  | M7     | M7                  | —                                | P803      | —           | IRQ19        | SCK2_A/DE2/OM_0_SIO1                                                                    | GTETRG6/GTIOC12B         | —                   | —                     |
| P8     | P8                  | —      | —                   | —                                | P812      | —           | IRQ23        | CTS_RTS7_B/SS7_B/DE7/USB_EXICEN/SD1DAT4_A/PDMCLK2                                       | GTIOC11A                 | AN022               | —                     |
| P9     | P9                  | —      | —                   | —                                | P012      | —           | IRQ15        | —                                                                                       | —                        | AN012               | —                     |
| P10    | P10                 | —      | —                   | —                                | P010      | —           | IRQ14        | —                                                                                       | —                        | AN010               | —                     |
| P11    | P11                 | M8     | M8                  | —                                | P009      | —           | IRQ13-DS     | —                                                                                       | —                        | AN009/IVREF1        | —                     |
| P12    | P12                 | —      | —                   | —                                | P805      | —           | IRQ30        | TXD8_A/SDA8_A/MOSI8_A/ET1_MDIO                                                          | —                        | AN017/IVCMP0        | LCD_TC ON1_B/VIO_D15  |
| P13    | P13                 | P13    | P13                 | —                                | P512      | —           | IRQ14        | CTS8_A/SCL1_A <sup>1</sup> /CTX1/ET1_INT                                                | GTIOC0A                  | —                   | —                     |
| P14    | P14                 | —      | —                   | —                                | P413      | A22         | IRQ18        | ET_TAS_STA3                                                                             | GTOWUP/GTCTPPO7/ULPTEE1  | —                   | —                     |
| P15    | P15                 | —      | —                   | —                                | P515      | —           | IRQ12        | CTS_RTS4_C/SS4_C/DE4/SCL2_B <sup>1</sup> /ET_TAS_STA0                                   | GTIOC13A                 | —                   | LCD_CLK_B             |
| P16    | P16                 | M13    | M13                 | —                                | P709      | CS4         | IRQ10        | CTS_RTS4_B/SS4_B/DE4/SCL2_A <sup>1</sup> /MISOB_B/ET0_MDIO                              | GTCPPO5                  | —                   | VIO_D13               |
| P17    | P17                 | N15    | N15                 | —                                | P400      | —           | IRQ0         | TXD1_A/SDA1_A/MOSI1_A/I3C_SCL0/AUDIO_CLK/SD1CLK_B                                       | GTIOC6A/AGTIO1           | ADTRG1              | VIO_D0                |
| R1     | R1                  | —      | P4                  | VCC2_13                          | —         | —           | —            | —                                                                                       | —                        | —                   | —                     |
| R2     | —                   | N2     | —                   | VCC18_MIPI                       | —         | —           | —            | —                                                                                       | —                        | —                   | —                     |
| R3     | —                   | N3     | —                   | VSS_MIPI                         | —         | —           | —            | —                                                                                       | —                        | —                   | —                     |
| —      | R2                  | —      | N2                  | —                                | P315      | —           | IRQ29        | SCK3_C/DE3/SSLA3_A                                                                      | —                        | —                   | —                     |
| —      | R3                  | —      | —                   | —                                | P900      | —           | IRQ30        | CTS3_C                                                                                  | GTADSM0                  | —                   | —                     |
| R4     | R4                  | N4     | N4                  | —                                | P103      | —           | IRQ16        | CTS_RTS9_A/SS9_A/DE9/SSLB0_A/CTX0/OM_0_SIO2/GPTP_PPS0                                   | GTOWUP/GTIOC2A           | AD1FLAG1            | —                     |
| R5     | R5                  | N5     | N5                  | —                                | P101      | —           | IRQ1         | RXD9_A/SCL9_A/MISO9_A/MOSIB_A/OM_0_SIO3/GPTP_CAPTURE1                                   | GTETRGB/GTIOC8A/AGTEE0   | —                   | —                     |

Table 1.17 Pin list for the Standard product (8 of 9)

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC       | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU       |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|------------------------|---------------------|----------------------|
| R6     | R6                  | N6     | N6                  | —                                | P802      | —           | IRQ18        | RXD2_A/SCL2_A/MISO2_A/OM_0_SIO6                                                         | GTIW/GTIOC12A          | —                   | —                    |
| R7     | R7                  | N7     | N7                  | —                                | P804      | —           | IRQ14        | CTS_RTS2_A/SS2_A/DE2/OM_0_SIO7                                                          | GTETRGD/GTIOC13A       | —                   | DSL_TE               |
| R8     | R8                  | —      | —                   | —                                | P501      | —           | IRQ25        | TXD8_B/SDA8_B/MOSI8_B/USB_OVRCURA/SD1DAT6_A/PDMDAT1                                     | GTIOC12A               | AN020               | —                    |
| R9     | R9                  | N8     | N8                  | AVCC0                            | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| R10    | R10                 | N9     | N9                  | AVSS0                            | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| R11    | R11                 | N10    | N10                 | —                                | P005      | —           | IRQ10-DS     | —                                                                                       | —                      | AN005/IVCMP3        | —                    |
| R12    | R12                 | P12    | P12                 | —                                | P003      | —           | IRQ29        | —                                                                                       | —                      | AN003/IVCMP3        | —                    |
| R13    | R13                 | —      | —                   | —                                | P513      | —           | IRQ31        | SCK8_A/DE8/ET0_INT                                                                      | GTIOC13B               | AN016/IVCMP0        | LCD_TC ON3_B/VIO_FLD |
| R14    | R14                 | —      | —                   | —                                | P514      | —           | IRQ13        | SCK4_C/DE4/SDA2_B <sup>*1</sup> /ET_TAS_STA1                                            | GTIOC13B               | —                   | LCD_EX TCLK_B        |
| R15    | R15                 | L13    | L13                 | —                                | P415      | WAIT        | IRQ8         | TXD4_B/SDA4_B/MOSI4_B/RSPCKB_B/CTX1/ET1_MDC                                             | GTIOC0A                | —                   | VIO_HD               |
| R16    | R16                 | L10    | L10                 | —                                | P409      | A18         | IRQ6         | TXD3_A/SDA3_A/MOSI3_A/SDA0_A <sup>*1</sup> /USB_OVRCURADS/USBHS_OVRCURA/GPTP_CAPTURE0   | GTOWUP/ULPTOA0         | ADST1               | —                    |
| R17    | R17                 | K12    | K12                 | —                                | P407      | CS6         | IRQ22        | SCK1_C/DE1/SDA0_B <sup>*1</sup> /USB_VBUS/USBHS_VBUSEN/GPTP_PTPOUT3                     | GTIOC10B/AGTIO0/RTCOUT | ADTRG0              | —                    |
| T1     | —                   | P1     | —                   | MIPI_DL0_P                       | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T2     | —                   | P2     | —                   | MIPI_CL_P                        | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T3     | —                   | P3     | —                   | MIPI_DL1_P                       | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T4     | —                   | P4     | —                   | AVCC_MIPI                        | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| —      | T1                  | —      | P1                  | CLKOUT                           | P205      | —           | IRQ1-DS      | TXD4_A/SDA4_A/MOSI4_A/SCL1_B <sup>*1</sup> /SSLA1_A/USB_OVRCURA/SD1CD                   | GTIV/GTIOC4A/AGTO1     | —                   | —                    |
| —      | T2                  | —      | P2                  | —                                | P203      | —           | IRQ2-DS      | RXD4_A/SCL4_A/MISO4_A/RSPCKA_A/CTX0/USB_VBUSEN/SD1CLK_A                                 | GTIOC5A/ULPTOA1        | —                   | —                    |
| —      | T3                  | —      | P3                  | —                                | P313      | —           | IRQ27        | TXD3_C/SDA3_C/MOSI3_C/MISOA_A/USB_ID/SD1DAT0_A                                          | —                      | —                   | —                    |
| —      | T4                  | —      | —                   | —                                | P901      | —           | IRQ31        | CTS_RTS3_C/SS3_C/DE3                                                                    | GTADSM1/AGTIO1         | —                   | —                    |
| T5     | T5                  | P5     | P5                  | —                                | P809      | —           | IRQ20        | TXD7_B/SDA7_B/MOSI7_B/OM_0_SCLKN                                                        | —                      | —                   | —                    |
| T6     | T6                  | P6     | P6                  | —                                | P800      | —           | IRQ11        | CTS2_A/OM_0_SIO5                                                                        | GTIU/GTIOC11A/AGTOA0   | —                   | —                    |
| T7     | T7                  | —      | —                   | —                                | P502      | —           | IRQ26        | SCK8_B/DE8/USB_OVRCURB/SD1DAT7_A/PDMDAT2                                                | GTIOC12B               | AN019               | —                    |
| T8     | T8                  | R7     | R7                  | —                                | P014      | —           | IRQ27        | —                                                                                       | —                      | AN014/D A0/IVCMP0   | —                    |
| T9     | T9                  | P8     | P8                  | VREFL                            | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T10    | T10                 | P9     | P9                  | VREFL0                           | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T11    | T11                 | R11    | R11                 | —                                | P004      | —           | IRQ9-DS      | —                                                                                       | —                      | AN004/IVCMP2        | —                    |
| T12    | T12                 | M9     | M9                  | —                                | P007      | —           | IRQ28        | —                                                                                       | —                      | AN007/IVCMP3        | —                    |
| T13    | T13                 | N11    | N11                 | —                                | P001      | —           | IRQ7-DS      | —                                                                                       | —                      | AN001/IVCMP3        | —                    |
| T14    | T14                 | —      | —                   | —                                | P806      | —           | IRQ0         | RXD8_A/SCL8_A/MISO8_A/ET1_MDC                                                           | —                      | AN018               | LCD_TC ON0_B/VIO_D14 |
| T15    | T15                 | —      | —                   | —                                | P715      | —           | IRQ12        | RXD4_C/SCL4_C/MISO4_C/ET_TAS_STA2                                                       | GTIOC12A               | —                   | LCD_DATA23_B         |
| T16    | T16                 | P14    | P14                 | —                                | P815      | —           | IRQ15        | CTX0/USB_DM                                                                             | GTIOC8A                | —                   | —                    |
| T17    | T17                 | P15    | P15                 | VSS_USB                          | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| U1     | —                   | R1     | —                   | MIPI_DL0_N                       | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |

**Table 1.17 Pin list for the Standard product (9 of 9)**

| BGA289 | BGA289 without MIPI | BGA224 | BGA224 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC       | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU |
|--------|---------------------|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|------------------------|---------------------|----------------|
| U2     | —                   | R2     | —                   | MIPI_CL_N                        | —         | —           | —            | —                                                                                       | —                      | —                   | —              |
| U3     | —                   | R3     | —                   | MIPI_DL1_N                       | —         | —           | —            | —                                                                                       | —                      | —                   | —              |
| —      | U1                  | —      | R1                  | CACREF                           | P204      | —           | IRQ26        | SCK4_A/DE4/SDA1_B <sup>1</sup> /SSLA0_A/USB_OVRCURB/SD1WP                               | GTIW/GTIOC4B/AGTIO1    | —                   | —              |
| —      | U2                  | —      | R2                  | —                                | P202      | —           | IRQ3-DS      | CTS_RTS4_A/SS4_A/DE4/MOSIA_A/CRX0/USB_EXICEN/SD1CMD_A                                   | GTIOC5B/ULPTOB1        | —                   | —              |
| —      | U3                  | —      | R3                  | —                                | P314      | —           | IRQ28        | RXD3_C/SCL3_C/MISO3_C/SSLA2_A/SD1DAT1_A                                                 | —                      | ADTRG0              | —              |
| U4     | U4                  | —      | N3                  | VSS_13                           | —         | —           | —            | —                                                                                       | —                      | —                   | —              |
| U5     | U5                  | R5     | R5                  | —                                | P808      | —           | IRQ15        | RXD7_B/SCL7_B/MISO7_B/OM_0_SCLK                                                         | GTIOC13B               | —                   | —              |
| U6     | U6                  | R6     | R6                  | —                                | P100      | —           | IRQ2         | SCK9_A/DE9/MISOB_A/OM_0_SIO0/GPTP_MATCH1                                                | GTETRGA/GTIOC8B/AGTIO0 | —                   | —              |
| U7     | U7                  | —      | —                   | CACREF                           | P500      | —           | IRQ24        | RXD8_B/SCL8_B/MISO8_B/USB_VBUSEN/SD1DAT5_A/PDMDAT0                                      | GTIOC11B               | AN021               | —              |
| U8     | U8                  | P7     | P7                  | —                                | P015      | —           | IRQ13        | —                                                                                       | —                      | AN015/D A1/IVCMP0   | —              |
| U9     | U9                  | R8     | R8                  | VREFH                            | —         | —           | —            | —                                                                                       | —                      | —                   | —              |
| U10    | U10                 | R9     | R9                  | VREFH0                           | —         | —           | —            | —                                                                                       | —                      | —                   | —              |
| U11    | U11                 | R10    | R10                 | —                                | P008      | —           | IRQ12-DS     | —                                                                                       | —                      | AN008/IVREF0        | —              |
| U12    | U12                 | P10    | P10                 | —                                | P006      | —           | IRQ11-DS     | —                                                                                       | —                      | AN006/IVCMP2        | —              |
| U13    | U13                 | R12    | R12                 | —                                | P000      | —           | IRQ6-DS      | —                                                                                       | —                      | AN000/IVCMP2        | —              |
| U14    | U14                 | P11    | P11                 | —                                | P002      | —           | IRQ8-DS      | —                                                                                       | —                      | AN002/IVCMP2        | —              |
| U15    | U15                 | R13    | R13                 | —                                | P511      | —           | IRQ15        | CTS_RTS8_A/SS8_A/DE8/SDA1_A <sup>1</sup> /CRX1/ET1_LINKSTA                              | GTIOC0B                | —                   | —              |
| U16    | U16                 | R14    | R14                 | —                                | P814      | —           | IRQ16        | CRX0/USB_DP                                                                             | GTIOC8B                | —                   | —              |
| U17    | U17                 | R15    | R15                 | VCC_USB                          | —         | —           | —            | —                                                                                       | —                      | —                   | —              |

Note: Several pin names have the added suffix of \_A, \_B, and \_C. These suffixes have special conditions for electrical characteristics. See [section 70, Electrical Characteristics](#) for detail.

Note 1. There are two types for IIC function, one is a simple IIC by SCI and another is a dedicated IIC. This terminal is for a dedicated IIC.

**Table 1.18 Pin list for the SiP product (1 of 9)**

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC                 | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/CEU        |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|----------------------------------|---------------------|-----------------------|
| A1     | A1                  | VSS                              | —         | —           | —            | —                                                                                       | —                                | —                   | —                     |
| A2     | A2                  | —                                | P114      | D5/DQ5      | IRQ30-DS     | CTS_RTS0_A/SS0_A/DE0/SSLA0_B/SSIRXD0_B/SD0DAT6_B                                        | GTETRG/GTIOC2B                   | ADSYNC              | LCD_DAT A8_A          |
| A3     | A3                  | —                                | P609      | D7/DQ7      | IRQ29        | TXD0_C/SDA0_C/MOSI0_C/MISOA_B/CTX1                                                      | GTIU/GTIOC5B/ULPTOA1-DS          | AD1FLAG1            | LCD_DAT A6_A          |
| A4     | A4                  | —                                | P113      | D4/DQ4      | IRQ28        | RXD0_A/SCL0_A/MISO0_A/SSLA1_B/SSILRCK0_B/SSIFS0_B/SD0DAT5_B                             | GTETRGB/GTIOC2A/ULPTOA0-DS       | ADST1               | LCD_DAT A9_A          |
| A5     | A5                  | —                                | P301      | D1/DQ1      | IRQ6         | TXD6_B/SDA6_B/MOSI6_B/SD0DAT2_B                                                         | GTOULO/GTIOC4B/AGTIO0/ULPTEE0-DS | —                   | LCD_DAT A12_A         |
| A6     | A6                  | TDI                              | P208      | —           | IRQ3         | RXD9_B/SCL9_B/MISO9_B/CRX1                                                              | GTOVLO/GTIOC1B                   | VCOUT               | —                     |
| A7     | A7                  | TMS/SWDIO                        | P210      | —           | IRQ24        | CTS_RTS9_B/SS9_B/DE9                                                                    | GTOULO/GTIOC0B                   | —                   | —                     |
| A8     | A8                  | VLO                              | —         | —           | —            | —                                                                                       | —                                | —                   | —                     |
| A9     | A9                  | VLO                              | —         | —           | —            | —                                                                                       | —                                | —                   | —                     |
| A10    | A10                 | VSS_DCDC                         | —         | —           | —            | —                                                                                       | —                                | —                   | —                     |
| A11    | A11                 | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                                | —                   | —                     |
| A12    | A12                 | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                                | —                   | —                     |
| A13    | A13                 | —                                | P309      | —           | IRQ25-DS     | CTS9_B/ET1_GTX_CLK/RGMII1_TXC                                                           | GTCPP08                          | VCOUT               | LCD_DAT A15_A/VIO_D10 |

Table 1.18 Pin list for the SiP product (2 of 9)

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC           | ADC16H/DAC12/ACMPHS | MIPI/ GLCDC/C EU     |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|----------------------------|---------------------|----------------------|
| A14    | A14                 | —                                | P906      | —           | IRQ9         | CTS6_A/USB_ID/SSILRCK1_A/SSIFS1_A/ET1_RXD0/RGMII1_RXD0/RMII1_RXD0/PDMDAT0               | GTIOC13B/ULPTO1            | AD0FLAG1            | LCD_DAT A20_A/VIO_D5 |
| A15    | A15                 | —                                | P905      | —           | IRQ8         | RXD3_B/SCL3_B/MISO3_B/ET1_RX_CLK/RGMII1_RXC/RMII1_REF50CK/PDMDAT1                       | GTCPP013                   | AD1FLAG1            | LCD_DAT A19_A/VIO_D6 |
| A16    | A16                 | —                                | P907      | —           | IRQ10        | SCK6_A/DE6/USB_EXICEN/SSIBCK1_A/ET1_RXD1/RGMII1_RXD1/RMII1_RXD1/PDMCLK2                 | GTIOC13A/ULPTEE1           | ADSYNC              | LCD_DAT A21_A/VIO_D4 |
| A17    | A17                 | —                                | P207      | —           | IRQ25        | ET1_RXD5                                                                                | GTCPP03                    | —                   | LCD_DAT A9_B         |
| A18    | A18                 | VSS                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| B1     | B1                  | —                                | P813      | SDCS        | IRQ15        | SCK7_A/DE7/PDMCLK2                                                                      | GTETRG/AGTIOC7B            | —                   | VIO_D13              |
| B2     | B2                  | —                                | PA12      | D9/DQ9      | IRQ11        | RXD9_C/SCL9_C/MISO9_C                                                                   | GTIW/GTIOC6B               | —                   | —                    |
| B3     | B3                  | —                                | P115      | D6/DQ6      | IRQ31-DS     | CTS0_A/MOSIA_B/SSITXD0_B/SD0DAT7_B                                                      | GTETRGD/GTIOC5A            | AD0FLAG1            | LCD_DAT A7_A         |
| B4     | B4                  | —                                | PA11      | D8/DQ8      | IRQ10        | SCK9_C/DE9                                                                              | GTIV/GTIOC6A               | —                   | —                    |
| B5     | B5                  | —                                | P112      | D3/DQ3      | IRQ27        | TXD0_A/SDA0_A/MOSIO_A/SSLA2_B/SSIBCK0_B/SD0DAT4_B                                       | GTETRG/AGTIOC3B/ULPTOB0-DS | ADST0               | LCD_DAT A10_A        |
| B6     | B6                  | TD0/SWO/CLKOUT                   | P209      | —           | IRQ25        | TXD9_B/SDA9_B/MOSI9_B/CTX1                                                              | GTOVUP/GTIOC1A             | —                   | —                    |
| B7     | B7                  | TCK/SWCLK                        | P211      | —           | IRQ23        | SCK9_B/DE9                                                                              | GTOUUP/GTIOC0A             | —                   | —                    |
| B8     | B8                  | VLO                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| B9     | B9                  | VLO                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| B10    | B10                 | VSS_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| B11    | B11                 | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| B12    | B12                 | VCC_DCDC                         | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| B13    | B13                 | —                                | P311      | —           | IRQ23-DS     | SCK3_B/DE3/CRX0/ET1_TX_CLK                                                              | GTADSM1/GTCPP06/AGTOB1     | —                   | LCD_DAT A17_A/VIO_D8 |
| B14    | B14                 | —                                | P908      | —           | IRQ11        | TXD6_A/SDA6_A/MOSI6_A/CRX1/USB_OVRCURB/USBHS_ID/ET1_RXD2/RGMII1_RXD2/PDMCLK1            | GTIOC12B/ULPTEV11          | ADST1               | LCD_DAT A22_A/VIO_D3 |
| B15    | B15                 | —                                | P909      | —           | IRQ21-DS     | RXD6_A/SCL6_A/MISO6_A/CTX1/USB_OVRCUR/USBHS_EXICEN/ET1_RXD3/RGMII1_RXD3/PDMCLK0         | GTIOC12A/ULPTOA1           | ADST0               | LCD_DAT A23_A/VIO_D2 |
| B16    | B16                 | —                                | P904      | —           | IRQ2         | ET1_RXD4                                                                                | GTIOC11B                   | —                   | LCD_DAT A8_B         |
| B17    | B17                 | —                                | PD01      | —           | IRQ22        | SCK8_C/DE8/SD0DAT2_C/ET1_RXD6                                                           | GTCPP02                    | —                   | —                    |
| B18    | B18                 | —                                | PD02      | —           | IRQ21        | TXD8_C/SDA8_C/MOSI8_C/SD0DAT1_C/ET1_RXD7                                                | GTCPP01                    | —                   | —                    |
| C1     | C1                  | —                                | PA06      | CS1/CKE     | IRQ17        | CTS2_C/SD0DAT1_A/PDMDAT1                                                                | GTETRG/AGTIOC7B            | —                   | VIO_D11              |
| C2     | C2                  | —                                | P613      | D15/DQ15    | IRQ19        | CTS0_C/USBHS_OVRCURB                                                                    | GTETRG/AGTIOC9B/AGTO1      | —                   | LCD_DAT A2_A         |
| C3     | C3                  | —                                | PA13      | D10/DQ10    | IRQ12        | CTS_RTS9_C/SS9_C/DE9                                                                    | GTOVUP/GTIOC10A            | —                   | —                    |
| C4     | C4                  | —                                | P300      | D2/DQ2      | IRQ4         | SCK0_A/DE0/SSLA3_B/SD0DAT3_B                                                            | GTIOC3A/ULPTEV10-DS        | —                   | LCD_DAT A11_A        |
| C5     | C5                  | —                                | P302      | D0/DQ0      | IRQ5         | RXD6_B/SCL6_B/MISO6_B/SD0DAT1_B                                                         | GTOUUP/GTIOC4A/ULPTO0-DS   | —                   | LCD_DAT A13_A        |
| C6     | C6                  | —                                | P200      | —           | NMI          | —                                                                                       | —                          | —                   | —                    |
| C7     | C7                  | RES                              | —         | —           | —            | —                                                                                       | —                          | —                   | —                    |
| C8     | C8                  | —                                | P110      | —           | IRQ20        | SD0DAT4_C                                                                               | GTIOC9B                    | —                   | —                    |
| C9     | C9                  | —                                | P903      | —           | IRQ1         | —                                                                                       | GTIOC11A                   | —                   | LCD_DAT A2_B         |
| C10    | C10                 | TCLK                             | P308      | —           | IRQ26-DS     | CTS3_B/SD0CLK_B/ET1_TX_ER/ETHPHYCLK                                                     | GTIU/GTCPP09/ULPTOB1       | —                   | VIO_D11              |
| C11    | C11                 | TDATA2                           | P305      | —           | IRQ8         | SD0WP/ET1_TXD2/RGMII1_TXD2                                                              | GTOVUP/GTCPP012/ULPTEE1    | —                   | VIO_D14              |
| C12    | C12                 | TDATA0                           | P307      | —           | IRQ27-DS     | CTS_RTS6_A/SS6_A/DE6/SD0CMD_B/ET1_TXD0/RGMII1_TXD0/RMII1_TXD0                           | GTIV/GTCPP010/ULPTOA1      | —                   | VIO_D12              |
| C13    | C13                 | —                                | P911      | —           | IRQ6         | ET1_TXD5                                                                                | GTIOC3B                    | —                   | LCD_DAT A5_B         |

Table 1.18 Pin list for the SiP product (3 of 9)

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC          | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/C EU       |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|---------------------------|---------------------|-----------------------|
| C14    | C14                 | CLKOUT                           | P206      | CS7         | IRQ0-DS      | USB_VBUS/SSIDATA1_A/SD0DAT7_C/ET1_RX_DV/RGMII1_RX_CTL/RMII1_CRS_DV                      | GTIU/GTCCPO0/ULPTOB1      | —                   | VIO_D0                |
| C15    | C15                 | —                                | PD04      | —           | IRQ20        | CTS_RTS8_C/SS8_C/DE8/USBHS_ID/SD0CMD_C/ET0_RXD5                                         | GTIOC3A                   | —                   | —                     |
| C16    | C16                 | —                                | PD03      | —           | IRQ21        | RXD8_C/SCL8_C/MISO8_C/USBHS_EXICEN/SD0DAT0_C/ET0_RXD4                                   | GTIOC3B                   | —                   | —                     |
| C17    | C17                 | —                                | PD05      | —           | IRQ19        | CTS8_C/USBHS_OVRCURB/SD0CLK_C/ET0_RXD6                                                  | GTIOC2B                   | —                   | —                     |
| C18    | C18                 | —                                | PD06      | —           | IRQ18        | USBHS_OVRCURA/SD0WP/ET0_RXD7                                                            | GTIOC2A                   | —                   | —                     |
| D1     | D1                  | —                                | PA04      | A1          | IRQ19        | SCK2_C/DE2/SD0DAT3_A                                                                    | GTIU/GTIOC4B              | ADST0               | VIO_D9                |
| D2     | D2                  | CACREF/CLKOUT                    | P611      | D13/DQ13    | IRQ17        | SCK0_C/DE0/MOSIA_B/USBHS_VBUS                                                           | GTOULO/GTIOC4B            | —                   | LCD_DAT A4_A          |
| D3     | D3                  | —                                | P610      | D12/DQ12    | IRQ16        | RXD0_C/SCL0_C/MISO0_C/RSPCKA_B/CRX1                                                     | GTOUUP/GTIOC4A/ULPTOB1-DS | —                   | LCD_DAT A5_A          |
| D4     | D4                  | —                                | PA14      | D11/DQ11    | IRQ13        | TXD9_C/SDA9_C/MOSI9_C                                                                   | GTOVLO/GTIOC10B           | —                   | —                     |
| D5     | D5                  | —                                | P303      | —           | IRQ29-DS     | SCK6_B/DE6                                                                              | GTIOC7B                   | —                   | LCD_DAT A14_A         |
| D6     | D6                  | —                                | P915      | —           | IRQ8         | CTS6_B                                                                                  | GTIOC5A                   | —                   | LCD_DAT A1_B          |
| D7     | D7                  | —                                | P108      | —           | IRQ24        | SD0DAT6_C                                                                               | GTIOC10B                  | —                   | —                     |
| D8     | D8                  | —                                | P111      | —           | IRQ19        | SD0DAT3_C                                                                               | GTIOC9A                   | —                   | —                     |
| D9     | D9                  | —                                | P109      | —           | IRQ23        | SD0DAT5_C                                                                               | GTIOC10A                  | —                   | —                     |
| D10    | D10                 | —                                | P310      | —           | IRQ24-DS     | TXD3_B/SDA3_B/MOSI3_B/ET1_TX_EN/RGMII1_TX_CTL/RMII1_TX_EN                               | GTCPP07/AGTEE1            | —                   | LCD_DAT A16_A/VIO_D9  |
| D11    | D11                 | TDATA3                           | P304      | —           | IRQ9         | SD0DAT0_B/ET1_TXD3/RGMII1_TXD3                                                          | GTOVLO/GTIOC7A/ULPTO1     | —                   | VIO_D15               |
| D12    | D12                 | TDATA1                           | P306      | —           | IRQ28-DS     | SD0CD/ET1_TXD1/RGMII1_TXD1/RMII1_TXD1                                                   | GTIW/GTCCPO11/ULPTEV1     | —                   | VIO_D13               |
| D13    | D13                 | —                                | P912      | —           | IRQ5         | ET1_TXD6                                                                                | GTIOC3A                   | —                   | LCD_DAT A6_B          |
| D14    | D14                 | —                                | PB04      | —           | IRQ9         | SCK5_C/DE5/ET0_TXD3/RGMII0_TXD3                                                         | GTCPP03                   | AD0FLAG1            | LCD_DAT A14_B/VIO_CLK |
| D15    | D15                 | —                                | PB07      | —           | IRQ1         | ET0_TXD5                                                                                | GTIOC9B                   | —                   | LCD_DAT A10_B         |
| D16    | D16                 | —                                | PB05      | —           | IRQ15        | CTS5_C/ET0_TXD7                                                                         | GTCPP04                   | —                   | LCD_DAT A12_B         |
| D17    | D17                 | —                                | PB03      | —           | IRQ13        | TXD5_C/SDA5_C/MOSI5_C/ET0_TXD2/RGMII0_TXD2                                              | GTCPP01                   | ADSYNC              | LCD_DAT A15_B/VIO_HD  |
| D18    | D18                 | —                                | PB01      | ALE         | IRQ12        | CTS_RTS1_B/SS1_B/DE1/ET0_TX_CLK                                                         | GTCPP02                   | AD1FLAG1            | LCD_DAT A13_B/VIO_FLD |
| E1     | E1                  | —                                | PA15      | EBCLK/SDCLK | IRQ14        | CTS9_C/PDMCLK1                                                                          | GTIOC7A                   | —                   | VIO_D14               |
| E2     | E2                  | —                                | P615      | —           | IRQ7         | TXD7_A/SDA7_A/MOSI7_A/USBHS_EXICEN                                                      | GTETRGC/GTCCPO10          | —                   | LCD_DAT A0_A          |
| E3     | E3                  | —                                | P614      | WR/WR0/DQM0 | IRQ20        | RXD7_A/SCL7_A/MISO7_A/USBHS_ID                                                          | GTETRGC/GTCCPO9/AGTO0     | —                   | LCD_DAT A1_A          |
| E4     | E4                  | —                                | P612      | D14/DQ14    | IRQ18        | CTS_RTS0_C/SS0_C/DE0/SSLA0_B/USBHS_OVRCURA                                              | GTIOC9A                   | —                   | LCD_DAT A3_A          |
| E5     | E5                  | —                                | P914      | —           | IRQ9         | CTS_RTS6_B/SS6_B/DE6                                                                    | GTIOC5B                   | —                   | LCD_DAT A0_B          |
| E6     | E6                  | MD                               | P201      | —           | IRQ4         | —                                                                                       | —                         | —                   | —                     |
| E10    | E10                 | —                                | P902      | ALE         | IRQ0         | AUDIO_CLK/ETHPHYCLK                                                                     | GTCPP013                  | —                   | LCD_DAT A3_B/VIO_D1   |
| E11    | E11                 | —                                | P312      | —           | IRQ22-DS     | CTS_RTS3_B/SS3_B/DE3/CTX0/ET1_RX_ER/RMII1_RX_ER/PDMDAT2                                 | GTADSM0/GTCCPO5/AGTOA1    | —                   | LCD_DAT A18_A/VIO_D7  |
| E12    | E12                 | —                                | P910      | —           | IRQ7         | ET1_TXD4                                                                                | GTCPP012                  | —                   | LCD_DAT A4_B          |
| E13    | E13                 | CLKOUT                           | P913      | —           | IRQ3         | ET1_TXD7                                                                                | GTCPP011                  | —                   | LCD_DAT A7_B          |

Table 1.18 Pin list for the SiP product (4 of 9)

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF    | GPT/AGT/ULPT/RTC       | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/C EU       |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|--------------------------------------------------------------------------------------------|------------------------|---------------------|-----------------------|
| E14    | E14                 | —                                | PB02      | —           | IRQ11        | RXD5_C/SCL5_C/MISO5_C/ET0_TXD1/RGMII0_TXD1/RMII0_TXD1                                      | GTCPP00                | ADST1               | LCD_DAT A16_B/VIO_VD  |
| E15    | E15                 | —                                | PB06      | —           | IRQ0         | CTS_RTS5_C/SS5_C/DE5/ET0_TXD6                                                              | GTIOC9A                | —                   | LCD_DAT A11_B         |
| E16    | E16                 | —                                | PD07      | —           | IRQ17        | USBHS_VBUSEN/SD0CD/ET0_TXD4                                                                | GTCPP00                | —                   | —                     |
| E17    | E17                 | —                                | PB00      | —           | IRQ10        | SCK1_B/DE1/USBHS_VBUSEN/ET0_TXD0/RGMII0_TXD0/RMII0_TXD0/PDMDAT2                            | GTCPP04                | ADST0               | LCD_DAT A17_B         |
| E18    | E18                 | —                                | P706      | —           | IRQ7         | RXD1_B/SCL1_B/MISO1_B/USBHS_OVRCURB-DS/ET0_GTX_CLK/RGMII0_TXC/ETHPHYCLK/PDMDAT0            | GTCPP02/AGTIO0         | —                   | VIO_D10               |
| F1     | F1                  | —                                | PA02      | A3          | IRQ31        | RXD2_C/SCL2_C/MISO2_C/SD0DAT5_A                                                            | GTIW/GTCPP09           | ADSYNC              | VIO_D7                |
| F2     | F2                  | —                                | PA10      | CS2/RAS     | IRQ4         | SCK5_B/DE5/PDMCLK0                                                                         | GTCPP013               | —                   | LCD_TCO N1_A/VIO_D15  |
| F3     | F3                  | —                                | PA08      | CS0/WE      | IRQ6         | RXD5_B/SCL5_B/MISO5_B                                                                      | GTETRGD/GTCPP011       | —                   | LCD_TCO N3_A          |
| F4     | F4                  | —                                | PA09      | CS3/CAS     | IRQ5         | TXD5_B/SDA5_B/MOSI5_B                                                                      | GTCPP012               | —                   | LCD_TCO N2_A          |
| F5     | F5                  | —                                | PC14      | —           | IRQ0         | TXD6_C/SDA6_C/MOSI6_C/ET0_WOL                                                              | GTADSM1/GTCPP09        | —                   | —                     |
| F7     | F7                  | VCC_08                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| F8     | F8                  | VSS_08                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| F9     | F9                  | VSS3                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| F10    | F10                 | VCL3                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| F11    | F11                 | VSS_07                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| F12    | F12                 | VCC_07                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| F13    | F13                 | —                                | P700      | —           | IRQ16-DS     | RXD2_B/SCL2_B/MISO2_B/MISOA_C/SSIDATA1_B/SD1WP/ET0_RXD2/RGMII0_RXD2                        | GTIOC5A                | —                   | VIO_D4                |
| F14    | F14                 | —                                | P702      | —           | IRQ18-DS     | CTS2_B/RSPCKA_C/SSIBCK1_B/SD1DAT5_B/ET0_RXD0/RGMII0_RXD0/RMII0_RXD0                        | GTIOC6A/ULPT00         | —                   | VIO_D6                |
| F15    | F15                 | —                                | P406      | —           | IRQ31        | TXD2_B/SDA2_B/MOSI2_B/SSLA3_C/SSIRXD0_A/SD1CD/ET0_RXD3/RGMII0_RXD3                         | GTIOC1B                | —                   | VIO_D3                |
| F16    | F16                 | —                                | P701      | —           | IRQ17-DS     | CTS_RTS2_B/SS2_B/DE2/MOSIA_C/SSILRCK1_B/SSIFS1_B/SD1DAT4_B/ET0_RXD1/RGMII0_RXD1/RMII0_RXD1 | GTIOC5B/ULPT01         | —                   | VIO_D5                |
| F17    | F17                 | —                                | P707      | —           | IRQ8         | TXD1_B/SDA1_B/MOSI1_B/USBHS_OVRCURA-DS/ET0_TX_ER/ETHPHYCLK/PDMDAT1                         | GTCPP03                | —                   | LCD_DAT A18_B/VIO_D11 |
| F18    | F18                 | —                                | P705      | —           | IRQ19        | CTS1_B/SSLA2_C/CRX0/ET0_TX_EN/RGMII0_TX_CTL/RMII0_TX_EN/PDMCLK2                            | GTADSM1/GTCPP01/AGTIO0 | —                   | VIO_D9                |
| G1     | G1                  | —                                | PA00      | A5          | IRQ22        | CTS_RTS5_B/SS5_B/DE5/SD0DAT7_A                                                             | GTOVLO/GTCPP07         | AD1FLAG1            | LCD_CLK A/VIO_D5      |
| G2     | G2                  | —                                | PA03      | A2          | IRQ20        | TXD2_C/SDA2_C/MOSI2_C/SD0DAT4_A                                                            | GTIV/GTCPP010          | ADST1               | VIO_D8                |
| G3     | G3                  | —                                | PA05      | A0/BC0/DQM1 | IRQ18        | CTS_RTS2_C/SS2_C/DE2/SD0DAT2_A/PDMDAT2                                                     | GTETRGD/GTIOC4A        | —                   | VIO_D10               |
| G4     | G4                  | —                                | PA07      | RD          | IRQ16        | CTS7_A/SD0DAT0_A/PDMDAT0                                                                   | GTETRGB/GTIOC7A        | VCOUT               | VIO_D12               |
| G5     | G5                  | —                                | PC12      | —           | IRQ2         | SCK6_C/DE6/ET0_MDIO                                                                        | GTCPP011               | —                   | —                     |
| G7     | G7                  | VCC_09                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| G8     | G8                  | VSS_09                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| G9     | G9                  | VSS4                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| G10    | G10                 | VCL4                             | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| G11    | G11                 | VSS_06                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| G12    | G12                 | VCC_06                           | —         | —           | —            | —                                                                                          | —                      | —                   | —                     |
| G13    | G13                 | —                                | P405      | —           | IRQ30        | SCK2_B/DE2/SSITXD0_A/SD1DAT3_B/ET0_RX_DV/RGMII0_RX_CTL/RMII0_CRS_DV                        | GTIOC1A/AGTIO1         | —                   | VIO_D2                |

Table 1.18 Pin list for the SiP product (5 of 9)

| BGA303 | BGA303 without MIP1 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC      | ADC16H/DAC12/ACMPHS | MIP1/GLCDC/C EU             |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|-----------------------|---------------------|-----------------------------|
| G14    | G14                 | —                                | P704      | —           | IRQ26        | SSLA1_C/CTX0/SD1DAT7_B/ET0_RX_ER/RMII0_RX_ER/PDMCLK1                                    | GTADSM0/GTCPP00/AGT00 | —                   | VIO_D8                      |
| G15    | G15                 | —                                | P703      | —           | IRQ19-DS     | SSLA0_C/SD1DAT6_B/ET0_RX_CLK/RGMII0_RXC/RMII0_REF50CK/PDMCLK0                           | GTIOC6B/AGT01         | VCOUT               | VIO_D7                      |
| G16    | G16                 | VSS_03                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| G17    | G17                 | VCC_05                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| G18    | G18                 | VSS_05                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H1     | H1                  | —                                | P504      | A7          | IRQ7         | SD0WP                                                                                   | GTOULO/GTCPP01        | —                   | VIO_D3                      |
| H2     | H2                  | —                                | P503      | A6          | IRQ6         | SD0CD                                                                                   | GTOUUP/GTCPP06        | —                   | VIO_D4                      |
| H3     | H3                  | —                                | P505      | A8          | IRQ8         | SD0CLK_A                                                                                | GTOWUP/GTCPP02        | —                   | VIO_D2                      |
| H4     | H4                  | —                                | PA01      | A4          | IRQ21        | CTS5_B/SD0DAT6_A                                                                        | GTOVUP/GTCPP08        | AD0FLAG1            | LCD_TCO<br>NO_A/<br>VIO_D6  |
| H5     | H5                  | —                                | PC11      | —           | IRQ3         | CTS_RTS6_C/SS6_C/DE6/ET0_MDC                                                            | GTCPP012              | —                   | —                           |
| H7     | H7                  | VCC_10                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H8     | H8                  | VSS_10                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H9     | H9                  | VSS7                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H10    | H10                 | VCL5                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H11    | H11                 | VSS5                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H12    | H12                 | VCC_04                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H13    | H13                 | VSS_04                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H15    | H15                 | VCC_03                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H16    | H16                 | VCC_USBHS                        | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H17    | H17                 | USBHS_DP                         | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| H18    | H18                 | USBHS_DM                         | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J1     | J1                  | —                                | P506      | A9          | IRQ9         | SD0CMD_A                                                                                | GTOWLO/GTCPP03        | —                   | VIO_D1                      |
| J2     | J2                  | —                                | P507      | A10         | IRQ10        | CTS_RTS7_A/SS7_A/DE7/ET_TAS_STA0                                                        | GTADSM0/GTIOC0A       | —                   | LCD_EXT<br>CLK_A/<br>VIO_D0 |
| J3     | J3                  | —                                | P508      | A11         | IRQ1         | CTS5_A/ET_TAS_STA1                                                                      | GTADSM1/GTIOC0B       | —                   | VIO_VD                      |
| J4     | J4                  | —                                | P509      | A12         | IRQ2         | CTS_RTS5_A/SS5_A/DE5/ET_TAS_STA2                                                        | GTIOC1A/ULPTEV1       | —                   | VIO_HD                      |
| J5     | J5                  | —                                | PC13      | —           | IRQ1         | RXD6_C/SCL6_C/MISO6_C/ET0_INT                                                           | GTCPP010              | —                   | —                           |
| J7     | J7                  | VCC2_11                          | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J8     | J8                  | VSS_11                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J9     | J9                  | VCL7                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J10    | J10                 | VCL6                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J11    | J11                 | VSS6                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J12    | J12                 | VCL2                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J13    | J13                 | VSS2                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J15    | J15                 | VSS_02                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J16    | J16                 | USBHS_RREF                       | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J17    | J17                 | VSS2_USBHS                       | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| J18    | J18                 | VSS1_USBHS                       | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| K1     | K1                  | —                                | PC15      | A16         | IRQ30        | CTS6_C/CRX1                                                                             | GTADSM0               | —                   | —                           |
| K2     | K2                  | CACREF                           | P608      | A14         | IRQ22        | TXD5_A/SDA5_A/MOSI5_A                                                                   | GTOWUP/GTCPP04        | —                   | VIO_FLD                     |
| K3     | K3                  | —                                | P510      | A13         | IRQ3         | RXD5_A/SCL5_A/MISO5_A/ET_TAS_STA3                                                       | GTIOC1B/ULPTEV10      | —                   | VIO_CLK                     |
| K4     | K4                  | —                                | PD00      | A15         | IRQ23        | SCK5_A/DE5/CTX1                                                                         | GTOWLO/GTCPP05        | —                   | —                           |
| K5     | K5                  | VSS                              | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| K6     | K6                  | VSS                              | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| K7     | K7                  | VSS_12                           | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| K8     | K8                  | VSS9                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |
| K9     | K9                  | VCL9                             | —         | —           | —            | —                                                                                       | —                     | —                   | —                           |

**Table 1.18 Pin list for the SiP product (6 of 9)**

| BGA303 | BGA303 without MIP1 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC       | ADC16H/DAC12/ACMPHS | MIP1/GLCDC/C EU |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|------------------------|---------------------|-----------------|
| K10    | K10                 | VCL8                             | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| K11    | K11                 | VSS8                             | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| K12    | K12                 | VCL1                             | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| K13    | K13                 | VSS1                             | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| K15    | K15                 | VCC_02                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| K16    | K16                 | AVCC_USBHS                       | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| K17    | K17                 | XTAL                             | P213      | —           | IRQ2         | TXD1_C/SDA1_C/MOSI1_C                                                                   | GTETRG/GTIOC0A/ULPTEE0 | ADTRG1              | —               |
| K18    | K18                 | EXTAL                            | P212      | —           | IRQ3         | RXD1_C/SCL1_C/MISO1_C                                                                   | GTETRGD/GTIOC0B/AGTEE1 | —                   | —               |
| L1     | L1                  | —                                | PC10      | —           | IRQ4         | —                                                                                       | GTCPP013               | —                   | —               |
| L2     | L2                  | VSS                              | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L3     | L3                  | PUP                              | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L4     | L4                  | VCC2_16                          | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L5     | L5                  | VSS_16                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L7     | L7                  | VCC2_12                          | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L8     | L8                  | VSS_14                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L9     | L9                  | VSS_15                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L10    | L10                 | VSS10                            | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L11    | L11                 | VCL10                            | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L12    | L12                 | VCL0                             | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L13    | L13                 | VSS0                             | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L14    | L14                 | —                                | P403      | —           | IRQ14-DS     | CTS_RTS1_A/SS1_A/DE1/SSIBCK0_A/SD1DAT1_B/ET1_WOL                                        | GTIOC3A/RTCIC1         | AD0FLAG1            | —               |
| L15    | L15                 | —                                | P404      | —           | IRQ15-DS     | CTS1_A/SSILRCK0_A/SSIFS0_A/SD1DAT2_B/ET0_WOL                                            | GTIOC3B/RTCIC2         | AD1FLAG1            | —               |
| L16    | L16                 | VCC_01                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| L17    | L17                 | XCOUT                            | P214      | —           | IRQ21        | —                                                                                       | —                      | —                   | —               |
| L18    | L18                 | XCIN/EXCIN                       | P215      | —           | IRQ20        | —                                                                                       | —                      | —                   | —               |
| M1     | M1                  | —                                | PC09      | —           | IRQ5         | —                                                                                       | —                      | —                   | —               |
| M2     | M2                  | VSS                              | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M3     | M3                  | VSS                              | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M4     | M4                  | VCC2_17                          | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M5     | M5                  | VSS_17                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M8     | M8                  | VCC2_14                          | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M9     | M9                  | VCC2_15                          | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M11    | M11                 | VSS11                            | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M12    | M12                 | VCL11                            | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M14    | M14                 | —                                | P414      | A23         | IRQ9         | RXD4_B/SCL4_B/MISO4_B/SSLB0_B/CRX1/ET1_MDIO                                             | GTIOC0B                | —                   | VIO_CLK         |
| M15    | M15                 | CACREF                           | P402      | —           | IRQ4-DS      | SK1_A/DE1/CRX0/AUDIO_CLK/SD1DAT0_B/ET0_LINKSTA                                          | RTCIC0                 | —                   | —               |
| M16    | M16                 | —                                | P410      | A19         | IRQ5         | SK3_A/DE3/SCL0_A*/USB_OVRCURB-DS/USBHS_OVRCURB/GPTP_MATCH0                              | GTOVLO/GTIOC9B/AGTOB1  | ADST0               | —               |
| M17    | M17                 | VBATT                            | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| M18    | M18                 | VSS_01                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| N1     | N1                  | —                                | PC08      | —           | IRQ29        | —                                                                                       | GTCPP08                | —                   | —               |
| N2     | N2                  | VSS                              | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| N3     | N3                  | VSS                              | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| N4     | N4                  | VCC2_18                          | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| N5     | N5                  | VSS_18                           | —         | —           | —            | —                                                                                       | —                      | —                   | —               |
| N7     | N7                  | —                                | P105      | —           | IRQ0         | CTS_RTS8_B/SS8_B/DE8/SSLB2_A/OM_0_ECSINT1                                               | GTIOC1A/ULPT01-DS      | ADSYNC              | —               |
| N10    | N10                 | —                                | P810      | —           | IRQ21        | SK7_B/DE7/SD1DAT2_A/PDMCLK0                                                             | GTIOC10A/ULPT0A0       | —                   | —               |

Table 1.18 Pin list for the SiP product (7 of 9)

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC                  | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/C EU        |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|-----------------------------------|---------------------|------------------------|
| N14    | N14                 | —                                | P710      | CS5         | IRQ17        | CTS4_B/SSLB3_B/ET0_LINKSTA                                                              | GTIOC11B                          | —                   | LCD_EXT CLK_B/ VIO_D12 |
| N15    | N15                 | CACREF                           | P411      | A20         | IRQ4         | CTS_RTS3_A/SS3_A/DE3/USB_ID/ USBHS_ID/GPTP_PTPOUT1                                      | GTOVUP/GTIOC9A/AGTOA1             | —                   | DSI_TE                 |
| N16    | N16                 | —                                | P408      | A17         | IRQ7         | RXD3_A/SCL3_A/MISO3_A/SCL0_B <sup>1</sup> / USB_VBUS/USBHS_VBUS/ GPTP_PTPOUT2           | GTOWLO/GTIOC10A/ ULPTOB0          | ADSYNC              | —                      |
| N17    | N17                 | —                                | P412      | A21         | IRQ20-DS     | CTS3_A/USB_EXICEN/ USBHS_EXICEN/GPTP_PTPOUT0                                            | GTOULO/GTCPP08/AGTEE1             | —                   | —                      |
| N18    | N18                 | —                                | P401      | —           | IRQ5-DS      | RXD1_A/SCL1_A/MISO1_A/ I3C_SDA0/CTX0/SD1CMD_B                                           | GTETRG/AGTIOC6B                   | —                   | VIO_D1                 |
| P1     | P1                  | VSS                              | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| P2     | P2                  | VSS                              | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| P3     | P3                  | VSS                              | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| P4     | P4                  | VCC2_19                          | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| P5     | P5                  | VSS_19                           | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| P6     | P6                  | —                                | P104      | —           | IRQ1         | CTS9_A/SSLB1_A/OM_0_CS1                                                                 | GTETRGB/GTIOC1B                   | AD0FLAG1            | —                      |
| P7     | P7                  | —                                | P107      | —           | IRQ31        | CTS4_A/OM_0_CS0                                                                         | GTOWUP/GTIOC8A/AGTOA0             | ADST0               | —                      |
| P8     | P8                  | —                                | P106      | —           | IRQ16        | CTS8_B/SSLB3_A/OM_0_RESET                                                               | GTOWLO/GTIOC8B/AGTOB0/ ULPTEE1-DS | ADST1               | —                      |
| P9     | P9                  | —                                | P811      | —           | IRQ22        | CTS7_B/USB_ID/SD1DAT3_A/ PDMCLK1                                                        | GTIOC10B/ULPTOB0                  | —                   | —                      |
| P10    | P10                 | —                                | P013      | —           | IRQ14        | —                                                                                       | —                                 | AN013               | —                      |
| P11    | P11                 | —                                | P011      | —           | IRQ16        | —                                                                                       | —                                 | AN011               | —                      |
| P12    | P12                 | —                                | P807      | —           | IRQ11        | —                                                                                       | GTIOC13A                          | —                   | LCD_TCO N2_B           |
| P13    | P13                 | CACREF                           | P708      | WR1/BC1     | IRQ11        | SCK4_B/DE4/SDA2_A <sup>1</sup> /MOSIB_B/ AUDIO_CLK/ET0_MDC                              | GTCPP06                           | —                   | VIO_VD                 |
| P14    | P14                 | —                                | P712      | —           | IRQ2         | CTS1_C/SSLB1_B/GPTP_CAPTURE1                                                            | GTIOC2B/AGTOB0                    | —                   | LCD_DAT A20_B          |
| P15    | P15                 | —                                | P714      | —           | IRQ13        | TXD4_C/SDA4_C/MOSI4_C/ GPTP_PPS1                                                        | GTIOC12B                          | —                   | DSI_TE/ LCD_DAT A22_B  |
| P16    | P16                 | —                                | P711      | —           | IRQ3         | CTS_RTS1_C/SS1_C/DE1/SSLB2_B/ GPTP_PPS0                                                 | GTIOC11A/AGTEE0                   | —                   | LCD_DAT A19_B          |
| P17    | P17                 | —                                | P713      | —           | IRQ14        | CTS4_C/GPTP_MATCH1                                                                      | GTIOC2A/AGTOA0                    | —                   | LCD_DAT A21_B          |
| P18    | P18                 | —                                | P400      | —           | IRQ0         | TXD1_A/SDA1_A/MOSI1_A/I3C_SCL0/ AUDIO_CLK/SD1CLK_B                                      | GTIOC6A/AGTIO1                    | ADTRG1              | VIO_D0                 |
| R1     | R1                  | —                                | P602      | —           | IRQ28        | RXD0_B/SCL0_B/MISO0_B                                                                   | GTIOC7B/ULPTEE0                   | —                   | —                      |
| R2     | R2                  | VSS                              | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| R3     | R3                  | VSS                              | —         | —           | —            | —                                                                                       | —                                 | —                   | —                      |
| R4     | R4                  | CACREF                           | P600      | —           | IRQ30        | OM_0_RST01                                                                              | GTIOC6B/ULPTEVI1-DS               | —                   | —                      |
| R5     | R5                  | —                                | P601      | —           | IRQ29        | SCK0_B/DE0/OM_0_WP1                                                                     | GTIOC6A/ULPTEVI0/RTCOUT           | —                   | —                      |
| R6     | R6                  | —                                | P102      | —           | IRQ17        | TXD9_A/SDA9_A/MOSI9_A/ RSPCKB_A/CRX0/OM_0_SIO4                                          | GTOWLO/GTIOC2B/AGTO0              | ADTRG0              | —                      |
| R7     | R7                  | —                                | P801      | —           | IRQ12        | TXD2_A/SDA2_A/MOSI2_A/ OM_0_DQS                                                         | GTIV/GTIOC11B/AGTOB0              | —                   | —                      |
| R8     | R8                  | —                                | P803      | —           | IRQ19        | SCK2_A/DE2/OM_0_SIO1                                                                    | GTETRG/AGTIOC12B                  | —                   | —                      |
| R9     | R9                  | —                                | P812      | —           | IRQ23        | CTS_RTS7_B/SS7_B/DE7/ USB_EXICEN/SD1DAT4_A/PDMCLK2                                      | GTIOC11A                          | AN022               | —                      |
| R10    | R10                 | —                                | P012      | —           | IRQ15        | —                                                                                       | —                                 | AN012               | —                      |
| R11    | R11                 | —                                | P010      | —           | IRQ14        | —                                                                                       | —                                 | AN010               | —                      |
| R12    | R12                 | —                                | P009      | —           | IRQ13-DS     | —                                                                                       | —                                 | AN009/ IVREF1       | —                      |
| R13    | R13                 | —                                | P805      | —           | IRQ30        | TXD8_A/SDA8_A/MOSI8_A/ ET1_MDIO                                                         | —                                 | AN017/ IVCMP0       | LCD_TCO N1_B/ VIO_D15  |
| R14    | R14                 | —                                | P512      | —           | IRQ14        | CTS8_A/SCL1_A <sup>1</sup> /CTX1/ET1_INT                                                | GTIOC0A                           | —                   | —                      |
| R15    | R15                 | —                                | P413      | A22         | IRQ18        | ET_TAS_STA3                                                                             | GTOUUP/GTCPP07/ ULPTEE1           | —                   | —                      |

Table 1.18 Pin list for the SiP product (8 of 9)

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC       | ADC16H/DAC12/ACMPHS | MIPI/GLCDC/C EU      |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|------------------------|---------------------|----------------------|
| R16    | R16                 | —                                | P515      | —           | IRQ12        | CTS_RTS4_C/SS4_C/DE4/SCL2_B*/ET_TAS_STA0                                                | GTIOC13A               | —                   | LCD_CLK_B            |
| R17    | R17                 | —                                | P709      | CS4         | IRQ10        | CTS_RTS4_B/SS4_B/DE4/SCL2_A*/MISOB_B/ET0_MDIO                                           | GTCPPO5                | —                   | VIO_D13              |
| R18    | R18                 | —                                | P407      | CS6         | IRQ22        | SCK1_C/DE1/SDA0_B*/USB_VBUS/USBHS_VBUSEN/GPTP_PTPOUT3                                   | GTIOC10B/AGTIO0/RTCOUT | ADTRG0              | —                    |
| T1     | T1                  | Vpp                              | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T2     | T2                  | VSS                              | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T3     | —                   | VCC18_MIPI                       | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T4     | —                   | VSS_MIPI                         | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| —      | T3                  | —                                | P315      | —           | IRQ29        | SCK3_C/DE3/SSLA3_A                                                                      | —                      | —                   | —                    |
| —      | T4                  | —                                | P900      | —           | IRQ30        | CTS3_C                                                                                  | GTADSM0                | —                   | —                    |
| T5     | T5                  | —                                | P103      | —           | IRQ16        | CTS_RTS9_A/SS9_A/DE9/SSLB0_A/CTX0/OM_0_SIO2                                             | GTOWUP/GTIOC2A         | AD1FLAG1            | —                    |
| T6     | T6                  | —                                | P101      | —           | IRQ1         | RXD9_A/SCL9_A/MISO9_A/MOSIB_A/OM_0_SIO3                                                 | GTETRGB/GTIOC8A/AGTEE0 | —                   | —                    |
| T7     | T7                  | —                                | P802      | —           | IRQ18        | RXD2_A/SCL2_A/MISO2_A/OM_0_SIO6                                                         | GTIW/GTIOC12A          | —                   | —                    |
| T8     | T8                  | —                                | P804      | —           | IRQ14        | CTS_RTS2_A/SS2_A/DE2/OM_0_SIO7                                                          | GTETRGD/GTIOC13A       | —                   | DSI_TE               |
| T9     | T9                  | —                                | P501      | —           | IRQ25        | TXD8_B/SDA8_B/MOSI8_B/USB_OVRCURA/SD1DAT6_A/PDMDAT1                                     | GTIOC12A               | AN020               | —                    |
| T10    | T10                 | AVCC0                            | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T11    | T11                 | AVSS0                            | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| T12    | T12                 | —                                | P005      | —           | IRQ10-DS     | —                                                                                       | —                      | AN005/IVCMP3        | —                    |
| T13    | T13                 | —                                | P003      | —           | IRQ29        | —                                                                                       | —                      | AN003/IVCMP3        | —                    |
| T14    | T14                 | —                                | P513      | —           | IRQ31        | SCK8_A/DE8/ET0_INT                                                                      | GTIOC13B               | AN016/IVCMP0        | LCD_TCO N3_B/VIO_FLD |
| T15    | T15                 | —                                | P514      | —           | IRQ13        | SCK4_C/DE4/SDA2_B*/ET_TAS_STA1                                                          | GTIOC13B               | —                   | LCD_EXT CLK_B        |
| T16    | T16                 | —                                | P415      | WAIT        | IRQ8         | TXD4_B/SDA4_B/MOSI4_B/RSPCKB_B/CTX1/ET1_MDC                                             | GTIOC0A                | —                   | VIO_HD               |
| T17    | T17                 | —                                | P409      | A18         | IRQ6         | TXD3_A/SDA3_A/MOSI3_A/SDA0_A*/USB_OVRCURA-DS/USBHS_OVRCURA/GPTP_CAPTURE0                | GTOWUP/ULPTOA0         | ADST1               | —                    |
| T18    | T18                 | VCC_USB                          | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| U1     | U1                  | VCC2_13                          | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| U2     | —                   | MIPI_DL0_P                       | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| U3     | —                   | MIPI_CL_P                        | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| U4     | —                   | MIPI_DL1_P                       | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| U5     | —                   | AVCC_MIPI                        | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |
| —      | U2                  | CLKOUT                           | P205      | —           | IRQ1-DS      | TXD4_A/SDA4_A/MOSI4_A/SCL1_B*/SSLA1_A/USB_OVRCURA/SD1CD                                 | GTIV/GTIOC4A/AGTO1     | —                   | —                    |
| —      | U3                  | —                                | P203      | —           | IRQ2-DS      | RXD4_A/SCL4_A/MISO4_A/RSPCKA_A/CTX0/USB_VBUSEN/SD1CLK_A                                 | GTIOC5A/ULPTOA1        | —                   | —                    |
| —      | U4                  | —                                | P313      | —           | IRQ27        | TXD3_C/SDA3_C/MOSI3_C/MISOA_A/USB_ID/SD1DAT0_A                                          | —                      | —                   | —                    |
| —      | U5                  | —                                | P901      | —           | IRQ31        | CTS_RTS3_C/SS3_C/DE3                                                                    | GTADSM1/AGTIO1         | —                   | —                    |
| U6     | U6                  | —                                | P809      | —           | IRQ20        | TXD7_B/SDA7_B/MOSI7_B/OM_0_SCLKN                                                        | —                      | —                   | —                    |
| U7     | U7                  | —                                | P800      | —           | IRQ11        | CTS2_A/OM_0_SIO5                                                                        | GTIU/GTIOC11A/AGTOA0   | —                   | —                    |
| U8     | U8                  | —                                | P502      | —           | IRQ26        | SCK8_B/DE8/USB_OVRCURB/SD1DAT7_A/PDMDAT2                                                | GTIOC12B               | AN019               | —                    |
| U9     | U9                  | —                                | P014      | —           | IRQ27        | —                                                                                       | —                      | AN014/DA0/IVCMP0    | —                    |
| U10    | U10                 | VREFL                            | —         | —           | —            | —                                                                                       | —                      | —                   | —                    |

**Table 1.18 Pin list for the SiP product (9 of 9)**

| BGA303 | BGA303 without MIPI | Power, System, Clock, Debug, CAC | I/O ports | ExBus/SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/USBHS/OSPI/SSIE/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/PDMIF | GPT/AGT/ULPT/RTC      | ADC16H/DAC12/ACMPHS  | MIPI/GLCDC/C EU             |
|--------|---------------------|----------------------------------|-----------|-------------|--------------|-----------------------------------------------------------------------------------------|-----------------------|----------------------|-----------------------------|
| U11    | U11                 | VREFL0                           | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| U12    | U12                 | —                                | P004      | —           | IRQ9-DS      | —                                                                                       | —                     | AN004/IVCMP2         | —                           |
| U13    | U13                 | —                                | P007      | —           | IRQ28        | —                                                                                       | —                     | AN007/IVCMP3         | —                           |
| U14    | U14                 | —                                | P001      | —           | IRQ7-DS      | —                                                                                       | —                     | AN001/IVCMP3         | —                           |
| U15    | U15                 | —                                | P806      | —           | IRQ0         | RXD8_A/SCL8_A/MISO8_A/ET1_MDC                                                           | —                     | AN018                | LCD_TCO<br>N0_B/<br>VIO_D14 |
| U16    | U16                 | —                                | P715      | —           | IRQ12        | RXD4_C/SCL4_C/MISO4_C/<br>ET_TAS_STA2                                                   | GTIOC12A              | —                    | LCD_DAT<br>A23_B            |
| U17    | U17                 | —                                | P815      | —           | IRQ15        | CTX0/USB_DM                                                                             | GTIOC8A               | —                    | —                           |
| U18    | U18                 | VSS_USB                          | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V1     | V1                  | VSS                              | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V2     | —                   | MIPI_DL0_N                       | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V3     | —                   | MIPI_CL_N                        | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V4     | —                   | MIPI_DL1_N                       | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| —      | V2                  | CACREF                           | P204      | —           | IRQ26        | SCK4_A/DE4/SDA1_B*/SSLA0_A/<br>USB_OVRCURB/SD1WP                                        | GTIW/GTIOC4B/AGTIO1   | —                    | —                           |
| —      | V3                  | —                                | P202      | —           | IRQ3-DS      | CTS_RTS4_A/SS4_A/DE4/MOSIA_A/<br>CRX0/USB_EXICEN/SD1CMD_A                               | GTIOC5B/ULPTOB1       | —                    | —                           |
| —      | V4                  | —                                | P314      | —           | IRQ28        | RXD3_C/SCL3_C/MISO3_C/SSLA2_A/<br>SD1DAT1_A                                             | —                     | ADTRG0               | —                           |
| V5     | V5                  | VSS_13                           | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V6     | V6                  | —                                | P808      | —           | IRQ15        | RXD7_B/SCL7_B/MISO7_B/<br>OM_0_SCLK                                                     | GTIOC13B              | —                    | —                           |
| V7     | V7                  | —                                | P100      | —           | IRQ2         | SCK9_A/DE9/MISOB_A/OM_0_SIO0                                                            | GTETRG/GTIOC8B/AGTIO0 | —                    | —                           |
| V8     | V8                  | CACREF                           | P500      | —           | IRQ24        | RXD8_B/SCL8_B/MISO8_B/<br>USB_VBUSEN/SD1DAT5_A/PDMDAT0                                  | GTIOC11B              | AN021                | —                           |
| V9     | V9                  | —                                | P015      | —           | IRQ13        | —                                                                                       | —                     | AN015/DA<br>1/IVCMP0 | —                           |
| V10    | V10                 | VREFH                            | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V11    | V11                 | VREFH0                           | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |
| V12    | V12                 | —                                | P008      | —           | IRQ12-DS     | —                                                                                       | —                     | AN008/<br>IVREF0     | —                           |
| V13    | V13                 | —                                | P006      | —           | IRQ11-DS     | —                                                                                       | —                     | AN006/<br>IVCMP2     | —                           |
| V14    | V14                 | —                                | P000      | —           | IRQ6-DS      | —                                                                                       | —                     | AN000/<br>IVCMP2     | —                           |
| V15    | V15                 | —                                | P002      | —           | IRQ8-DS      | —                                                                                       | —                     | AN002/<br>IVCMP2     | —                           |
| V16    | V16                 | —                                | P511      | —           | IRQ15        | CTS_RTS8_A/SS8_A/DE8/SDA1_A*/<br>CRX1/ET1_LINKSTA                                       | GTIOC0B               | —                    | —                           |
| V17    | V17                 | —                                | P814      | —           | IRQ16        | CRX0/USB_DP                                                                             | GTIOC8B               | —                    | —                           |
| V18    | V18                 | VSS                              | —         | —           | —            | —                                                                                       | —                     | —                    | —                           |

Note: Several pin names have the added suffix of \_A, \_B, and \_C. These suffixes have special conditions for electrical characteristics. See [section 70, Electrical Characteristics](#) for detail.

Note 1. There are two types for IIC function, one is a simple IIC by SCI and another is a dedicated IIC. This terminal is for a dedicated IIC.